A Survey of On-Policy Distillation for Large Language Models

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Abstract

As Large Language Models (LLMs) continue to grow in both capability and cost, transferring frontier capabilities into smaller, deployable students has become a central engineering problem, and knowledge distillation remains the dominant technique for this transfer. The prevailing recipe in industrial pipelines, static imitation of teacher-generated text, carries a structural weakness that grows more severe as tasks become longer and more reasoning-intensive. Because the student is trained on flawless teacher prefixes but must generate its own at inference, small errors tend to accumulate into trajectories it has rarely been trained to recover from, and the resulting exposure bias has been shown to scale roughly with the square of sequence length. On-Policy Distillation (OPD) reorganizes the training loop around this observation by having the teacher provide feedback on what the student *actually produces*, with the goal of reducing the compounding term toward linear and reframing distillation as an iterative correction process rather than single-pass imitation. The resulting literature has expanded along divergence design, reward-guided optimization, and self-play, yet contributions remain scattered across the knowledge distillation, RLHF, and imitation learning communities without a unified treatment. This survey provides such a treatment. We formalize OPD as f-divergence minimization over student-sampled trajectories, organize the field along three design axes (what to optimize, where the signal comes from, and how to stabilize training in practice), and consolidate success conditions, recurring failure modes, and the connection between OPD and KL-constrained reinforcement learning. We close with open problems that emerge from this synthesis, including distillation scaling laws, uncertainty-aware feedback, agent-level distillation, and the growing overlap between knowledge distillation and RL.

Awesome List & Paper Collection
https://github.com/nick7nlp/Awesome-LLM-On-Policy-Distillation

1 Introduction

The capability of Large Language Models (LLMs) has advanced rapidly over the past two years, with recent systems reaching state-of-the-art performance on reasoning, code generation, and multilingual benchmarks (Yang et al., 2025a; DeepSeek-AI et al., 2025; Gemma Team et al., 2024). This progress has come at a training cost that only a few organizations can sustain. Transferring the resulting capabilities from large frontier models into smaller deployable students has therefore become a core step in the modern LLM pipeline. Knowledge distillation, first formalized by Hinton et al. (2015) as a way for a student to inherit the soft output structure of a teacher, has broadened in scope over this period. What began as a model compression technique for transferring knowledge within the same architecture now serves as a general mechanism for moving capabilities across model scales and architectural families. The release of DeepSeek-R1 (DeepSeek-AI et al., 2025) made this shift concrete, distilling a 671B mixture-of-experts teacher into dense students spanning 1.5B to 70B parameters while preserving long chain-of-thought reasoning largely intact.

The dominant recipe behind these successes, however, has a known structural weakness. In almost all large-scale pipelines, distillation proceeds *off-policy*. The student is trained to match the teacher's next-token distribution over a fixed corpus, typically the original pre-training data or traces generated in advance by the teacher, and every gradient step conditions on flawless teacher prefixes. At inference, this assumption no longer holds. The student must generate autoregressively from its own partial outputs, and any divergence between the training states and the states it visits at deployment translates into compounding error along the trajectory. This corresponds to the *compounding error* identified in interactive imitation learning (Ross et al., 2011), where autoregressive generation amplifies the distribution mismatch quadratically in sequence length, producing an expected discrepancy of $O(\epsilon T^2)$ over a horizon T. The consequences are most pronounced on reasoning tasks, where a single early misstep can derail an entire proof or program. Gudibande et al. (2023) report that off-policy imitation of proprietary LLMs often yields students that reproduce surface style without acquiring the underlying reasoning competence.

Addressing this mismatch requires changing *where* the training data comes from rather than *what* is being matched. **On-Policy Distillation (OPD)** addresses this by shifting the sampling distribution from a static corpus to the student's own evolving policy. The student proposes trajectories, populates the states it will visit at deployment, and receives teacher feedback on those states rather than on idealized expert prefixes. Depending on the available teacher access, this feedback ranges from full token-level distributions in white-box settings (Agarwal et al., 2024; Gu et al., 2024), through scalar rewards or pairwise preferences when only black-box APIs are available (Ye et al., 2025; Jiang et al., 2023), to contrastive signals the model derives from its own outputs in teacher-free settings (Chen et al., 2024). The theoretical payoff mirrors the classical DAgger result (Ross et al., 2011), where querying an expert on the learner's own visited states reduces the $O(\epsilon T^2)$ compounding of off-policy imitation to $O(\epsilon T)$. Translated into autoregressive language modeling, this turns distillation from a single-pass matching problem into an iterative optimization loop in which the training distribution co-evolves with the model, allowing distributional gaps between training and deployment to be corrected during training rather than deferred to inference.

The body of work that has emerged from this reformulation can be organized around a small number of recurring design choices. Early methods focused on *how* to sample and *which* divergence to apply on student trajectories (Agarwal et al., 2024; Gu et al., 2024; Ko et al., 2024), establishing that the optimal divergence depends on where in the sequence one is training and what kind of supervision the teacher can supply. A parallel line of work established that the resulting objective is formally equivalent to a KL-constrained form of reinforcement learning (Yang et al., 2026c), bridging two previously separate research communities. A third branch established that the "teacher" need not be external at all, with methods such as SPIN (Chen et al., 2024) and OPSD (Zhao et al., 2026b) showing that a model can bootstrap improvement from its own generations by exploiting asymmetries in its output distribution across contexts. These ideas have since moved from papers into production pipelines, with Qwen3 (Yang et al., 2025a), DeepSeek-V4 (DeepSeek-AI, 2026), Gemma 2 (Gemma Team et al., 2024), and MiMo-V2 (Xiaomi LLM-Core Team et al., 2026) all adopting OPD as a core training ingredient. The adoption cuts across architectural lines, covering both mixture-of-experts and dense systems and spanning proprietary and open-weight release strategies, because all of these systems face the same structural problem once reasoning chains grow long enough for compounding error to dominate. DeepSeek-V4 went the furthest, replacing its mixed RL stage with pure multi-teacher OPD for model consolidation.

GKD (Agarwal et al., 2024), alongside the concurrent MiniLLM (Gu et al., 2024), brought on-policy distillation into the LLM era in mid-2023, and in less than two years the literature has expanded to over one hundred papers spanning divergence design, reward-guided optimization, self-play, multi-teacher debate, agentic trajectory distillation, and cross-modal transfer. The research focus has shifted accordingly. Where early work asked whether on-policy training helps at all, the current frontier asks how to make it efficient enough for trillion-parameter teacher inference, how to extend it beyond single-turn generation into multi-step agent trajectories, and how to unify it with the reinforcement learning pipelines that increasingly share its computational infrastructure. This trajectory parallels a broader shift in LLM development. As models become more capable and reasoning chains grow longer, the gap between off-policy training states and deployment-time states widens, which makes on-policy correction increasingly attractive for the next generation of reasoning-capable systems (Fu et al., 2026; DeepSeek-AI, 2026).

Despite this rapid adoption, the research literature on OPD has yet to catch up with the practice. Methods approach the same problem through the separate lenses of knowledge distillation, RLHF, and interactive imitation learning, inheriting different notations, benchmarks, and failure taxonomies from their parent communities. Existing surveys of LLM distillation (Xu et al., 2024) generally retain the classical compression framing, treating off-policy and on-policy methods as interchangeable variants instead of as regimes with materially different theoretical guarantees and failure modes. As a result, no current treatment we are aware of offers a unified mathematical account of the on-policy dynamics that connect these threads, and a systematic comparison across the white-box, black-box, and teacher-free instantiations that researchers and engineers must now weigh against one another in practice remains missing.

This survey addresses that gap through four contributions that also structure the rest of the paper.

- **A unified theoretical framework.** We recast the transition from off-policy to on-policy distillation as a sequential decision-making problem and show that the core OPD algorithms are instances of f-divergence minimization over student-sampled trajectories (Section 2). This framework supplies a common analytical vocabulary for methods previously studied in isolation and makes their relationships explicit in terms of divergence choice, argument ordering, and sampling mixture.
- **A design-centric taxonomy.** Rather than grouping methods by surface similarity, we organize them along three core design axes (Section 3), namely objective function design (Section 4), signal source architecture (Section 5), and training dynamics optimization (Section 6). The resulting one-method-one-category classification makes the relationships among methods explicit in terms of the specific design choices each paper optimizes.
- **A theoretical account of failure.** We consolidate the empirical record into a coherent view of when and why OPD breaks down (Section 7), covering the flawed prefix trap, self-play saturation, diversity collapse, the calibration-capability gap, and emerging multi-turn agentic failure modes, and we state formal conditions under which OPD provably improves over off-policy alternatives.
- **Comprehensive coverage and comparative analysis.** We survey the white-box, black-box, and self-distillation regimes within this common framework, summarize them in three comparison tables (Tables 1, 2, and 3), and document method selection factors (Section 3.3) that map deployment constraints to applicable methods and compute budgets.

The structure of the paper follows the same design-axis logic. Section 2 develops the mathematical preliminaries, formalizes autoregressive generation as a sequential decision problem, and derives the exposure bias that motivates on-policy training. Section 3 presents the method landscape and selection considerations. Sections 4–6 then traverse the three design axes from "what to optimize" through "where the signal comes from" to "how to stabilize training in practice," with each section building on the choices surfaced in the previous one. Section 7 consolidates the theoretical perspectives, showing that the patterns observed across Sections 4–6 follow from a small number of unifying principles. Section 8 discusses industrial deployments, system-level optimization, and emerging domains, and Section 9 lays out the open problems that emerge from this synthesis.

Our scope is centered on the on-policy regime. We cover methods in which the student generates its own training data during distillation, and we include work from adjacent fields such as imitation learning, online RL, and preference optimization whenever it operates on student-generated data under teacher or verifier supervision. We exclude generic off-policy KD, compression techniques orthogonal to the training regime such as pruning and quantization, and inference-time methods that leave model weights unchanged.

2 Background and Unified Math

To understand why on-policy training is often preferable, we formalize the compounding geometry of autoregressive generation. We begin with classical KD and its limitations, then establish the unified f-divergence framework that subsumes modern OPD objectives.

Defining "on-policy." A distillation method is *on-policy* if the training data for the student is sampled from the student's own current policy p_θ at training time, rather than from a fixed external corpus $\mathcal{D}$ or from the teacher's generation distribution p_T. Formally, on-policy training optimizes:

$$\min_\theta \mathbb{E}_{x\sim\mathcal{D}}\, \mathbb{E}_{y\sim p_\theta(\cdot|x)} \left[\mathcal{L}(y, x; \theta, T)\right] \tag{1}$$

where $\mathcal{L}$ is the distillation loss (divergence, reward, or hybrid). The key distinction is that the *outer* expectation is over the student's own generations, not a static dataset. This creates a non-stationary optimization landscape because as θ updates, the data distribution p_θ shifts, requiring fresh rollouts at each training step. The computational cost of this fresh generation is the central systems-level challenge of OPD (Section 6.3).

Notation. Throughout this survey, we use lowercase p for token-level conditional distributions ($p_T(\cdot|x, y_{<t})$ for teacher, $p_\theta(\cdot|x, y_{<t})$ for student) and uppercase P for sequence-level distributions ($P_T(y|x)$, $P_\theta(y|x)$). $\mathcal{D}$ denotes the training dataset, p_{data} the data distribution, and $|V|$ the vocabulary size. D_{KL} and D_f denote KL divergence and general f-divergence respectively. In the RL formalization, π denotes policies and $R(x, y)$ the outcome reward. When discussing individual methods, we follow each paper's conventions where clarity demands.

2.1 Classical Knowledge Distillation

The original KD formulation (Hinton et al., 2015) trains the student to match the teacher's temperature-softened output distribution:

$$p_T^{(\tau)}(y|x) = \frac{\exp(z_y/\tau)}{\sum_j \exp(z_j/\tau)} \tag{2}$$

where z_y are the teacher's logits and $\tau > 1$ is a temperature parameter. When $\tau \to \infty$, the softened distribution approaches a uniform distribution over the vocabulary, exposing the teacher's "dark knowledge" about inter-class similarities that hard labels hide. The gradient of the distillation loss with respect to the student's logits $z_i^{\mathcal{S}}$ shows why temperature matters:

$$\frac{\partial \mathcal{L}_{\text{KD}}}{\partial z_i^{\mathcal{S}}} = \frac{1}{\tau}(p_i^{\mathcal{S}} - p_i^{\mathcal{T}}) \tag{3}$$

In the high-temperature limit ($\tau \to \infty$), a Taylor expansion $\exp(z_i/\tau) \approx 1 + z_i/\tau$ together with the zero-mean assumption $\sum_i z_i^{\mathcal{S}} = \sum_i z_i^{\mathcal{T}} = 0$ reduces the gradient to $\frac{1}{|V|\tau^2}(z_i^{\mathcal{S}} - z_i^{\mathcal{T}})$ (Hinton et al., 2015), so classical KD in this regime is equivalent to minimizing the mean squared error between raw logits. This forces the student to replicate the teacher's *entire* logit structure, transferring both primary predictive modes and nuanced sub-optimal probability masses (the "dark knowledge"). The $1/\tau^2$ factor explains why the soft-target term is commonly upweighted by τ^2 relative to the hard-label cross-entropy in the combined KD objective. In practice, LLM distillation operates at $\tau = 1$ or low-to-moderate temperatures (Zhong et al., 2024) because the large vocabulary ($|V| > 30{,}000$) already produces richly structured non-peak probabilities without explicit softening, and higher temperatures may amplify noise in the teacher's poorly calibrated tail distribution.

For autoregressive LLMs, the token-level distillation loss factors as:

$$\mathcal{L}_{\text{Token-KD}} = \mathbb{E}_{x,y\sim\mathcal{D}}\left[\sum_{t=1}^{|y|} D_{\text{KL}}(p_T(\cdot|x,y_{<t}) \parallel p_\theta(\cdot|x,y_{<t}))\right] \tag{4}$$

This is computationally tractable because it relies on static, pre-computed prefixes from the dataset, but it optimizes next-token accuracy while assuming an error-free history $y_{<t}$. Kim & Rush (2016) extended this to Sequence-Level Knowledge Distillation (Seq-KD), which instead minimizes the global KL divergence over the entire sequence space:

$$\mathcal{L}_{\text{Seq-KD}} = D_{\text{KL}}(P_T(y|x) \parallel P_\theta(y|x)) = \sum_{y\in\mathcal{Y}} P_T(y|x)\log\frac{P_T(y|x)}{P_\theta(y|x)} \tag{5}$$

Since $|\mathcal{Y}| = |V|^T$ grows exponentially with length, exact computation is intractable. Seq-KD approximates $P_T(y|x)$ with a Dirac delta at the beam-search output $\hat{y} = \arg\max_y P_T(y|x)$, reducing to standard negative log-likelihood on teacher-generated sequences:

$$\mathcal{L}_{\text{Seq-KD}} \approx -\log P_\theta(\hat{y}|x) \tag{6}$$

This approximation works well when the teacher is highly confident but discards all information about the teacher's uncertainty and alternative modes. The practical consequence is twofold. Seq-KD collapses the teacher's distributional richness into a single point estimate, and it still trains on a *static* dataset disconnected from the student's own generation behavior. These two limitations (loss of distributional signal and off-policy training) motivate the richer objectives below.

Subsequent work (Zhong et al., 2024) revisited these foundations for modern LLMs and showed that classical assumptions (small teacher-student gap, shared vocabulary, similar architectural inductive biases) frequently fail when distilling across heterogeneous model families. The gap between classical assumptions and modern reality motivates the more general f-divergence framework below.

From classical KD to on-policy. The historical progression from Hinton's classical KD to modern OPD amounts to a series of relaxations of four classical assumptions: (1) shared vocabulary, (2) i.i.d. data, (3) static teacher, and (4) off-policy training data. Seq-KD (Kim & Rush, 2016) relaxes assumption (2) by operating at the sequence level but retains the static data assumption. GKD (Agarwal et al., 2024) relaxes assumption (4) by training on student-generated sequences. DSKD (Zhang et al., 2025b) relaxes assumption (1) through dual-space alignment. G-OPD (Yang et al., 2026c) relaxes assumptions (2)–(4) simultaneously while adding RL-augmented reward extrapolation that pushes the student beyond the teacher's frontier. Each relaxation targets a specific failure mode that intensifies with model scale: vocabulary mismatch is irrelevant for same-family distillation but catastrophic for cross-family transfer (relaxation 1), while exposure bias is negligible for short sequences but devastating for multi-step reasoning (relaxation 4). This progression is not merely historical but informative: the relevant assumptions in a given deployment scenario determine which level of relaxation is necessary, and unnecessary relaxation introduces avoidable complexity.

2.2 Off-Policy Exposure Bias

The token-level loss in Eq. (4) conditions on ground-truth prefixes $y_{<t}$ drawn from a static dataset. During inference, the student generates its own prefixes, creating a distribution mismatch between training and deployment that compounds over sequence length. We now formalize why this mismatch is catastrophic.

Standard knowledge distillation minimizes the KL divergence over states drawn from the dataset distribution $d_{\mathcal{D}}(s)$:

$$\mathcal{L}_{\text{Off-Policy}} = \mathbb{E}_{x,y\sim p_{\text{data}}}\left[\sum_{t=1}^{|y|} D_{\text{KL}}(p_T(\cdot|x,y_{<t}) \parallel p_\theta(\cdot|x,y_{<t}))\right] \tag{7}$$

At inference, the student acts according to its own policy, inducing a different state visitation distribution $d_{\pi_\theta}(s)$. By the DAgger theorem (Ross et al., 2011), if a policy mimics an expert with per-step error ϵ under the training distribution, the expected total discrepancy over a trajectory of length T under the learner's own state visitation scales as $O(\epsilon T^2)$.

This quadratic compounding has severe practical consequences for reasoning tasks. Even under the simplest independent-error assumption, a mathematical proof requiring 10 reasoning steps with per-step accuracy 95% yields only $(0.95)^{10} \approx 60\%$ trajectory-level correctness. The DAgger bound predicts something strictly worse. Because off-policy training never exposes the student to its own error states, each mistake pushes the model into regions where its next-step accuracy degrades further (distributional shift), producing $O(\epsilon T^2)$ total error rather than the $O(\epsilon T)$ of independent compounding. On-policy training offers a way to mitigate this vicious cycle by exposing the student to its own error states during training, so that it can learn recovery strategies that keep distributional drift from amplifying local mistakes into trajectory-level collapse.

Remark: The DAgger bound in LLMs. While the theoretical reduction from $O(\epsilon T^2)$ to $O(\epsilon T)$ is appealing, applying the DAgger bound to LLMs demands careful qualification. The original theorem assumes an *interactive expert* that emits optimal actions in any state. In white-box OPD, the "expert" emits a next-token distribution $p_T(y_t|\hat{y}_{<t})$ conditioned on the student's prefix $\hat{y}_{<t}$. If the student hallucinates a severely out-of-distribution prefix, the teacher's conditional distribution may itself become poorly calibrated, because the teacher was never trained on such inputs. Forcing the student to match this noisy distribution violates the core assumption of interactive imitation learning, potentially destabilizing training rather than recovering the $O(\epsilon T)$ bound. Jeong (2026) offer direct empirical evidence of this failure. Unconditional token-level matching on student-generated prefixes leads to abrupt output collapse (KL divergence dropping from 2.637 to 0.343 at a single teacher-reset event), directly illustrating the predicted failure mode. This observation motivates the adaptive trust mechanisms surveyed in Section 6.1 and explains why unconditional token-level matching is insufficient for robust OPD.

2.3 The Unified f-Divergence Framework

OPD resolves exposure bias by altering the training expectation to sample from the student's own rollouts, or a mixture policy π_{mix}. The generalized on-policy objective decouples the sampling trajectory from the local matching metric:

$$\mathcal{L}_{\text{OPD}}(\theta) = \mathbb{E}_{y \sim \pi_{\text{mix}}} \left[\sum_{t=1}^{|y|} \mathcal{D}_f \left(p_T(\cdot|x, y_{<t}),\ p_\theta(\cdot|x, y_{<t}) \right) \right] \tag{8}$$

Here, $\mathcal{D}_f$ represents a divergence from the f-divergence family (Wen et al., 2023). Formally, given two distributions P and Q, an f-divergence is defined as:

$$D_f(P \parallel Q) = \mathbb{E}_{y \sim Q} \left[f\left(\frac{P(y)}{Q(y)} \right) \right] \tag{9}$$

where $f : (0, \infty) \to \mathbb{R}$ is a convex generator with $f(1) = 0$. The choice of f determines the implicit weighting of likelihood ratios $p_T(y)/p_\theta(y)$:

- **Forward KL** ($f(u) = u \log u$): Mode-covering (zero-avoiding). The student must allocate probability mass wherever the teacher does, often bridging distinct teacher modes and hallucinating in the inter-mode space.
- **Reverse KL** ($f(u) = -\log u$): Mode-seeking (zero-forcing). The student collapses its mass onto the teacher's single highest peak, ignoring secondary acceptable outputs but guaranteeing high precision.
- **JSD** ($f(u) = u \log u - (u+1) \log \frac{u+1}{2}$): Symmetric, bounded, and smoothly interpolates between mode-covering and mode-seeking behavior.
- **α-divergence**: Parameterized family that continuously interpolates between Forward KL ($\alpha \to 1$) and Reverse KL ($\alpha \to 0$), allowing fine-grained control over the mode-seeking/covering tradeoff.

In practice, these divergences each suit different task geometries. For tasks with a *unique* correct answer (mathematical proofs, code correctness), Reverse KL's mode-seeking behavior concentrates the student on the single optimal solution path. For tasks with *many* acceptable outputs (creative writing, open-ended QA), Forward KL's mode-covering behavior preserves output diversity. JSD offers a compromise but lacks theoretical optimality guarantees for either extreme. All these divergences are computable as expectations under the student's own policy (since $D_f(P_T \parallel P_\theta) = \mathbb{E}_{y \sim p_\theta}[f(p_T(y)/p_\theta(y))]$), making them directly amenable to on-policy optimization. The sampling distribution p_θ is the policy being optimized, so gradients flow through the reparameterized samples without importance weights or off-policy corrections that would add additional variance.

This framework subsumes all modern OPD objectives. The choice of f determines the geometric behavior, while the choice of π_{mix} controls the degree of on-policy exploration. We illustrate by mapping the three foundational methods into this space.

GKD (Agarwal et al., 2024) defines $\pi_{\text{mix}} = \lambda p_\theta + (1-\lambda) p_{\text{data}}$ as an explicit interpolation, and is agnostic to $\mathcal{D}_f$, empirically testing Forward KL, Reverse KL, and JSD. Setting $\lambda \to 1$ makes GKD purely on-policy, while $\lambda = 0$ reduces to standard off-policy KD. The λ parameter offers a smooth dial between the extremes, and moderate values ($\lambda \approx 0.5$) tend to combine the benefits of on-policy exposure with the stability of grounded, dataset-derived prefixes. GKD's experiments report that on-policy sampling ($\lambda = 1$) consistently outperforms off-policy across the divergence choices they tested. JSD performs best on translation tasks, where the output space has moderate diversity (neither a single correct answer nor fully open-ended generation) and the symmetric nature of JSD naturally balances mode-covering and mode-seeking pressures appropriate for this intermediate regime.

Empirically, all three divergences yield competitive results on summarization and instruction-following, validating the exposure bias hypothesis. This suggests that the sampling policy (on-policy vs. off-policy) matters more than the specific divergence choice when task geometry does not strongly favor one extreme.

MiniLLM (Gu et al., 2024) selects Reverse KL as $\mathcal{D}_f = D_{\text{KL}}(p_\theta \parallel p_T)$ and employs $\pi_{\text{mix}} = (1-\alpha) p_\theta + \alpha p_T$ with $\alpha = 0.2$ for stability. Because Reverse KL places the student in both the expectation and the log-ratio, MiniLLM reformulates optimization via REINFORCE, treating $\log(p_T/p_\theta)$ as a per-step reward.

DistiLLM (Ko et al., 2024) computes its token-level loss on student-generated prefixes but uses an adaptive scheduler that gradually increases the fraction of student-generated outputs while caching earlier rollouts in a replay buffer for sample efficiency (an "adaptive off-policy" strategy in the authors' terminology). Its core contribution is a skewed mixture target $\tilde{p} = \alpha p_T + (1-\alpha) p_\theta$, computing $D_{\text{KL}}(p_T \parallel \tilde{p})$. This avoids zero-division instability when $p_\theta(y) \approx 0$, because $\tilde{p}(y) \geq \alpha p_T(y) > 0$ always, achieving tractability and stability without policy gradients.

Comparative synthesis. These three methods expose a core tradeoff triangle in OPD design. GKD prioritizes *generality* (divergence-agnostic) and *simplicity* (no REINFORCE variance) at the cost of limited theoretical guidance on divergence selection. MiniLLM commits to Reverse KL for its mode-seeking precision but inherits REINFORCE's high variance, requiring reward baselines, length penalties, and careful α tuning. DistiLLM avoids both the divergence selection problem and the variance problem by engineering a mixture target that is numerically stable by construction, but at the cost of adding a replay buffer (mixing on-policy and off-policy data) and an adaptive scheduler that adds hyperparameter complexity. The progression from GKD through MiniLLM to DistiLLM thus traces a shift from algorithmic simplicity toward computational and engineering sophistication, with each method solving a specific failure mode of its predecessor.

Modern OPD, viewed through this decomposition, is not a series of disparate heuristics but a systematic exploration of a design space governed by three interacting choices, namely trajectory sampling (π_{mix}), divergence generator (f), and argument ordering within the chosen divergence.

2.4 Distillation Scaling Laws

The unified f-divergence framework characterizes *what* to optimize, but leaves open *how performance scales* with computational investment. Unlike pre-training, where Hoffmann et al. (2022) established compute-optimal scaling laws (the Chinchilla laws), distillation creates a qualitatively different resource allocation problem: the budget must be split between teacher inference, student rollouts, and gradient updates, with non-trivial interactions between these axes.

Busbridge et al. (2025) initiated the study of distillation-specific scaling and found that the teacher-student relationship obeys qualitatively different dynamics from pre-training alone. Their key finding is that a *capacity gap* emerges when the teacher becomes too powerful relative to the student. The student's loss follows a power law in teacher cross-entropy that transitions between two regimes: initially, stronger teachers improve the student (denser gradient signal from better-calibrated next-token probabilities), but beyond a critical teacher capacity the student's ability to absorb the signal saturates. This capacity-gap phenomenon connects directly to the divergence choice in Section 2.3: Forward KL's mode-covering pressure exacerbates the gap (the student must spread mass across all teacher modes, including those beyond its capacity), while Reverse KL's mode-seeking behavior partially mitigates it (the student selectively matches its highest-confidence subset).

No comprehensive framework yet predicts how distillation quality scales jointly with teacher size N_T, student size N_S, data volume D, and on-policy rollout budget R. A natural extension of the Busbridge law might take the form Quality $\propto N_T^{\alpha} \cdot N_S^{\beta} \cdot D^{\gamma} \cdot R^{\delta}$, but pinning down the exponents and interaction terms for on-policy methods remains open (Section 9). The rollout budget R is unique to on-policy distillation, adding a new scaling axis absent from both pre-training and off-policy distillation. Each rollout step requires a full forward pass through the student, making compute cost scale linearly with R while the learning benefit likely exhibits diminishing returns.

3 Landscape and Method Selection

Before examining individual methods in detail, this section provides a structured overview of the field: the organizational taxonomy, a comparative table of representative methods, and a discussion of method selection considerations.

3.1 Method Landscape

Existing OPD methods can be organized along three axes that correspond to sequential design decisions in the training pipeline: (1) the objective function to optimize, (2) the source of the supervisory signal, and (3) the mechanism for stabilizing training dynamics. Figure 1 illustrates this organization.

To illustrate the taxonomy in practice, GKD (Agarwal et al., 2024) selects forward KL (or JSD) as its divergence objective, relies on white-box teacher logits as its signal source, and interpolates between student-generated and ground-truth sequences (π_{mix}) to stabilize on-policy training dynamics. The DeepSeek-R1 distilled models (DeepSeek-AI et al., 2025) follow the same three-stage logic at industrial scale, adopting cross-entropy as the objective, 800K reasoning chains generated by the R1 teacher as the signal source, and cosine learning-rate decay with difficulty-aware curricula for stability.

The three stages interact in both constraining and reinforcing ways. Some combinations are incompatible: forward KL in its exact token-level form requires the teacher's full output distribution, ruling out API-constrained settings where only generated text is available. Other combinations reinforce each other, as RL-augmented objectives naturally couple with external feedback sources (verifiers, reward models), while fixed divergences align with white-box logit access that permits exact gradient computation. An objective choice therefore constrains the viable signal sources and dynamics strategies.

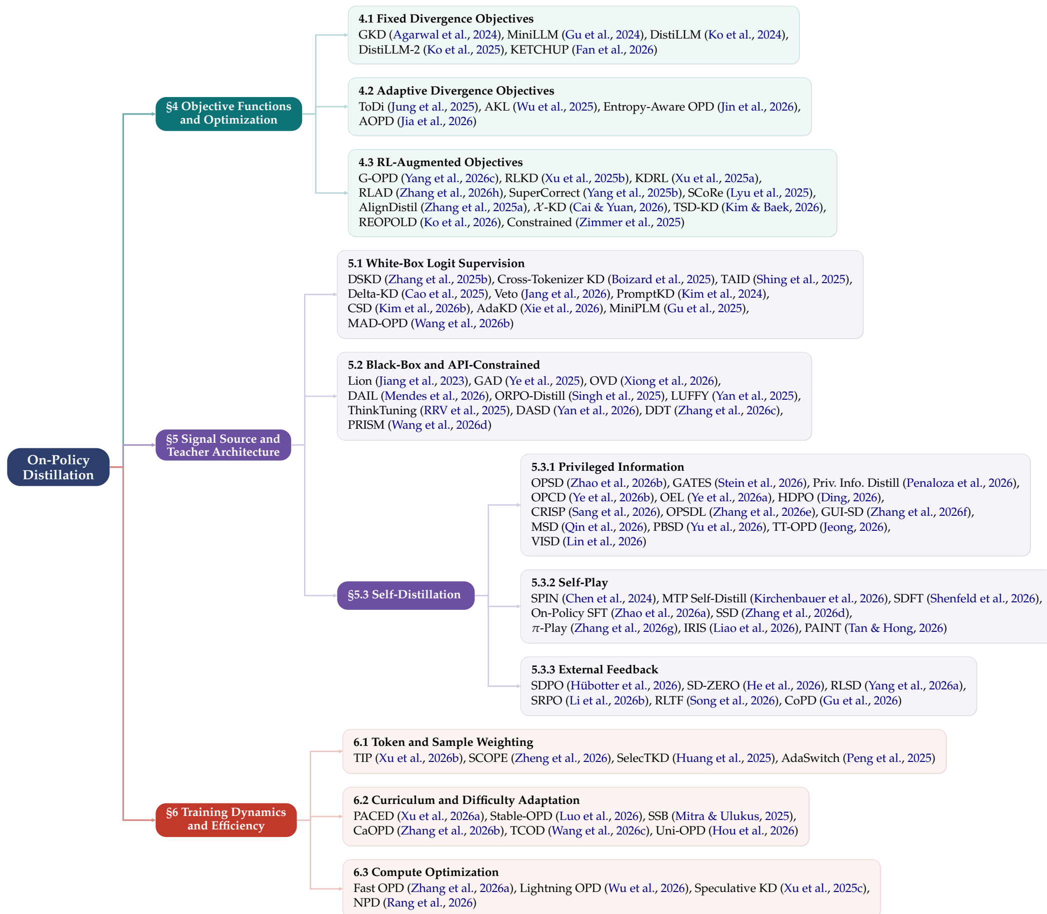

Figure 1: Taxonomy of On-Policy Distillation methods organized along three design axes: (1) Objective function design (§4), (2) Signal source and teacher architecture (§5), and (3) Training dynamics and efficiency (§6). Each leaf lists representative methods with a badge showing the method count. Methods are placed under their primary contribution; those contributing to secondary dimensions (e.g., PAINT and PRISM contributing implicit curricula) are discussed in the relevant section prose rather than duplicated in the tree.

Chronologically, the distribution of methods reflects shifting research emphasis. Early work (2023–2024) concentrated on the objective axis, debating forward KL, reverse KL, and JSD (Agarwal et al., 2024; Gu et al., 2024). By mid-2025, the focus shifted to signal architecture, particularly self-distillation methods that eliminate external teacher dependence. The most recent work (late 2025–2026) addresses training dynamics, specifically the instabilities that on-policy sampling introduces but static-dataset distillation avoids. Industrial systems such as DeepSeek-V4 and Qwen3 increasingly combine all three axes, pairing adaptive or RL-augmented objectives with multi-teacher signals and curriculum-based stabilization (Yang et al., 2026c; 2025a).

3.2 Method Comparison Table

Classification methodology. Each method is assigned to exactly one primary category based on its *core contribution dimension*. A paper whose novelty lies in the loss function (e.g., G-OPD's KL-constrained RL reformulation) belongs to "Objective." A new signal architecture (e.g., DSKD's dual-space projection) places a method under "Signal." A training stability or efficiency contribution (e.g., TIP's token weighting) maps to "Dynamics." Methods that contribute to multiple dimensions are classified by their *most distinctive* contribution, reducing the categorization ambiguity that arises when a single method spans multiple dimensions (Xu et al., 2024).

Table organization. Table 1 provides a compact reference mapping 15 representative methods to their primary category, signal source, and loss granularity. Table 2 extends this to the full inventory of 61 methods with year, divergence type, and key innovation. Table 3 complements both by listing concrete teacher-student pairings and benchmark results across six application domains.

Distributional observations. Self-distillation is the largest single category with 18 methods (30% of all entries), nearly all published in 2026. Training Dynamics follows with 13 methods (21%), reflecting concentrated recent work on stabilizing and accelerating on-policy rollouts. In contrast, fixed-divergence objectives (the earliest OPD formulations from 2024) account for only 6 entries, consistent with a gradual shift from the foundational "which KL direction" question toward adaptive or RL-augmented alternatives. These proportions also match a broader trend in LLM research in which the model's own rollouts increasingly serve as the primary data source, whether or not an external teacher is present.

Table 1: Comparison of key OPD methods categorized by their primary contribution.

Method	Category	Core Contribution	Signal Source	Loss Granularity
GKD (Agarwal et al., 2024)	Objective (Fixed)	DAgger for LLMs, interpolation	White-Box Logits	Token
DistiLLM (Ko et al., 2024)	Objective (Fixed)	Skewed KL mixtures	White-Box Logits	Token
MiniLLM (Gu et al., 2024)	Objective (Fixed)	Sequence-level Reverse KL via RL	White-Box Logits	Sequence
ToDi (Jung et al., 2025)	Objective (Adaptive)	Log-ratio-routed token divergence	White-Box Logits	Token
Entropy-Aware (Jin et al., 2026)	Objective (Adaptive)	Entropy-based KL interpolation	White-Box Logits	Token
G-OPD (Yang et al., 2026c)	Objective (RL)	OPD equivalence to KL-RL	White-Box Logits	Sequence/Token
RLKD (Xu et al., 2025b)	Objective (RL)	KD augmented with reward model	Reward Model	Sequence
DSKD (Zhang et al., 2025b)	Signal (White-Box)	Cross-architecture alignment	White-Box Logits	Token
Delta-KD (Cao et al., 2025)	Signal (White-Box)	Base-to-Instruct delta isolation	White-Box Logits	Token
Lion (Jiang et al., 2023)	Signal (Black-Box)	Verbal feedback curriculum	Black-Box API	Sequence
GAD (Ye et al., 2025)	Signal (Black-Box)	Adversarial distribution matching	Black-Box API	Sequence
OPSD (Zhao et al., 2026b)	Signal (Self)	Privileged Info (PI) conditioning	Self-Generated	Token
SPIN (Chen et al., 2024)	Signal (Self)	Self-play vs reference responses	Self-Generated	Sequence
TIP (Xu et al., 2026b)	Dynamics (Weight)	Token importance profiling via diff	White-Box Logits	Token
PACED (Xu et al., 2026a)	Dynamics (Weight)	Beta-kernel curriculum pacing	White-Box Logits	Token

3.3 Method Selection Considerations

Table 3 suggests that the most successful deployments tend to make deliberate choices at each of the three stages rather than optimizing a single axis in isolation. The constraints and interactions among these axes appear to reduce the space of viable configurations substantially, and we summarize the key selection factors below.

Teacher access constraints. The available signal source determines which objectives are feasible. White-box logit access enables exact token-level divergence computation, supporting methods such as DistiLLM (Ko et al., 2024), G-OPD (Yang et al., 2026c), and cross-tokenizer approaches like DSKD (Zhang et al., 2025b) and Cross-Tok. KD (Boizard et al., 2025). API-only access restricts supervision to generated text or scalar scores, favoring methods designed for sparse feedback such as OVD (Xiong et al., 2026), GAD (Ye et al., 2025), and LUFFY (Yan et al., 2025). When no external teacher is available, self-distillation methods operate either through privileged information (OPSD (Zhao et al., 2026b), GATES (Stein et al., 2026)) or through verifier-guided self-play (SD-ZERO (He et al., 2026), SSD (Zhang et al., 2026d)).

Task characteristics. Reasoning and mathematical tasks benefit from mode-seeking objectives (reverse KL or RL-augmented variants) that concentrate probability mass on correct solutions (Gu et al., 2024; Yang et al., 2026c; Xu et al., 2025a). Open-ended generation tasks instead require mode-covering objectives (forward KL or adaptive entropy-aware diver-

Table 2: Comprehensive comparison of on-policy distillation methods organized by primary contribution dimension.

Method	Year	Category	Divergence/Objective	Signal	Granularity	Key Innovation
Objective: Fixed Divergences						
GKD (Agarwal et al., 2024)	2024	Objective	F-KL / R-KL / JSD	White-box	Token	Unified OPD framework, DAgger for LLMs
MiniLLM (Gu et al., 2024)	2024	Objective	Reverse KL	White-box	Sequence	REINFORCE with teacher as reward
DistiLLM (Ko et al., 2024)	2024	Objective	Skew KL (SKL+SRKL)	White-box	Token	α-smoothing for KL stability
DistiLLM-2 (Ko et al., 2025)	2025	Objective	Contrastive SKL/SRKL	White-box	Token	Asymmetric loss per data source
KETCHUP (Fan et al., 2026)	2025	Objective	Reverse KL	White-box	Sequence	K-step truncated REINFORCE
Constrained (Zimmer et al., 2025)	2025	Objective	Constrained KL	White-box	Sequence	State-augmented CMDP
Objective: Adaptive Divergences						
AKL (Wu et al., 2025)	2025	Objective	Adaptive FKL+RKL	White-box	Token	Head/tail gap-weighted blending
ToDi (Jung et al., 2025)	2025	Objective	Log-ratio-weighted FKL/RKL	White-box	Token	Sigmoid per-token blending
Entropy-Aware (Jin et al., 2026)	2026	Objective	Continuous FKL+RKL blend	White-box	Token	Smooth entropy interpolation
AOPD (Jia et al., 2026)	2026	Objective	Asymmetric PG/F-KL switch	White-box	Token	Advantage-sign token-level switching
Objective: RL-Augmented						
G-OPD (Yang et al., 2026c)	2026	Objective	KL-Constrained RL	White-box	Token/Seq	OPD $\equiv$ dense KL-RL, reward extrapolation
RLKD (Xu et al., 2025b)	2025	Objective	KD + Reward Model	Reward Model	Sequence	Structure-aware reward model (GSRM)
KDRL (Xu et al., 2025a)	2025	Objective	Joint KD + RL	White-box	Token/Seq	On-policy KL regularizer during RL
RLAD (Zhang et al., 2026h)	2026	Objective	Trust Region Ratio	White-box	Token/Seq	PPO-style selective teacher following
AlignDistil (Zhang et al., 2025a)	2025	Objective	DPO + Reverse DPO	White-box	Sequence	Synthetic preference from rollouts
SuperCorrect (Yang et al., 2025b)	2025	Objective	Template + Cross-DPO	White-box	Sequence	Hierarchical thought scaffolding
SCoRe (Lyu et al., 2025)	2025	Objective	Earliest-error correction	White-box	Sequence	Root-cause error targeting
Signal: White-Box Innovations						
DSKD (Zhang et al., 2025b)	2025	Signal	Dual-Space KL	White-box	Token	Cross-vocabulary projection
Cross-Tok. KD (Boizard et al., 2025)	2025	Signal	Latent OT alignment	White-box	Token	Optimal transport for tokenizer mismatch
Delta-KD (Cao et al., 2025)	2025	Signal	Base-to-Instruct delta	White-box	Token	Signal isolation
PromptKD (Kim et al., 2024)	2024	Signal	Prompt-based elicitation	White-box	Token	Input-side distillation
TAID (Shing et al., 2025)	2025	Signal	Temporal target interpolation	White-box	Token	Progressive target revelation
MiniPLM (Gu et al., 2025)	2025	Signal	Pre-training KD	White-box	Token	OPD during pre-training
MAD-OPD (Wang et al., 2026b)	2026	Signal	Multi-agent debate ensemble	White-box	Token	Confidence-weighted multi-teacher
Training Dynamics						
TIP (Xu et al., 2026b)	2026	Dynamics	Soft-OR token weighting	White-box	Token	Entropy$\times$divergence quadrant scoring
SCOPE (Zheng et al., 2026)	2026	Dynamics	Rollout routing	White-box	Token	Redirect on flawed prefix
SelecTKD (Huang et al., 2025)	2025	Dynamics	Confidence filtering	White-box	Token	Top-1 probability threshold
AdaSwitch (Peng et al., 2025)	2025	Dynamics	Exploration/guidance switch	White-box	Token	Binary mode switching
PACED (Xu et al., 2026a)	2026	Dynamics	Beta-kernel curriculum	White-box	Sample	Frontier difficulty sampling
Fast OPD (Zhang et al., 2026a)	2026	Dynamics	Prefix truncation	White-box	Hybrid	Early-token focus
Lightning OPD (Wu et al., 2026)	2026	Dynamics	Offline caching	White-box	Token	$4\times$ cost reduction
Speculative KD (Xu et al., 2025c)	2025	Dynamics	Draft-verify	White-box	Block	Amortized teacher cost
NPD (Rang et al., 2026)	2026	Dynamics	Async gen-train decoupling	White-box	Token	$8.1\times$ speedup via Δ-IFD filtering
Revisiting OPD (Fu et al., 2026)	2026	Dynamics	Top-p rollout + masking	White-box	Token	Empirical failure mode fixes
Stable-OPD (Luo et al., 2026)	2026	Dynamics	R-KL + reference div.	White-box	Token	Rollout mixing for stability
Veto (Jang et al., 2026)	2026	Dynamics	Geometric bridge KL	White-box	Token	Adaptive gradient veto
TCOD (Wang et al., 2026c)	2026	Dynamics	Temporal curriculum	White-box	Trajectory	Multi-turn KL stability
Uni-OPD (Hou et al., 2026)	2026	Dynamics	Dual-perspective data balancing	White-box (multi-teacher)	Token	Unified LLM+MLLM OPD
Self-Distillation						
SPIN (Chen et al., 2024)	2024	Self-Play	Self-play DPO	Self	Sequence	Iterative self-vs-reference game
IRIS (Liao et al., 2026)	2026	Self-Play	Interpolative Rényi self-play	Self	Sequence	Unifies SPIN/SPACE/SPIF
OPSD (Zhao et al., 2026b)	2026	Self (PI)	KL on GT-conditioned	Self	Token	Ground-truth as privileged info
CRISP (Sang et al., 2026)	2026	Self (PI)	Conciseness PI	Self	Token	57% token reduction, +9% accuracy
GATES (Stein et al., 2026)	2026	Self (PI)	Gated consensus KL	Self	Token	Document as PI, gated supervision
OPSDL (Zhang et al., 2026e)	2026	Self (PI)	Short-ctx $\rightarrow$ long-ctx	Self	Token	Long-context self-distillation
SD-ZERO (He et al., 2026)	2026	Self (EF)	Self-revision preference	Self + Verifier	Sequence	Binary reward $\rightarrow$ dense self-supervision
π-Play (Zhang et al., 2026g)	2026	Self (PI+SP)	Multi-agent co-evolution	Self	Token	Internally generated PI from self-play
SSD (Zhang et al., 2026d)	2026	Self (SP)	Temperature-sampled SFT	Self	Sequence	Minimalist self-play
SDPO (Hübotter et al., 2026)	2026	Self (EF)	Textual feedback + DPO	Self + Verifier	Sequence	Credit assignment from outcomes
RLTF (Song et al., 2026)	2026	Self (EF)	Critique-conditioned	Self + Verifier	Sequence	Text feedback for self-play
SRPO (Li et al., 2026b)	2026	Self (EF)	GRPO + SDPO routing	Self + Verifier	Sequence	Sample-routed multi-objective
CoPD (Gu et al., 2026)	2026	Self (EF)	Co-evolving bidirectional OPD	Mutual (parallel)	Sequence	Multi-capability consolidation
PAINT (Tan & Hong, 2026)	2026	Self (PI)	Partial-solution masking	Self (re-score)	Token	Overlap-adaptive self-distillation
MSD (Qin et al., 2026)	2026	Self (PI)	Cross-lingual safety distill	Self (English PI)	Token	Multilingual safety alignment
GUI-SD (Zhang et al., 2026f)	2026	Self (PI)	Visual spatial PI for GUI	Self (Gaussian mask)	Token	Multimodal GUI grounding
PBSD (Yu et al., 2026)	2026	Self (PI)	Pref.-based reward-reg. KL	Self (ctx-augmented)	Sequence	DPO-style loss, preference margin
TT-OPD (Jeong, 2026)	2026	Self (PI)	Turn-level truncated KL	Self (EMA + outcome hints)	Turn	Multi-turn agentic OPD
VISD (Lin et al., 2026)	2026	Self (PI)	Direction-magnitude decouple	Self (structured judge)	Token	Structured PI for video reasoning
Black-Box						
GAD (Ye et al., 2025)	2025	Signal	Adversarial matching	Black-box	Sequence	Discriminator as proxy reward
Lion (Jiang et al., 2023)	2023	Signal	Verbal feedback	Black-box	Sequence	NL critique curriculum
OVD (Xiong et al., 2026)	2026	Signal	Verbal scores (0–9)	Black-box	Sequence	Score-based trajectory matching
DAIL (Mendes et al., 2026)	2026	Signal	Mixed-policy decoding	Black-box	Sequence	Didactic-to-constructive learning
LUFFY (Yan et al., 2025)	2025	Signal	Mixed-policy GRPO	Off-policy + RL	Sequence	Importance-weighted off-policy
ORPO-Distill (Singh et al., 2025)	2025	Signal	Preference optimization	Black-box	Sequence	Teacher-ranked pairs
PRISM (Wang et al., 2026d)	2026	Signal	Black-box pre-alignment	Black-box (API)	Sequence	Multimodal OPD before RLVR

gences) that preserve output diversity. Instruction-following tasks occupy a middle ground where moderate on-policy mixing ($\lambda \geq 0.5$) with JSD provides strong quality-compute tradeoffs (Agarwal et al., 2024). Multi-step agentic tasks demand sequence-level objectives with trajectory-aware credit assignment (Lyu et al., 2025; Zhang et al., 2026h).

Compute budget. Under constrained compute (<500 GPU-hours), prefix-truncated rollouts (Zhang et al., 2026a) and offline caching strategies (Wu et al., 2026) reduce on-policy overhead substantially. At medium scale, full on-policy training with adaptive divergences (ToDi (Jung et al., 2025), AKL (Wu et al., 2025)) becomes feasible. Large-scale deployments (>5000 GPU-hours) typically adopt multi-stage pipelines that begin with off-policy warmup, transition to on-policy refinement, and optionally add RL-based exploration, as reported for Qwen3 (Yang et al., 2025a).

Stability requirements. Common failure modes motivate specific stabilization choices. Length inflation during on-policy training can be mitigated by reference divergence constraints and rollout mixing (Luo et al., 2026). The flawed prefix trap, where teacher feedback on student-generated errors produces misleading gradients, is addressed by token-level

Table 3: Representative experimental configurations of OPD methods across different application domains.

Method	Teacher	Student	Training Signal	Key Benchmarks
Mathematical Reasoning				
G-OPD (Yang et al., 2026c)	Qwen3-30B-A3B-Instruct	Qwen3-1.7B/4B	Dense logit + reward extrap.	MATH, GSM8K, Code
OPSD (Zhao et al., 2026b)	Self (GT-conditioned)	Qwen3-1.7B/4B/8B-Inst	GT-PI self-distillation	AIME, HMMT
PACED (Xu et al., 2026a)	Qwen3-14B	Qwen3-8B	Beta-kernel curriculum	MATH-500, AIME
KDRL (Xu et al., 2025a)	Skywork-OR1-Math-7B	R1-Distill-Qwen-1.5B	Joint KD + RL	AIME, MATH
LUFFY (Yan et al., 2025)	DeepSeek-R1 (off-policy traces)	Qwen2.5-Math-7B/1.5B	Mixed-policy GRPO	AIME, AMC, MATH-500
AOPD (Jia et al., 2026)	Qwen2.5-Math-7B (teacher)	Qwen2.5-Math-1.5B/7B	Asymmetric PG + F-KL	AIME'24/'25, MATH-500
SCoRe (Lyu et al., 2025)	Qwen2.5-72B-Inst	Qwen2.5-7B/3B, Llama-3.1-8B	Earliest-error correction	12 Agent benchmarks
Instruction Following & General				
GKD (Agarwal et al., 2024)	T5-XL (3B)	T5-Small/Base	On-policy FKL/RKL/JSD	XSum, WMT
DistiLLM (Ko et al., 2024)	GPT-2 XL (1.5B)	GPT-2 (124M)	Skewed KL	ROUGE-L, Dolly
AlignDistil (Zhang et al., 2025a)	Qwen2.5-72B-Inst	Qwen2.5-1.5B-Inst	DPO + token-level KD	AlpacaEval, MT-Bench
Industrial Scale				
Qwen3 (Yang et al., 2025a)	Qwen3-32B/235B-A22B	Qwen3-0.6B–14B	On-policy logit distillation	AIME, MATH, LiveCode
Gemma 2 (Gemma Team et al., 2024)	Gemma 2 27B	Gemma 2 9B / 2B	KD in pre-training	MMLU, HellaSwag
MiMo-V2 (Xiaomi LLM-Core Team et al., 2026)	Domain specialists	MiMo-V2-Flash (309B/15B MoE)	Multi-teacher logit + reward	AIME, MATH, GPQA
KAT-Coder-V2 (Li et al., 2026a)	5 domain specialists	Unified agentic coder	Specialize-then-Unify OPD	SWE-bench (79.6%)
Nemotron-Cascade 2 (Yang et al., 2026d)	Domain RL/SFT teachers	30B MoE (3B active)	Cascade RL + domain OPD	IMO, IOI, ICPC
DeepSeek-V4 (DeepSeek-AI, 2026)	10+ domain experts (1.6T)	DeepSeek-V4-Pro (1.6T MoE)	Full-vocab multi-teacher R-KL	AIME, LiveCode, HMMT
ORBIT (Liang et al., 2026)	Multi-mode RL experts	DeepSeek-Distill-Qwen-1.5B, Qwen3-4B, Openmath-Nemotron-7B	Multi-teacher mode fusion	AIME, GPQA, MMLU-Pro
Multimodal & Domain-Specific				
VOLD (Bousselham et al., 2025)	Qwen3-8B (text-only)	Qwen2.5-VL-3B	GRPO + token-level KL	MMMU-Pro, MathVision
CORD (Hu et al., 2026)	Text-mode LALM	Audio-mode LALM	Cross-modal R-KL + GRPO	Audio QA/reasoning
Speculative KD (Xu et al., 2025c)	Gemma-7B-IT, Qwen2-7B	Gemma-2B, Qwen2-0.5B	Block-verified on-policy KL	Translation, Dialogue
Cross-Tok. KD (Boizard et al., 2025)	Llama-2-7B, Mistral-7B	OPT-350M, Pythia-160M–1B	Latent OT alignment	QA, Summarization
DP-OPD (Khadem et al., 2026)	GPT-2 Large 774M	DistilGPT-2 82M	GKD + DP-SGD (student only)	Yelp, BigPatent
Veto (Jang et al., 2026)	Qwen2-7B-IT	Qwen2-0.5B-IT	Geometric bridge KL	Reasoning, Generation
TAID (Shing et al., 2025)	GPT-2 XL, Llama-3-8B	GPT-2 Base/Med, Llama-3-2B	Temporal interpolation	Summarization, QA
GUI-SD (Zhang et al., 2026f)	Self (visual PI)	Qwen3-VL-Inst-8B	Entropy-guided KL	ScreenSpot-v2/Pro, OSWorld
MAD-OPD (Wang et al., 2026b)	Multi-teacher debate	Qwen3/3.5 (1.7B–14B)	Confidence-weighted JSD/R-KL	Agentic, Code
MSD (Qin et al., 2026)	Self (English CoT PI)	Qwen2.5-7B, Llama-3-8B	DPSW cross-lingual distill	Multilingual Safety
NPD (Rang et al., 2026)	Teacher (parallel prefill)	openPangu-Embedded-1B	Async CE+KD + Δ-IFD	GSM8K, MATH (8.1× speedup)
VISD (Lin et al., 2026)	Self (video-aware judge)	VideoLLM	Dir.-mag. decoupled KL + RL	Open-o3-Video benchmarks
Uni-OPD (Hou et al., 2026)	Multi-teacher (LLM+MLLM)	Qwen3 (1.7B–4B), Qwen3-VL-4B	Dual-perspective OPD	Math, Code, MLLM (16 benchmarks)
Self-Distillation				
SPIN (Chen et al., 2024)	Self (iter $t-1$)	Zephyr-7B-SFT	Self-play DPO	Open LLM, MT-Bench
CRISP (Sang et al., 2026)	Self (concise prompt)	Qwen3-8B/14B	Per-token R-KL	AIME, MATH-500
SD-ZERO (He et al., 2026)	Self (revision-conditioned)	Qwen3-4B-Inst, OLMo-3-7B	On-policy self-distill	AIME, MATH, Codeforces
π-Play (Zhang et al., 2026g)	Self (QCP-conditioned)	Qwen3-4B/8B	Multi-agent self-distill	NQ, TriviaQA, HotpotQA
SSD (Zhang et al., 2026d)	Self (temperature-sampled)	Qwen3-30B-Inst	On-policy SFT	LiveCodeBench v6
SDPO (Hübotter et al., 2026)	Self (textual feedback)	Qwen3-8B	Credit-assignment self-distill	Code, Math
SRPO (Li et al., 2026b)	Self (SDPO+GRPO)	Qwen3-8B	Sample-routed dual-objective	Chem, Phys, Bio, Mat, ToolUse
PBSD (Yu et al., 2026)	Self (ctx-augmented ref.)	Qwen3-1.7B/4B/8B-Inst	Pref.-based reward-reg. KL	AIME'24/'25, HMMT'25, Tool Use
TT-OPD (Jeong, 2026)	Self (EMA + outcome hints)	Qwen3.5-9B	Turn-level truncated KL	Healthcare AI Gym (18 benchmarks)
OPSD-Comp. (Kim & Lee, 2026)	Self (GT-conditioned)	Qwen3-8B, R1-Distill-7B, AceReason-7B	Correct-only OPSD	MATH-500, AIME'24/'25
Black-Box				
GAD (Ye et al., 2025)	GPT-5-Chat (API only)	Qwen2.5-14B-Inst	Adversarial minimax	LMSYS-Chat, Dolly
Lion (Jiang et al., 2023)	ChatGPT (API only)	LLaMA-7B/13B	NL critique curriculum	BIG-Bench Hard, AGIEval
OVD (Xiong et al., 2026)	QwQ-32B / Env. Agent	Qwen2.5-3B, Llama-3.2-3B	Verbal scores (0–9)	Web QA, MATH

reliability filtering (Xu et al., 2026b; Zheng et al., 2026). Thinking-pattern mismatch between teacher and student is typically resolved through an off-policy cold-start phase (Li et al., 2026d) before transitioning to on-policy training.

4 Objective Functions and Optimization

The objective function determines what mathematical quantity the student optimizes at each training step, governing gradient geometry, mode coverage, and ultimately the achievable performance ceiling. Methods in this category have progressed from fixed divergences that apply a single metric uniformly, through adaptive divergences that modulate the metric per-token based on local distributional geometry, to RL-augmented objectives that incorporate reward signals beyond the teacher's output distribution.

4.1 Fixed Divergence Objectives

GKD (Agarwal et al., 2024) established the canonical OPD framework by adapting the DAgger algorithm (Ross et al., 2011) for autoregressive language models. It samples trajectories from a mixture policy $\pi_{\text{mix}} = \lambda p_\theta + (1-\lambda) p_{\text{data}}$ and applies standard f-divergences (Forward KL, Reverse KL, or JSD) at each token position. The mixture coefficient λ interpolates between fully off-policy ($\lambda = 0$) and fully on-policy ($\lambda = 1$), serving as a single control for exposure bias mitigation. On instruction-following and summarization benchmarks, moderate on-policy mixing ($\lambda \geq 0.5$) consistently outperforms pure off-policy training, validating the distributional alignment hypothesis.

Pure KL divergences, though elegant, exhibit numerical instability during on-policy exploration. When the student generates tokens to which the teacher assigns near-zero probability, Forward KL produces unbounded gradients. Reverse KL ignores these regions entirely, creating blind spots in the loss landscape. DistiLLM (Ko et al., 2024) tackles this through

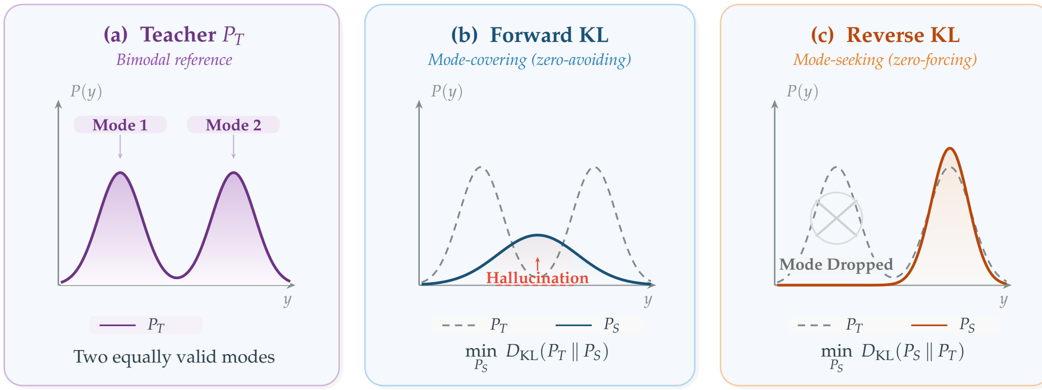

Figure 2: Forward KL vs. Reverse KL divergence for fitting a student distribution P_S to a bimodal teacher P_T. **(a)** The teacher places equal probability mass on two valid modes (e.g., two correct answers or two stylistic variants). **(b)** Forward KL minimizes $D_{\mathrm{KL}}(P_T \parallel P_S)$ and is mode-covering (zero-avoiding): the student spreads a single broad Gaussian over both modes and places substantial probability in the inter-mode hallucination zone (red dashed boundary), corresponding to outputs that are a blend of the two valid answers and therefore neither. **(c)** Reverse KL minimizes $D_{\mathrm{KL}}(P_S \parallel P_T)$ and is mode-seeking (zero-forcing): the student concentrates sharply onto a single peak (Mode 2), dropping Mode 1 entirely but producing crisp, coherent generations within the chosen mode. This trade-off motivates the adaptive-divergence methods in §4 (ToDi, AKL, Entropy-Aware OPD), which interpolate between the two regimes based on per-token teacher entropy or log-ratio statistics.

Skewed KL (SKL), constructing a smoothed target $\tilde{p} = \alpha p_T + (1-\alpha)p_\theta$.

$$\mathcal{L}_{\mathrm{SKL}} = \mathbb{E}_{y\sim p_\theta}\left[\sum_{t=1}^{|y|} D_{\mathrm{KL}}\big(p_T(\cdot|y_{<t}) \parallel \alpha p_T(\cdot|y_{<t}) + (1-\alpha)p_\theta(\cdot|y_{<t})\big)\right] \tag{10}$$

Skewing ensures the target density is bounded below by $(1-\alpha)p_\theta$, which avoids gradient explosion without requiring RL policy gradients. DistiLLM further proposes Skewed Reverse KL (SRKL) for mode-seeking applications, creating a symmetric pair of stabilized divergences. Its successor, DistiLLM-2 (Ko et al., 2025), recognizes that teacher-generated and student-generated data carry qualitatively different learning signals and applies *asymmetric* objectives, using Forward SKL on teacher-generated data (where mode-covering teaches the student to explore the teacher's full distribution) and Reverse SRKL on student-generated data (where mode-seeking teaches the student to sharpen its own best responses). This source-aware asymmetry produces consistent improvements over symmetric single-divergence baselines across instruction following and summarization tasks, consistent with the view that the appropriate divergence is data-source-dependent.

DistiLLM-2 applies different divergences to different data sources, but operates at a coarse granularity (entire sequences grouped by source). The fixed-divergence family faces a more fundamental limitation rooted in the heterogeneity of autoregressive token distributions *within* a single sequence. The choice between Forward KL (mode-covering) and Reverse KL (mode-seeking) is a global decision applied uniformly across all token positions (Figure 2). The teacher's distributional structure, however, varies considerably within a single sequence. At a mathematical operator token, the teacher concentrates probability mass on one or two correct symbols, so mode-seeking is appropriate because the student must commit to the correct answer. At a conjunction or filler word, dozens of tokens share comparable probability, so mode-covering is appropriate because any synonym suffices. Applying a single divergence to both positions therefore wastes gradient signal in one regime or discards coverage information in the other. This per-position mismatch between the geometry of the loss and the geometry of the target distribution motivates the adaptive methods of Section 4.2.

Sequence-level objectives offer a complementary perspective. MiniLLM (Gu et al., 2024) elevates Reverse KL to the full sequence level, where the student p_θ appears inside the sampling expectation. The central challenge is computing the gradient of an expectation over p_θ when p_θ itself is parameterized by θ. Starting from $D_{\text{KL}}(p_\theta \parallel p_T) = \mathbb{E}_{y\sim p_\theta}[\log p_\theta(y|x) - \log p_T(y|x)]$ and applying the log-derivative trick:

$$\nabla_\theta \mathcal{L}_{\text{MiniLLM}} = \mathbb{E}_{y\sim p_\theta}\left[\Big(\log p_\theta(y|x) - \log p_T(y|x)\Big)\nabla_\theta \log p_\theta(y|x)\right] \tag{11}$$

Minimizing sequence-level Reverse KL is thus mathematically equivalent to policy gradient RL with reward $r(x, y) = \log p_T(y|x) - \log p_\theta(y|x)$. MiniLLM decomposes this into per-step rewards r_t and future returns $R_t = \sum_{t'=t}^{|y|} r_{t'}$, allowing variance-reducing baselines:

$$\nabla_\theta \mathcal{L}_{\text{MiniLLM}} = -\mathbb{E}_{y\sim p_\theta}\left[\sum_{t=1}^{|y|} (R_t - b)\, \nabla_\theta \log p_\theta(y_t|y_{<t})\right] \tag{12}$$

where b is a learned baseline. The per-step decomposition supports credit assignment at the token level. Per-step returns R_t attribute the trajectory's deviation from the teacher to specific token decisions rather than spreading uniform blame across the sequence, placing the optimization within the RL framework with the teacher's log-probability as a dense reward. This connection to policy optimization is consistent with MiniLLM's empirical strength on reasoning tasks where sequence-level coherence matters more than token-level accuracy, at the cost of higher training compute because REINFORCE estimation in combinatorial output spaces requires more iterations and careful baseline subtraction to converge. KETCHUP (Fan et al., 2026) addresses this variance problem by replacing the single-sample REINFORCE estimator with a K-step truncated return, reducing gradient variance while preserving the sequence-level objective. The constrained KD framework of Zimmer et al. (2025) further augments the sequence-level Reverse KL with a hard KL constraint implemented via state-augmented CMDP, ensuring that the student's divergence from the teacher remains bounded even during aggressive exploration.

The token-level vs. sequence-level tradeoff. The choice between token-level and sequence-level objectives represents a core bias-variance tradeoff whose resolution foreshadows the RL-augmented methods of §4.3. Token-level methods (GKD, DistiLLM) yield $|V|$-dimensional gradient signals at every position, yielding low variance but potentially biased gradients when the teacher's token-level distribution is poorly calibrated (the flawed prefix trap of Section 6.1). Sequence-level methods (MiniLLM, KETCHUP) yield unbiased gradients through the policy gradient theorem but suffer from high variance due to single-sample REINFORCE estimation over exponentially large sequence spaces. The practical resolution is task-dependent. For instruction following where token-level teacher quality is high, GKD-style methods tend to dominate. For reasoning tasks where sequence-level coherence matters more, MiniLLM-style methods are often preferred despite their variance, and KETCHUP's truncated returns offer an intermediate point on this spectrum. Recognizing that MiniLLM is already performing policy gradient RL (with the teacher's log-probability as reward) suggests that sequence-level distillation and RL-augmented distillation are not distinct paradigms but endpoints of a continuum parameterized by the reward source: pure teacher signal at one end, pure task reward at the other, and hybrid objectives (G-OPD, RLKD) interpolating between them.

Full-vocabulary vs. sampled-token KL. Orthogonal to the token-level vs. sequence-level choice, a practical implementation decision determines how much of the teacher's distribution is used at each position. Many RL-based OPD implementations approximate the per-position KL by computing $\log p_\theta(y_t|y_{<t}) - \log p_T(y_t|y_{<t})$ only at the *sampled* token y_t, treating this scalar as a per-token advantage within a policy gradient framework (Lu & Thinking Machines Lab, 2025). While computationally lightweight (requiring only one teacher log-probability per position), this single-sample estimate incurs high gradient variance and discards the distributional information across non-sampled tokens. DeepSeek-V4 (DeepSeek-AI, 2026) reports that *full-vocabulary* logit distillation, computing the exact $D_{\text{KL}}(p_\theta(\cdot|y_{<t}) \parallel p_T(\cdot|y_{<t}))$ over all $|V|$ tokens at each position, yields more stable gradients and more faithful knowledge transfer at the cost of transmitting or reconstructing the full

teacher logit vector. Their engineering solution (caching teacher hidden states and reconstructing logits on-the-fly through the prediction head) makes this feasible even with 10+ trillion-parameter teachers over vocabularies exceeding 100K tokens, which positions full-vocabulary KL as an attractive choice for industrial-scale OPD when infrastructure permits. Lightning OPD (Wu et al., 2026) further reports that the full teacher forward pass can be largely avoided by precomputing teacher log-probabilities over SFT rollouts, recording 4× speedup over standard OPD under a *teacher consistency* condition (using the same teacher for both SFT data generation and distillation).

4.2 Adaptive Divergence Objectives

Adaptive methods tackle this per-position mismatch by selecting or interpolating divergences based on local distributional geometry. When the teacher is certain (low entropy), mode-seeking Reverse KL preserves precision, and when uncertain (high entropy), mode-covering Forward KL preserves diversity.

ToDi (Jung et al., 2025) implements this insight through per-token weighting derived from the teacher-student probability log-ratio. A sigmoid-based function $w_t = \sigma(\log p_T(y_t|y_{<t}) - \log p_\theta(y_t|y_{<t}))$ dynamically blends Forward KL and Reverse KL for each token:

$$\mathcal{L}_{\text{ToDi}} = \mathbb{E}_{y \sim p_\theta} \left[\sum_{t=1}^{|y|} w_t \cdot D_{\text{KL}}(p_T \parallel p_\theta) + (1 - w_t) \cdot D_{\text{KL}}(p_\theta \parallel p_T) \right] \tag{13}$$

where w_t emphasizes Forward KL for tokens the student underestimates (boosting their probability) and Reverse KL for tokens it overestimates (suppressing excess probability). Entropy-Aware OPD (Jin et al., 2026) takes an alternative approach, using the teacher's per-token entropy $H(p_T(\cdot|y_{<t}))$ as a gating signal. The base objective is Reverse KL at all positions, with Forward KL *additively* activated at high-entropy tokens where the teacher distributes probability across multiple valid continuations:

$$\mathcal{L}_{\text{EA}} = \mathbb{E}_{y \sim p_\theta} \left[\sum_{t=1}^{|y|} D_{\text{KL}}(p_\theta \parallel p_T) + \mathbb{I}[H_t > \tau] \cdot D_{\text{KL}}(p_T \parallel p_\theta) \right] \tag{14}$$

Both methods consistently outperform any fixed divergence on math reasoning (MATH, GSM8K) and open-ended generation (AlpacaEval), consistent with the view that the appropriate divergence is position-dependent. The complementarity between these two methods is instructive. ToDi conditions adaptation on the teacher-student gap (a pairwise property), while Entropy-Aware OPD conditions on teacher uncertainty alone (a marginal property). The former adapts more aggressively when the student is far from the teacher, while the latter delivers a stable routing signal independent of student quality, making it more suitable for early training when the student's distribution is less reliable.

AKL (Wu et al., 2025) synthesizes elements of both approaches, dynamically interpolating between Forward KL and Reverse KL based on a per-token confidence metric derived from the teacher-student probability ratio. Unlike entropy-based routing, which depends only on the teacher's distributional shape, AKL considers the *relative* geometry of teacher and student, adapting more aggressively at positions where the gap is largest. All three adaptive methods admit a natural interpretation through information geometry. The Fisher information metric on the statistical manifold of autoregressive distributions varies across token positions. At reasoning-critical positions, the manifold is highly curved (the teacher concentrates probability on few tokens), making Reverse KL's mode-seeking behavior locally appropriate because it follows the manifold's natural gradient. At generation-flexible positions, the manifold is nearly flat (many synonyms are equally valid), and Forward KL's mode-covering behavior preserves useful entropy. Adaptive divergences implicitly navigate this varying curvature by selecting the locally best metric, which explains their consistent empirical superiority over any fixed choice.

AOPD (Jia et al., 2026) takes the per-token adaptive principle to its logical conclusion by diagnosing *why* standard OPD gradients fail at specific token positions. In non-positive advantage regions ($A_t \leq 0$), standard advantage-weighted policy gradients exhibit three pathologies that intensify under weak initialization. Heavy-tailed variance from unbounded negative advantages destabilizes updates, vanishing gradients at zero-advantage positions waste teacher signal, and "exploration black holes" trap learning because suppressing a wrong token only redistributes mass proportional to the student's own prior, unable to boost low-probability correct alternatives. AOPD resolves this by switching the objective at the token level. For positive-advantage tokens ($A_t > 0$), standard policy gradient exploitation is preserved. For non-positive tokens, the framework substitutes localized forward-KL guidance on the teacher's top-K support, creating an asymmetric exploitation-imitation mechanism. A threshold parameter τ (set to zero by default, corresponding to the probability difference between teacher and student) determines the switching point, with $\tau = -1$ recovering standard OPD and $\tau = 1$ recovering GKD. On competition-level mathematics (AIME 2024/2025, MATH-500), AOPD attains average gains of +4.09 with strong initialization and +8.34 with weak initialization over standard on-policy baselines, with the weak-initialization gap suggesting that the exploration black hole is among the dominant failure modes when the student starts far from the teacher. The relationship between AOPD and the earlier adaptive methods is one of complementary scope. ToDi, AKL, and Entropy-Aware OPD adapt the divergence *type* based on distributional geometry (teacher entropy, teacher-student gap), while AOPD adapts the *objective class itself* (policy gradient vs. divergence minimization) based on the gradient's structural health at each position.

The field has moved noticeably from "which divergence?" to "which divergence *where*?" The evidence from ToDi, AKL, Entropy-Aware OPD, and AOPD points to a consistent finding across evaluated settings, namely that no single fixed divergence dominates uniformly, and that even the simplest adaptive routing (a hard entropy threshold) often outperforms the best-tuned fixed alternative. Yet adaptive divergences, however sophisticated their per-token routing, share a structural limitation with their fixed counterparts. They optimize a distance to the teacher distribution. Perfect optimization under any f-divergence yields a student that matches the teacher but rarely exceeds it. When the teacher itself makes errors or lacks reasoning paths that a well-guided student could discover, divergence minimization can act more as a ceiling than as a floor.

4.3 RL-Augmented Objectives

Divergence matching, whether fixed or adaptive, tends to bound the student near the teacher's capability ceiling. In the limit of perfect optimization, the student matches the teacher's distribution and rarely exceeds it. RL-augmented objectives target this limitation by injecting an external performance signal that can steer the student toward solutions the teacher itself would be unlikely to produce.

OPD as KL-constrained RL. G-OPD (Yang et al., 2026c) formalizes an equivalence between standard OPD and dense KL-constrained reinforcement learning. It formulates distillation as maximizing a token-level reward derived from teacher logits, penalized by divergence from a reference policy:

$$\max_{\theta} \mathbb{E}_{y\sim p_\theta}\left[\sum_{t=1}^{|y|} \alpha \log p_T(y_t|y_{<t}) - D_{\mathrm{KL}}\big(p_\theta(\cdot|y_{<t}) \parallel p_{\mathrm{ref}}(\cdot|y_{<t})\big)\right] \tag{15}$$

When $\alpha = 1$, this reduces to standard Reverse KL distillation. When $\alpha > 1$, the objective forces the student to *extrapolate* beyond the teacher's probability mass, allowing a phenomenon documented as *Reward Extrapolation*, whereby the student discovers previously unseen, structurally sound reasoning paths in regions where the teacher's generation probability $p_T(y|x)$ is low but the outcome reward remains high. In the multi-teacher setting where several domain-specific RL experts share the same base model, reward extrapolation (ExOPD) produces a unified student that surpasses all its same-size domain teachers, showing that OPD need not be a lossy compression step.

This equivalence has a deeper theoretical consequence that unifies the entire section. It bridges the distillation and RL communities by showing that all objectives discussed here (GKD's token-level KL, MiniLLM's sequence-level Reverse KL, G-OPD's reward-augmented formulation) are special cases of KL-constrained policy optimization with different reward definitions and constraint strengths. Any advance in KL-constrained RL (better trust regions, adaptive penalty coefficients, variance reduction) immediately transfers to OPD, and vice versa.

Gradient decomposition and the KD+RL complementarity. The unified KD+RL framework of Li et al. (2025) decomposes the hybrid gradient into two orthogonal components, a dense KD gradient for token-level imitation and a Monte Carlo RL gradient for reward maximization. The KD component dampens the high variance inherent in policy gradient estimation, while the RL component prevents the student from collapsing onto teacher modes that are suboptimal for the downstream task. REOPOLD (Ko et al., 2026) instantiates this insight by interpreting on-policy distillation as policy optimization where the teacher-student log-likelihood ratio acts as a token reward. It stabilizes training through three mechanisms: mixture-based reward clipping to prevent over-trust of extreme teacher signals, entropy-based dynamic sampling to selectively leverage teacher feedback at high-uncertainty tokens, and a unified exploration-to-refinement schedule. Empirically, REOPOLD attains 6.7–12$\times$ greater sample efficiency than recent RL approaches and allows a 7B student to match a 32B teacher in visual reasoning with $\sim$3.3$\times$ inference speedup.

Structure-aware reward as an alternative to dense KL. While G-OPD and REOPOLD retain token-level KL as the primary distillation signal (differing only in how they augment or stabilize it), RLKD (Xu et al., 2025b) departs from this template entirely by replacing token-level KL with a Generative Structure Reward Model (GSRM). The GSRM parses both teacher and student reasoning paths into sequences of meta-reasoning and solving steps, then scores structural alignment between the two sequences. This structural reward is combined with a task-level outcome reward (e.g., math accuracy) and optimized via GRPO:

$$R_{\text{total}}(x,y) = \alpha\, R_{\text{GSRM}}(y, y^{\text{teach}}) + (1-\alpha)\, R_{\text{outcome}}(x,y) \tag{16}$$

The key insight is that reasoning distillation need not operate at the token level. By supervising at the *step structure* level, RLKD avoids the vocabulary-coupling assumption of logit-based methods while still transferring the teacher's multi-branch reasoning strategy. Empirically, RLKD trained on only 0.1% of the data under a pure RL regime surpasses standard SFT-RL pipelines, showing that structural reward can substitute for dense KL when the teacher's value lies in its reasoning organization rather than its token-level distribution. This progression from dense token-level KL (G-OPD) through variance-reduced token rewards (REOPOLD) to step-level structural matching (RLKD) traces an increasing abstraction of what "teacher knowledge" means, moving from distributional fidelity to behavioral organization.

Joint vs. sequential optimization. Given that KD and RL supply complementary gradient signals (dense imitation vs. sparse reward maximization), a persistent engineering challenge is *how* to combine them within a single training pipeline without mutual interference. Sequential approaches (distill-then-RL or RL-then-distill) suffer from catastrophic interference, as the RL stage erases distilled knowledge through policy drift, while the KD stage suppresses previously unseen solutions discovered through exploration. KDRL (Xu et al., 2025a) implements joint optimization by adding an on-policy KL regularizer between student and teacher *during* RL training, preventing policy drift while still allowing reward-driven exploration. RLAD (Zhang et al., 2026h) refines this further by recognizing that unconditional KL regularization is itself suboptimal. Its Trust Region Ratio Distillation (TRRD) replaces the standard KL penalty with a PPO-style likelihood-ratio objective anchored to a teacher-old-policy mixture. This selective strategy follows the teacher only when its signal improves the policy update, preventing blind KL minimization from dragging the student toward suboptimal trajectories on problems the teacher itself cannot solve. The progression from KDRL's uniform regularization to RLAD's selective trust region mirrors the fixed-to-adaptive evolution in divergence objectives (Section 4.1→4.2), echoing a recurring pattern across this survey that conditioning the learning signal on local geometry often works better than applying it uniformly.

Preference-based objectives as implicit RL. When logit access is unavailable or when sequence-level quality judgments are more natural than token-level distributions, several methods map the KD+RL problem onto Direct Preference Optimization (DPO). AlignDistil (Zhang et al., 2025a) constructs synthetic preference pairs from student rollouts scored by the teacher, applying both standard and reverse DPO objectives. OVD (Xiong et al., 2026) queries the teacher for pairwise preferences over student-generated responses, avoiding continuous probability extraction entirely. These preference-based objectives share a deep connection to token-level KL methods. Under DPO's closed-form solution, the optimal policy satisfies $\pi^*(y|x) \propto \pi_{\text{ref}}(y|x) \exp(r(y,x)/\beta)$, and substituting the teacher's log-probability as the implicit reward yields the geometric mixture distribution targeted by GKD with λ-mixing. The practical implication is that DPO-style methods are most useful when constructing preference pairs is cheaper than computing full teacher logits. PBSD (Yu et al., 2026) pushes this connection further in the self-distillation regime by deriving the reward-regularized optimum $\pi^* \propto \pi_{\text{teach}} \exp(r/\beta)$ against a context-augmented teacher rather than a static reference policy, and showing that the resulting pairwise preference loss provably dominates direct KL matching when the reward signal is informative (Section 5.3.1). The spectrum exposes a unifying gradient. Token-level KL (G-OPD, REOPOLD), step-level structure matching (RLKD), and sequence-level preference ranking (AlignDistil, OVD) all implement KL-constrained policy optimization at different abstraction levels, with the appropriate level determined by signal availability and the granularity at which teacher expertise manifests.

Fine-grained credit assignment. All methods above, whether operating at the token, step, or sequence level, share one limitation: they apply supervision uniformly across the student's trajectory. When the student makes a single error in step 3 of a 10-step proof, wholesale imitation retrains all 10 steps, diluting the corrective gradient on the actual failure point. SuperCorrect (Yang et al., 2025b) presents hierarchical thought templates with cross-model DPO for error localization, identifying which reasoning step diverged and concentrating feedback there. SCoRe (Lyu et al., 2025) pushes targeting to its logical extreme, correcting *only the earliest error* in the student's trajectory, concentrating all supervision at the single point of maximum leverage. This exploits the cascading nature of reasoning errors, where correcting the root cause eliminates all downstream consequences simultaneously. The contrast with RLKD's structural approach is instructive. RLKD rewards trajectories whose overall step organization matches the teacher's, while SCoRe surgically corrects the first point of divergence. The two are complementary, and combining global structural reward with local error correction remains unexplored.

From signal matching to environment reconstruction. The most radical extension asks whether the student can recover the teacher's *underlying optimization landscape*. $\mathcal{X}$-KD (Cai & Yuan, 2026) adopts the AVRIL framework to jointly model the teacher's latent reward function and perform policy distillation, encouraging the student to optimize in a reconstructed version of the teacher's training environment instead of mimicking its outputs. TSD-KD (Kim & Baek, 2026) combines indirect distillation (student generates candidates that the teacher re-ranks via preference feedback) with direct distillation (KL matching applied selectively to high-entropy tokens where teacher-student gaps are largest). These methods represent the current frontier, moving from matching the teacher's *behavior* to understanding its *incentives*.

Across objective types, a rough progression is visible. Fixed divergences deliver stable, well-understood optimization but tend to be bounded by the teacher's capability ceiling. Adaptive divergences can improve efficiency by matching the local geometry of the loss landscape. RL-augmented objectives offer a path around the teacher ceiling at the cost of added variance and instability. The appropriate choice depends on the specific application. For instruction following where teacher quality is high, fixed or adaptive divergences are often sufficient. For reasoning tasks where the student is expected to exceed the teacher, RL augmentation becomes attractive. For tasks where the teacher's training process is accessible, environment reconstruction offers a more complete form of knowledge transfer.

5 Signal Source and Teacher Architecture

With the objective function fixed (Section 4), the next design axis concerns *where the teacher signal comes from*: full logit access (white-box), API-only text outputs (black-box), or the model's own internal asymmetries (self-distillation). Each signal source imposes distinct constraints on the applicable objectives and achievable performance.

Historically, signal sources have evolved in step with model access constraints. Early OPD methods (GKD, DistiLLM) assumed full white-box access to model internals, a requirement that limits distillation to same-organization deployments. The rise of proprietary API-only models (GPT-4, Claude) necessitated black-box methods that can extract knowledge from text outputs alone. Most recently, self-distillation methods eliminate the external teacher entirely, permitting continuous improvement without access to any stronger model. This progression from full access through limited access to no external access represents increasing autonomy at the cost of signal density, a trade-off between autonomy and signal density that recurs throughout the design space.

Complex reasoning is among the settings where OPD shows its largest gains over off-policy alternatives. The reason is structural, because reasoning is *path-dependent* (Agarwal et al., 2024). A proof that begins "assume for contradiction..." requires entirely different subsequent tokens than one beginning "we proceed by induction..." If the student has never generated the inductive approach during training, it is unlikely to recover at inference time when that path would be optimal. Off-policy training over a fixed dataset of teacher reasoning traces only covers a finite set of solution paths, while the student needs to navigate its own terrain of partial solutions, dead ends, and recovery strategies. This observation motivates the on-policy formulation for reasoning transfer. The student generates its own intermediate reasoning steps $r^S \sim P_\theta(\cdot|x)$, and the teacher evaluates these student-generated chains instead of dictating its own:

$$\mathcal{L}_{\text{CoT-OPD}}(\theta) = \mathbb{E}_{x\sim\mathcal{D},\, r^S\sim P_\theta(\cdot|x)}\left[\sum_{t=1}^{|r^S|} D_{\text{KL}}\Big(P_T(\cdot|x, r^S_{<t}) \parallel P_\theta(\cdot|x, r^S_{<t})\Big)\right] \tag{17}$$

The Forward KL direction encourages mode-covering, so that the student allocates probability wherever the teacher does on the student's own prefixes, receiving corrective feedback on its actual generation patterns instead of memorizing the teacher's preferred trajectories. This self-consistent learning dynamic, where the student practices its own reasoning style under expert supervision, has made on-policy CoT distillation a common choice for transferring multi-step reasoning to smaller models. The principle that rationale transfer outperforms answer-only supervision was first established in the off-policy setting by Hsieh et al. (2023), and subsequent work showed that generating rationales on-policy (Agarwal et al., 2024) can yield further gains by reducing exposure bias, an effect that has also been observed at scale in DeepSeek-R1 (DeepSeek-AI et al., 2025) and its successors.

5.1 White-Box Logit Supervision

White-box access yields the densest possible signal, namely the full probability distribution across $|V|$ tokens at every decoding step. This exposes the teacher's implicit decision boundaries, including the relative probabilities among incorrect tokens (Hinton's "dark knowledge"), which carry information about similarity structure that binary labels cannot capture.

Standard logit distillation. The majority of methods surveyed in Section 4 assume white-box access. GKD, DistiLLM, ToDi, G-OPD, and their variants all require computing $p_T(\cdot|x, y_{<t})$ at each student-generated token position. The computational cost is dominated by the teacher forward pass, executed once per student-generated token (or once per sequence when caching is enabled).

Token-level adaptive supervision. Within the white-box setting, several methods refine *which* tokens receive supervision, and they can be organized along the axis of what drives the selection decision. SelecTKD (Huang et al., 2025) routes based on student-teacher agreement, applying a propose-and-verify mechanism where the student proposes tokens that the teacher verifies, and only positions where agreement holds receive full loss while rejected tokens are masked or down-weighted. AdaSwitch (Peng et al., 2025) routes based on student prefix reliability, maintaining a running error estimate and switching between exploration (student generates freely) and guidance (teacher supplies dense supervision) so that guidance kicks in only when the trajectory has drifted far enough to warrant teacher intervention. Token-Adaptive KD (Xie et al., 2026) routes based on gradient magnitude, computing per-token importance weights from the KL loss itself and concentrating compute where the teacher-student gap is largest. The three selection criteria, agreement, prefix quality, and gradient magnitude, are complementary rather than competing, each capturing a different signal about where supervision is most informative. They also compose naturally with entropy-based divergence routing (ToDi, Entropy-Aware), which selects *which loss function* applies at each position while token-level selection determines *how much gradient signal* propagates. The combination yields an orthogonal two-dimensional design space that current methods have only begun to explore jointly. These token-selection mechanisms complement the sample-level and token-level weighting schemes discussed in Section 6.

The temporal scope of white-box distillation is not limited to post-training. MiniPLM (Gu et al., 2025) extends on-policy distillation to the *pre-training* stage, showing that distilling during pre-training yields stronger foundations for downstream fine-tuning. This represents a qualitative shift, because instead of being a post-hoc compression step, OPD becomes an integral part of the model creation pipeline. Gemma 2 (Gemma Team et al., 2024) validates this approach at industrial scale, embedding KD directly into the pre-training loop for its 27B→9B→2B model cascade. The success of pre-training distillation raises a deeper question, namely whether OPD should be viewed as a separate post-processing step at all, or as an intrinsic component of the training objective alongside next-token prediction.

A critical challenge arises when teacher and student use different tokenizers or vocabulary sizes. Direct KL divergence over mismatched vocabularies is undefined. DSKD (Zhang et al., 2025b) addresses this through *dual-space* knowledge distillation, employing projectors that map between teacher and student representation spaces:

$$\mathcal{L}_{\text{DSKD}} = D_{\text{KL}}(P_{T\to S} \parallel P_S) + D_{\text{KL}}(P_{S\to T} \parallel P_T) \tag{18}$$

where $P_{T\to S}$ denotes the teacher's hidden states projected into the student's space and decoded through the student's prediction head. An exact token alignment algorithm handles vocabulary mismatches by identifying identical tokens across tokenizers. Vocabulary tokens with similar semantic embeddings should receive similar probability mass even when the tokenization schemes differ, allowing distillation across different architectures (e.g., Llama teacher to Qwen student) that share no vocabulary overlap.

Cross-Tokenizer KD (Boizard et al., 2025) takes an alternative approach based on optimal transport, computing an alignment matrix between teacher and student token sequences that minimizes the earth mover's distance in latent space:

$$\mathcal{L}_{\text{CrossTok}} = \inf_{\pi\in\Pi(P_S,P_T)} \mathbb{E}_{(z_S,z_T)\sim\pi} \left[\|W_{S\to T} z_S - z_T\|_2^2\right] \tag{19}$$

The distinction between hidden-state alignment (DSKD) and vocabulary-space alignment (Cross-Tokenizer KD) exposes a central design trade-off. Hidden-state alignment preserves richer representational structure but requires maintaining two projectors during training. Vocabulary-space alignment is more lightweight but operates at a level where information has already been compressed through the vocabulary bottleneck. In either case, the alignment maps must be *co-trained* with the student because the student's representation space shifts during on-policy training as its policy evolves, a form of representation drift absent in off-policy settings. Concrete Score Matching (Kim et al., 2026b) offers another solution, distilling via a continuous score function that bypasses the discrete vocabulary mismatch entirely. Together, these methods move OPD toward being more architecture-agnostic, allowing organizations to distill from teachers in one model family (e.g., Llama-based) into students in another (e.g., Qwen-based) without the historically restrictive requirement of shared vocabulary, and supporting heterogeneous teacher-student pairs that were previously difficult to handle.

Beyond selecting *which tokens* to supervise, a distinct question is *what information* the teacher should convey at each token. Not all content in the teacher's logits is equally useful. Delta-KD (Cao et al., 2025) formalizes the principle of *signal isolation*, distilling only the differential shift between a base model and its instruction-tuned variant. Since the base model's distribution captures generic language modeling knowledge the student already possesses through pre-training, distilling it is redundant at best and harmful at worst because it wastes gradient budget on information the student has already internalized. By extracting only the instruction-following delta, Delta-KD delivers a cleaner supervisory signal that transfers alignment without overwriting the student's foundation.

A related but distinct challenge is teacher *overconfidence*. When the teacher assigns extreme probability to tokens outside the student's reachable space, the resulting optimization targets are simultaneously precise and useless. Two solutions tackle this from opposite directions. Veto (Jang et al., 2026) (Adaptive Target Reformulation) refines the teacher's output *post-hoc*, constructing a geometric bridge in logit space with a parameter β that serves as both an Adaptive Gradient Veto (suppressing pathological gradients on low-confidence tokens) and a Decisiveness Knob (balancing performance with diversity), dynamically reformulating the teacher's target distribution based on the student's current capability. PromptKD (Kim et al., 2024) takes the opposite approach, adapting the teacher *pre-hoc* by prepending learnable prompt tokens (adding only 0.0007% of the teacher's parameters) that cause the teacher to produce "student-friendly" distributions more accessible to the student's limited capacity. Post-hoc methods (Veto, Delta-KD) operate on the supervisory signal after it is produced, while pre-hoc methods (PromptKD) modify the teacher's generation process itself. Both are effective, and they compose without interference.

TAID (Shing et al., 2025) adds yet another dimension of adaptation, namely temporal interpolation of the target itself. It constructs a time-dependent intermediate distribution $P_t = (1 - \lambda_t)P_\theta + \lambda_t P_T$ that starts close to the student and gradually approaches the full teacher distribution as training proceeds. This temporal curriculum over the target (as opposed to the input prompts as in PACED) offers an additional axis of control. PACED asks "which problems to train on," while TAID asks "how much of the teacher's knowledge to reveal at each stage." The progression from PACED (problem-level curriculum) through AdaSwitch (token-level timing) to TAID (target-level interpolation) and PromptKD (teacher-level adaptation) traces an important design spectrum, showing that curriculum can operate at four distinct granularities (prompt, token, target, teacher), all of which target the same underlying bottleneck of keeping the difficulty gap between teacher signal and student capability in the productive learning range. Because these four granularities are independent, they can in principle be combined within a single training loop, though no existing system integrates all four simultaneously.

5.2 Black-Box and API-Constrained Distillation

When only API access is available, the student observes the teacher's top-1 text output (or at best, top-k log-probabilities) rather than the full distribution. This information bottleneck rules out token-level divergence matching and forces methods to operate at the sequence level. Despite this limitation, black-box methods have been reported to work well in practice, suggesting that a substantial portion of the teacher's knowledge can be recovered from text-only observations.

Distilling Step-by-Step. Hsieh et al. (2023) demonstrated the foundational black-box approach, extracting not just the teacher's final answers but its intermediate reasoning steps (rationales) as additional supervision. Although technically off-policy (the student fine-tunes on static teacher-generated rationales rather than its own), this work is the conceptual ancestor of on-policy CoT distillation and established a key principle. The student is trained with a multi-task objective combining answer prediction and rationale generation, allowing a 770M-parameter model to outperform a 540B-parameter teacher with 500× fewer parameters and substantially less training data (50% fewer examples on average, up to 85% reduction on some tasks). This demonstrated that *how* the teacher reasons is often more valuable than *what* it concludes, a principle that pervades modern OPD and motivates the on-policy extensions below.

If extracting textual rationales recovers partial teacher reasoning, the natural next question is whether a richer proxy for the teacher's full distribution can be reconstructed from black-box outputs alone. The following methods attempt increasingly aggressive reconstructions, from adversarial discrimination through verbal scoring to preference ranking.

Adversarial distribution matching. GAD (Ye et al., 2025) reformulates the distillation problem as a generative adversarial game:

$$\max_G \min_D \; V(G, D) = \mathbb{E}_{(x,y^T)\sim\mathcal{T}} \left[-\log \sigma \Big(D(y^T) - D(G(x)) \Big) \right] \tag{20}$$

A discriminator D (initialized from the student with a scalar prediction head) distinguishes student rollouts from teacher API outputs via a Bradley-Terry preference model, while the generator G maximizes V via REINFORCE using the discriminator score as an evolving on-policy reward signal. Unlike white-box methods that match full distributions, GAD extracts knowledge through a scalar quality signal, trading distributional fidelity for API compatibility. It inherits the training instability of adversarial objectives, as the discriminator can overfit to stylistic cues rather than semantic quality, but produces consistent gains over SFT baselines.

Verbal feedback and curriculum. Lion (Jiang et al., 2023) exploits the teacher's capacity as a natural language critic through a three-stage adversarial loop. First, in *imitation*, the student is fine-tuned on teacher-generated responses. Second, in *discrimination*, the teacher identifies instructions on which the student still falls short. Third, in *generation*, the teacher creates harder instructions targeting the student's identified weaknesses. This closed-loop curriculum steadily narrows the teacher-student gap without requiring any gradient information from the teacher, making it applicable to any instruction-following API. Lion-13B reaches competitive performance on BIG-Bench Hard and AGIEval using only 70K training examples.

Didactic-to-constructive learning. DAIL (Mendes et al., 2026) tackles the gap between high-quality expert solutions (often out-of-distribution) and the student's actual reasoning space. Fine-tuning on expert solutions yields poor transfer because the student rarely learns to navigate its own reasoning space. DAIL uses a two-step process, transforming expert solutions into detailed, in-distribution reasoning traces via mixed policy decoding (the student generates while conditioned on the ground-truth solution), and applying a contrastive objective to focus learning on expert methodologies rather than surface patterns. Using fewer than 1,000 expert solutions, DAIL records 10–25% pass@k gains, consistent with the recurrent observation that the alignment of the supervision signal with the student's current distribution often matters more than its absolute quality.

Distribution-Aligned Sequence Distillation (DASD) (Yan et al., 2026) reexamines the dominant practice of SFT on teacher-generated reasoning traces, identifying three limitations. First, inadequate representation of the teacher's sequence-level distribution from single best-of-n samples. Second, misalignment between teacher output distribution and student capacity. Third, exposure bias from teacher-forced training. Its distribution-aligned pipeline resolves all three by generating multiple diverse teacher traces per prompt and aligning the student's generation distribution with the teacher's through sequence-level matching, reaching state-of-the-art performance using only 448K training samples. DDT (Zhang et al., 2026c) complements DASD's empirical findings with a theoretical framework explaining why RL generalizes better than SFT for reasoning. RL's on-policy data generation keeps training aligned with the model's evolving distribution instead of a static snapshot.

Verbal score distillation. OVD (Xiong et al., 2026) replaces token-level probability matching with trajectory matching using discrete verbal scores (0–9) from teacher models. This removes the requirement for token-level alignment between vocabularies, substantially reduces memory consumption by avoiding logit storage, and reports up to +12.9% absolute EM improvement on web QA and +25.7% on math benchmarks with a single rollout sample. The effectiveness of such coarse supervision suggests that on-policy exploration itself supplies a substantial fraction of the learning signal, with the teacher's role being closer to *selecting among student trajectories* than to *correcting individual tokens*.

These methods form a spectrum of signal richness versus access requirements. Adversarial methods recover the densest proxy for the teacher's distribution but require the most complex training dynamics (discriminator co-training, generator-discriminator instability).

Score-based methods are simpler to train but lose distributional information, recovering only a scalar quality signal. Preference-based methods occupy a principled middle ground, recovering relative quality without requiring calibrated absolute scores, and their connection to DPO-style objectives yields well-understood convergence guarantees. The signal type therefore maps naturally to API access tier, with full logit access enabling KL methods (Section 4), unavailable token-level logits motivating adversarial or score-based approaches, and ranking-only access favoring preference-based methods.

Preference-based formulations. ORPO-Distill (Singh et al., 2025) maps black-box distillation onto preference optimization, generating candidate responses from the student and querying the teacher API for pairwise rankings. The mixed-policy strategy, combining on-policy student traces with off-policy teacher traces, outperforms both pure off-policy and pure on-policy alternatives, suggesting that the contrastive structure of preference learning (learning what *not* to do alongside what to do) supplies complementary signal to the positive-only feedback of score-based methods. ThinkTuning (RRV et al., 2025) occupies an intermediate position between black-box distillation and RL. Rather than scoring after the fact, a same-size teacher reviews the student's incorrect answers *during* the GRPO training loop and delivers short corrective hints (rather than complete solutions). This interleaving of corrective feedback episodes with RL training acts as an implicit privileged signal that shapes policy gradient updates and is particularly effective for base models that lack strong prior reasoning behaviors, reaching +3.85% over zero-shot baselines across benchmarks (+2.08% on MATH-500, +2.23% on AIME, +3.99% on GPQA-Diamond over vanilla GRPO).

MoE discriminator for pipeline-staged OPD. PRISM (Wang et al., 2026d) contributes a distinct architectural innovation to black-box distillation by inserting an OPD-based *pre-alignment* stage between SFT and RLVR, tackling the observation that direct RLVR from SFT checkpoints suffers from cold-start instability due to the large distributional gap between SFT outputs and reward-aligned behavior. Rather than using a monolithic discriminator (as in GAD), PRISM employs a Mixture-of-Experts discriminator with specialized perception and reasoning expert heads, delivering *disentangled* corrective signals that distinguish visual grounding errors from logical reasoning failures. The adversarial game operates entirely at the response level without requiring teacher logits. The student generates on-policy responses, a black-box teacher (Gemini 3 Flash) generates reference demonstrations, and the MoE discriminator learns to distinguish them while delivering differentiated feedback through its expert routing. This disentangled feedback is critical for multimodal models where failure modes are heterogeneous, since a model may correctly perceive an image but reason incorrectly about it, or vice versa. On Qwen3-VL at 4B and 8B scales, the pre-alignment stage alone yields +4.4 and +6.0 points over direct SFT-to-RLVR pipelines, and the subsequent RLVR training converges faster and more stably, suggesting that black-box OPD can serve as a structured warm-up that shapes the policy landscape for downstream RL optimization.

Off-policy knowledge integration. Not all training signal must originate on-policy. Incorporating knowledge from off-policy sources (e.g., pre-existing teacher-generated datasets) into an on-policy framework presents its own difficulties. LUFFY (Yan et al., 2025) handles this by extending GRPO to a mixed-policy objective that incorporates off-policy reasoning traces alongside the student's on-policy rollouts. The key design challenge is preventing *superficial imitation*, because without correction the student tends to copy high-probability tokens from off-policy traces while ignoring low-probability but critical reasoning steps. LUFFY's policy shaping via regularized importance sampling reweights gradients to emphasize learning from unfamiliar yet effective actions, posting +6.4 average points over standard RLVR.

5.3 Self-Distillation

Self-distillation eliminates the need for an external teacher entirely. Instead, the model exploits asymmetries *within itself* to construct a training signal. Three families of methods have emerged, distinguished by the source of this internal asymmetry, namely privileged information that is available at training but not inference time, self-play between successive model checkpoints, and external feedback from verifiers or environments that the model itself has generated.

5.3.1 *Privileged Information*

The first family creates asymmetry through *privileged information* (PI), meaning training-time access to context unavailable at inference. Conditioning the same model on additional context produces a strictly stronger policy that can supervise the unconditioned version, turning a single network into both teacher and student.

Ground-truth as PI. OPSD (Zhao et al., 2026b) constructs the teacher policy by conditioning on both the problem x and the ground-truth answer $y^\star$, while the student conditions only on x. The student generates on-policy rollouts, and the teacher delivers dense token-level supervision.

$$\mathcal{L}_{\text{OPSD}}(\theta) = \mathbb{E}_{(x,y^\star)\sim\mathcal{S}}\mathbb{E}_{\hat{y}\sim p_\theta(\cdot|x)}\left[\sum_{n=1}^{|\hat{y}|} D(p_\theta(\cdot|x,y^\star,\hat{y}_{<n}) \parallel p_\theta(\cdot|x,\hat{y}_{<n}))\right] \tag{21}$$

On competition-level math benchmarks (AIME 2024/2025, HMMT), OPSD matches or exceeds GRPO at 4B and 8B scales while using only a single rollout per problem versus GRPO's eight, yielding a significant computational advantage. At 1.7B scale, however, the gains over GRPO are minimal, indicating that sufficient model capacity is necessary for the PI mechanism to yield useful supervision. The teacher's ground-truth conditioning must produce meaningfully different token distributions from the student, which requires enough parameters to represent the conditional shift.

A complementary limitation arises when all rollouts for a given prompt fail (the "cliff prompt" problem), causing the policy gradient to vanish entirely. This failure mode directly parallels PACED's finding (Section 6.1) that gradient SNR vanishes when student pass rate approaches zero, but with an important difference. PACED tackles this through curriculum selection (avoiding problematic prompts), while HDPO (Ding, 2026) tackles it through signal augmentation. HDPO augments GRPO with a JSD-based self-distillation term on ground-truth-conditioned rollouts, providing non-zero gradients even at the boundary of the student's competence, directly patching the cliff-prompt vulnerability without restricting the problem distribution.

Relaxing the ground-truth requirement. Ground-truth labels are not always available. GATES (Stein et al., 2026) shows that *any* training-time context can serve as PI by having a single model act as tutor (conditioned on a source document) and student (answering from the question alone). Because source documents do not guarantee correct answers, GATES presents a consensus-based gating mechanism that suppresses the gradient when tutor uncertainty is high, preventing the model from reinforcing unreliable self-supervision. Penaloza et al. (2026) generalize the OPSD and GATES paradigms into a unified theoretical framework, formalizing the conditions under which privileged self-distillation provably improves over standard training.

Behavioral PI: context distillation. The PI paradigm extends beyond factual ground truth to behavioral instructions. OPCD (Ye et al., 2026b) treats system prompts, exemplars, and retrieval context as privileged information, allowing a model to internalize behaviors that would otherwise require expensive in-context demonstrations at inference time. OPSDL (Zhang et al., 2026e) extends this to long-context settings, applying on-policy self-distillation to enhance long-context capabilities without relying on high-quality supervision data or sparse sequence-level rewards and enabling stable optimization for context lengths beyond what standard post-training methods reach. OEL (Ye et al., 2026a) extends to continuous post-deployment learning, using interaction outcomes as dynamic PI that updates the model's teacher policy as it accumulates experience, closing a loop that OPCD opened.

Reasoning compression via PI. CRISP (Sang et al., 2026) finds that a model prompted to "be concise" can serve as a teacher for its own verbose version, reducing chain-of-thought token count by 57–59% on MATH-500 while *improving* accuracy by 9–16 percentage points. Much of the verbosity in distilled reasoning models is trainable inefficiency rather than a necessary feature of competent reasoning.

Visual PI for GUI grounding. GUI-SD (Zhang et al., 2026f) extends the privileged information paradigm to multimodal GUI grounding, where the task is to predict click coordinates on a screen given a natural language instruction. The privileged context is a Gaussian soft mask centered on the target element's bounding box, giving the teacher version explicit spatial localization that the student must learn to infer from raw screenshots alone. This represents a qualitatively different form of PI from textual ground truth (OPSD) or document context (GATES), one that is *visual* and *spatial*, suggesting that the PI mechanism extends across modalities. Beyond the new PI type, GUI-SD proposes entropy-guided token weighting via a dual-criterion, where tokens receive higher loss weight based on both their digit significance (coordinate digits carry more semantic weight than filler tokens) and the teacher's confidence at that position. With only a single on-policy rollout per instance, GUI-SD outperforms GRPO (which requires multiple rollouts for reward estimation) on six GUI grounding benchmarks including ScreenSpot-v2, ScreenSpot-Pro, and OSWorld. The success of single-rollout PI distillation in the visual domain mirrors OPSD's finding in mathematical reasoning, offering further evidence that privileged information can deliver sufficiently dense gradient signal to reduce reliance on the sample-intensive reward estimation required by RL-based alternatives.

Cross-lingual safety via PI. MSD (Qin et al., 2026) extends the PI paradigm to multilingual safety alignment, where the same model's English-language reasoning serves as privileged information for its low-resource-language student, transferring safety boundaries without target-language safety data. Its Dual-Perspective Safety Weighting (DPSW) concentrates distillation gradients on safety-critical tokens by combining teacher confidence with student disagreement. We discuss MSD's broader implications for safety-aware OPD in Section 9.

Beyond KL matching via reward regularization. A recurring concern across the PI literature above is that directly matching the privileged teacher's distribution can destabilize training and, in the worst case, degrade reasoning capability as training proceeds. PBSD (Yu et al., 2026) diagnoses this pathology through a reward-regularized lens, replacing pure KL matching with the objective $\max_\pi \mathbb{E}[r(x,y)] - \beta D_{\mathrm{KL}}(\pi \,\|\, \pi_{\text{teach}})$, whose analytic optimum is a reward-reweighted teacher $\pi^* \propto \pi_{\text{teach}}(y|x)\exp(r(x,y)/\beta)$ that provably dominates the teacher itself whenever the reward is informative. Practically, PBSD realizes this optimum through a DPO-style pairwise loss where the preferred sample y^+ is drawn from the context-augmented teacher and the dispreferred sample y^- from the current student, preserving on-policy sampling while avoiding the forced imitation of every teacher token. On Qwen3-1.7B/4B/8B, PBSD is the only self-distillation method that both matches OPSD's token efficiency and surpasses its peak accuracy without the post-peak decline documented in Figure 2 of Yu et al. (2026), suggesting that preference-based targets are a more principled substitute for direct KL matching than curriculum or signal-augmentation patches alone. This post-peak decline is the same pathology that CaOPD (Zhang et al., 2026b) attributes to the training-deployment context mismatch and that Kim et al. (2026a) diagnose as epistemic suppression (Section 7.2), so PBSD can also be read as a self-distillation mechanism that converts these diagnostic insights into a corrective optimization target rather than a post-hoc recalibration step.

Outcome-conditioned PI at the turn level for agentic reasoning. The PI methods above all target single-turn generation. Extending PI to multi-turn agentic settings opens a separate question of granularity: at which temporal level should privileged information intervene? Existing agentic OPD work has explored the *trajectory* granularity (TCOD's temporal curriculum over turn prefixes, Section 6.2), the *step* granularity (MAD-OPD's step-level decomposition, Section 8), and the *skill* granularity (Skill-SD's skill-conditioned updates, Section 8), but none of them supply PI at the *turn* level itself. TT-OPD (Jeong, 2026) fills this gap. A gradient-free EMA teacher receives correctness-dependent cues (reinforcing phrases for trajectories scored correct, corrective phrases for those scored incorrect) that are inserted at the prompt-response boundary and then stripped from the teacher's output log-probabilities, so the student never observes the hints directly but still receives an outcome-aware KL signal at every conversation turn. Two aspects distinguish this instance from prior PI mechanisms. *First, the privileged context encodes trajectory-level correctness rather than problem-level ground truth*, which is the first PI realization we are aware of where the teacher is conditioned on a property of the student's own rollout rather than on external supervision. This generalizes the PI paradigm from single-response supervision (GATES, OPSD, GUI-SD) to multi-turn behavioral regularization. *Second, the ablation isolates why* *turn-level PI matters here*: EMA teacher updates alone eliminate the KL-collapse pathology but still allow the multi-turn structure to erode, and outcome hints without length control trigger response explosion, so neither EMA nor hints in isolation suffice. These pathologies are among the failure modes documented in Section 7.2, and the interaction between teacher dynamics and outcome-conditioned PI is what appears to keep agentic OPD stable in this setup.

Viewed together with TCOD, Skill-SD, and MAD-OPD, TT-OPD completes a granularity spectrum for agentic distillation in which the choice of temporal unit (trajectory, turn, step, skill) is matched to the level at which errors compound in the target task.

Structured PI for video reasoning. VISD (Lin et al., 2026) extends the privileged information paradigm from generic binary signals (OPSD's correctness conditioning) to *structured multi-dimensional* feedback for VideoLLM reasoning. A video-aware judge model evaluates rollouts along answer correctness, logical consistency, and spatio-temporal grounding quality, producing structured feedback that decomposes reasoning errors into diagnostically meaningful subtypes. The central mechanism is direction-magnitude decoupling. Rollout-level advantages from environmental rewards determine the *update direction* (which trajectories to reinforce or penalize), while the structured teacher-student discrepancy conditioned on judge feedback modulates *token-level update magnitudes*. This ensures RL stability while allowing fine-grained credit assignment aligned with specific reasoning error types rather than undifferentiated token-level matching. Combined with curriculum scheduling (distillation→RL transition) and EMA teacher stabilization for long video sequences, VISD attains approximately 2× faster convergence than RLVR baselines with consistent accuracy improvement on video reasoning benchmarks. The direction-magnitude decoupling parallels Self-Distilled RLVR's decomposition (Section 5.3.2), where self-distillation sets magnitudes and RL sets directions, but VISD's structured judge provides richer magnitude signals by distinguishing *why* a token matters (grounding error vs. logical inconsistency vs. factual mistake). VISD thus extends the PI spectrum from binary (OPSD) through spatial (GUI-SD) to multi-dimensional structured feedback, with domain-specific error taxonomies determining what the PI encodes.

5.3.2 Self-Play

A more radical question is whether a model can improve using nothing but its own rollouts, with no PI, no external feedback, and no additional context. The answer is a qualified yes, subject to important saturation limitations.

Self-play as a two-player game. SPIN (Chen et al., 2024) casts self-improvement as an adversarial game between successive checkpoints, training the updated model to distinguish its previous iteration's generations from human-written responses:

$$\mathcal{L}_{\text{SPIN}} = \mathbb{E}_{x,y\sim p_{\text{data}},y'\sim p_{\theta_t}}\left[\ell\left(\lambda\left(\log\frac{p_{\theta_{t+1}}(y|x)}{p_{\theta_t}(y|x)} - \log\frac{p_{\theta_{t+1}}(y'|x)}{p_{\theta_t}(y'|x)}\right)\right)\right] \tag{22}$$

SPIN carries a theoretical convergence guarantee. The global optimum of $\mathcal{L}_{\text{SPIN}}$ is achieved if and only if $p_{\theta_{t+1}} = p_{\text{data}}$, i.e., the student's distribution exactly matches the reference distribution. At this fixed point, the loss is zero and the gradient vanishes, so the iterative process is *stable*. Once the student reaches the target distribution, further iterations cannot destabilize it. The game-theoretic interpretation is that p_{data} is the unique Nash equilibrium of the two-player game, and SPIN's iterative updates correspond to fictitious play converging to this equilibrium. Starting from Zephyr-7B-SFT, it improves MT-Bench from 5.94 to 6.78 over 3 iterations, with diminishing returns at each round. This convergence guarantee comes with an inherent limitation, namely that SPIN converges to p_{data}, not to the optimal policy. If the reference data is sub-optimal (as human-written responses often are), SPIN will faithfully reproduce those sub-optimalities, which is consistent with the view that pure self-play largely recycles existing knowledge rather than generating new knowledge.

IRIS (Liao et al., 2026) offers a unifying framework for this family by replacing the implicit KL divergence in SPIN with an interpolative Rényi divergence parameterized by $\alpha \in (0, \infty)$, recovering SPIN ($\alpha \to 1$, KL regime), SPACE (Jensen-Shannon regime), and

SPIF ($\alpha = 2$, χ^2-regularized regime) as special cases. The Rényi parameter controls the exploration-exploitation tradeoff, with lower α encouraging broader distributional coverage (exploration) while higher α concentrates updates on the most likely improvements (exploitation). This unification shows that the various self-play methods in the literature differ only in their implicit divergence choice, not in their underlying mechanism, and suggests that α-tuning can be used as a meta-parameter for adapting self-play behavior to task requirements.

The trajectory from SPIN through IRIS to π-Play traces an escalating response to the saturation ceiling. SPIN accepts the ceiling (converging to p_{data}). IRIS pushes the ceiling by tuning the divergence's exploration-exploitation balance via the Rényi parameter. π-Play relaxes it further by injecting internal privileged information that generates genuinely new training signal. Each step suggests that self-play need not be inherently limited, but that breaking saturation tends to require increasingly creative sources of asymmetry. When internal asymmetry is exhausted, a natural remaining option is to inject external ground truth, motivating the external grounding mechanisms of Section 5.3.3.

Beyond adversarial games. The self-play paradigm extends beyond explicit two-player games to encompass several lighter-weight mechanisms. SDFT (Shenfeld et al., 2026) positions on-policy self-distillation as a mechanism for forgetting resistance via implicit KL regularization against the model's own prior checkpoint, converting the model's optimization trajectory into a regularization resource for continual learning. MTP via self-distillation (Kirchenbauer et al., 2026) targets *architectural* rather than behavioral improvement, converting a pre-trained autoregressive model into a multi-token predictor by training an auxiliary prediction head on the model's own next-token predictions, delivering $> 3\times$ faster decoding at typically 3–7% accuracy drop (model-dependent) and expanding the scope of what self-distillation can optimize.

On-Policy SFT (Zhao et al., 2026a) offers an instructive simplification. By filtering self-generated responses for correctness (via a verifier) and conciseness (via length truncation at the median successful length), standard SFT on this filtered on-policy data matches or exceeds full RL training. The combination of on-policy generation (which naturally produces diverse solution strategies) with outcome filtering (which selects the best strategies) implicitly performs a form of evolutionary optimization that requires no gradient through the reward function. This "generate-filter-train" recipe substantially reduces implementation complexity compared to full RL-based distillation while retaining the core benefit of on-policy data, namely distributional alignment. Self-Distilled RLVR (Yang et al., 2026a) further decomposes the gradient update into two components. Self-distillation determines the *magnitude* of per-token policy updates (how much to change each token's probability), while environmental RLVR feedback anchors the *direction* (which tokens should increase vs. decrease). At just 200 training steps, RLSD surpasses GRPO trained for 400 steps on Qwen3-VL-8B-Instruct, exhibiting $2\times$ sample efficiency from the dual signal structure.

Minimalist self-distillation. SSD (Zhang et al., 2026d) requires no RL, no verifier, no external teacher, and no code execution. It samples solutions at training-time temperature, fine-tunes via standard SFT, and reshapes token distributions in a context-dependent manner. On LiveCodeBench v6, SSD improves Qwen3-30B-Instruct from 42.4% to 55.3%, showing that when the base model is sufficiently capable, even the simplest self-play yields substantial gains. The requirement is not a sophisticated algorithm but sufficient *diversity* in the model's sampling distribution to produce high-quality variants that can serve as training targets.

Yet this also exposes the intrinsic boundary of pure self-play. When diversity is achieved through temperature sampling alone, the model's improvement is ultimately bounded by the information already present in its pre-training distribution. Bridging this boundary requires either connecting the model to external reality (Section 5.3.3) or internally generating privileged information through the self-play process itself (π-Play below).

Unifying PI and self-play. π-Play (Zhang et al., 2026g) unifies privileged information and self-play within a single multi-agent self-evolution framework. Self-play naturally produces a byproduct during task generation, the examiner agent constructs a question construction path (QCP) that provides high-quality PI at no additional cost. Three co-evolving agents (examiner π_ϕ^E, teacher π_ψ^T conditioned on QCP, student π_θ^S without it) are jointly optimized through alternating updates. As the student strengthens, the examiner generates harder tasks, and the teacher's behavior is softly constrained to track the student via a KL penalty, creating a self-adjusting curriculum that requires no external data. On Qwen3 (4B, 8B), data-free π-Play surpasses fully supervised search agents such as Search-R1 and delivers markedly higher evolutionary efficiency than conventional self-play (Dr.Zero), with the largest gains on multi-hop benchmarks where the teacher's token-level credit assignment delivers the most leverage.

Partial-solution PI. PAINT (Tan & Hong, 2026) occupies a distinctive position between pure self-play (no external information) and full PI (complete answer revealed). Rather than conditioning on the entire ground-truth solution (as in OPSD), PAINT exposes only a *partial* solution whose extent is adaptively controlled by the overlap between the student's rollout and the reference. It computes a recall-style overlap score based on the fraction of reference anchors (boxed answers, formulas, key numbers) that appear in the student's rollout, revealing more of the reference when the student is close (yielding fine-grained correction) and less when far (preventing trivial copying). This overlap-adaptive partial-solution masking creates a natural curriculum within the PI mechanism itself, where easy problems where the student nearly solves them receive precise corrective feedback, while hard problems where the student diverges completely receive only coarse directional guidance. PAINT further employs energy-space interpolation that applies the distillation loss only at positions where the teacher-student entropy mismatch exceeds a threshold, producing a sparse loss that concentrates gradient signal on the most informative positions. On competition-level mathematics (AIME 2024/2025 and HMMT 2025 average), PAINT records +2.1 over the OPSD baseline and +2.9 over GRPO, suggesting that partial revelation of privileged information can outperform both full revelation and no revelation, an information disclosure curve whose theoretical characterization remains open.

Together, π-Play and PAINT show that the self-play saturation problem can be mitigated *without* external feedback by internally generating privileged information, whether through multi-agent task construction or partial-solution conditioning. This mechanism is orthogonal to the external-verification strategies below, and combining both internal PI and external grounding remains an unexplored direction.

5.3.3 *External Feedback*

The self-play methods above require no external input but face an inherent ceiling. Without ground-truth or verifier access, the model cannot distinguish confident-but-wrong outputs from genuinely correct ones, leading to the saturation problem analyzed in Section 7.2. External feedback methods retain the self-distillation architecture (no separate teacher model) while injecting non-differentiable truth signals, typically code execution, symbolic verification, or outcome rewards, that anchor the self-play loop and prevent it from collapsing onto self-reinforcing errors. Unlike the internal-PI mechanisms of π-Play and PAINT, external feedback bridges self-distillation and reinforcement learning by treating environmental signals as the ultimate arbiter when internal asymmetries are exhausted.

From binary rewards to dense supervision. SD-ZERO (He et al., 2026) converts sparse verifier signals into dense self-supervision through a dual-role architecture. The Generator produces on-policy responses while the Reviser, conditioned on the Generator's output and its binary correctness signal, produces an improved version. The Generator then distills from the Reviser's token-level distribution, converting binary feedback into dense self-supervision. On Qwen3-4B-Instruct, SD-ZERO records 68.3% avg@8 on AIME 2024, outperforming GRPO (62.5%), showing that self-revision can be more effective than standard RL for converting sparse rewards into learning signal. SDPO (Hübotter et al., 2026) extends this beyond binary feedback to *structured textual* feedback, including runtime errors, failing unit tests, and LLM judge evaluations that construct fine-grained credit assignment attributing success or failure to specific reasoning steps rather than entire trajectories. RL from Text Feedback (RLTF) (Song et al., 2026) pushes further with free-form natural language critiques from an automated judge, training the model's single-turn policy to match its own feedback-conditioned second-turn generations.

From SD-ZERO's binary signal through SDPO's structured errors to RLTF's free-form critiques, a clear trajectory emerges. As external feedback becomes richer, the self-distillation objective becomes more precisely targeted, allowing the model to correct specific failure modes rather than globally adjusting its policy. Sample-Routed Policy Optimization (SRPO) (Li et al., 2026b) shows that these approaches can be composed by routing. Correct samples follow GRPO's reinforcement path (reward-weighted likelihood maximization) while failed samples follow SDPO's targeted logit-level correction. The insight is that correct and incorrect samples carry distinct types of learning signal, and treating both identically wastes the information each carries. SRPO raises the five-benchmark average on Qwen3-8B by 3.4% over GRPO and 6.3% over SDPO alone across science and tool-use tasks, consistent with the view that a useful external-feedback strategy involves routing each sample to its most informative objective.

Environmental grounding. RLTF and related methods use compiler outputs, unit test results, and symbolic math verifiers as external feedback that breaks self-reinforcing echo chambers. Even if the model becomes highly confident in a flawed derivation, the verifier assigns $R = 0$, delivering an unambiguous correction signal unavailable to pure self-play.

Saturation analysis. Kim et al. (2026a) trace self-distillation degradation in mathematical reasoning to the suppression of *epistemic verbalization*, the model's expression of uncertainty during reasoning. When self-distillation shortens reasoning traces, it disproportionately removes hedging phrases and uncertainty markers that serve as implicit "attention flags" for difficult sub-problems. The result is shorter but less calibrated reasoning, where the model confidently commits to flawed steps. This connects to the teacher uncertainty challenge in Section 7, where an overconfident student produces unreliable self-play targets, and the saturation problem ultimately requires either external grounding or the internally generated PI mechanism of π-Play to overcome.

Across signal sources, from white-box logits through black-box APIs to self-distillation, a fundamental trade-off between signal density and autonomy governs the design space. White-box methods supply the richest supervision ($|V|$-dimensional distributions at every token) but require same-organization deployment and co-location of teacher and student. Black-box methods sacrifice distributional information but operate across organizational boundaries, with recent methods (GAD, OVD) recovering unexpectedly effective proxy signals. Self-distillation eliminates external dependencies entirely but is bounded by the model's pre-existing capabilities unless augmented by external verification or internally generated PI. This continuum also reflects the field's maturity gradient. The white-box extreme is well-explored (GKD, DistiLLM, DSKD are mature), while the self-distillation extreme still lacks principled convergence guarantees for complex reasoning. Recent activity concentrates at the boundaries between categories, with methods like ThinkTuning blurring the line between black-box feedback and RL, and π-Play blurring the line between self-play and externally grounded verification.

6 Training Dynamics and Efficiency

Given an objective function (Section 4) and a signal source (Section 5), the final engineering decision is *how to stabilize and accelerate the on-policy training loop*. On-policy generation brings unique optimization challenges absent from standard supervised fine-tuning, including non-stationary data distributions (the student's policy shifts during training, making older rollouts stale), gradient signal-to-noise ratio (SNR) collapse on hard prompts (where all rollouts fail and no useful gradient emerges), and the computational overhead of autoregressive student rollouts followed by teacher scoring (several times more expensive than off-policy training).

These challenges are not independent. SNR collapse on hard prompts wastes compute, motivating curriculum strategies that avoid such prompts. Non-stationarity invalidates cached teacher signals, motivating efficient online scoring. And the compute overhead of autoregressive generation creates a systems-level bottleneck that requires architectural solutions. The methods in this section tackle these challenges through three complementary mechanisms.

6.1 Token and Sample Weighting

Not all teacher supervision is equally reliable in the on-policy setting. The core problem is the *flawed prefix trap*, where a student generating an erroneous token early in a sequence causes all subsequent teacher predictions to be conditioned on this out-of-distribution prefix. Because the teacher has never been trained on such prefixes, its conditional distribution $p_T(\cdot|x, \hat{y}^{\text{bad}}_{<t})$ becomes noisy and unreliable. Naively matching this corrupted signal propagates errors instead of correcting them.

Teacher reliability filtering. TIP (Xu et al., 2026b) (Token Importance Profiling) offers a principled framework for token weighting by organizing importance along two axes, student entropy h_t and teacher-student divergence δ_t. Crossing these axes yields four quadrants with qualitatively distinct learning roles. High-entropy, high-divergence tokens (Q1) carry the dominant corrective signal, high-entropy, low-divergence tokens (Q2) indicate agreement on uncertainty, low-entropy, high-divergence tokens (Q3) represent *overconfident errors* where the student is certain but wrong, and low-entropy, low-divergence tokens (Q4) are genuinely mastered. The theoretical result shows that Q3 tokens are structurally invisible to any entropy-only weighting scheme, because no non-decreasing function of entropy with $f(0) = 0$ can distinguish them from Q4. TIP's parameter-free Soft-OR score $s_t = \hat{h}_t + \hat{\delta}_t - \hat{h}_t \cdot \hat{\delta}_t$ recovers Q3 coverage while preserving Q1/Q2 sensitivity, reaching consistent improvements at 50% token retention across three model families with capacity gaps from $2\times$ to $9\times$.

SCOPE (Zheng et al., 2026) elevates this analysis from tokens to *rollouts*. Its dual-path architecture routes rollouts by correctness. Incorrect trajectories receive teacher-perplexity-weighted KL distillation (down-weighting unreliable guidance on flawed prefixes), while correct trajectories receive student-perplexity-weighted MLE (concentrating reinforcement at the capability boundary). This counters diversity collapse, the counterintuitive phenomenon where Pass@1 improves at the expense of Pass@k, yielding a 7.3% relative Pass@32 gain over competitive baselines across six reasoning benchmarks.

SelecTKD (Huang et al., 2025) applies a propose-and-verify mechanism at the token level, where the student proposes tokens that the teacher verifies through greedy Top-k or non-greedy Spec-k matching. Accepted tokens receive full loss while rejected tokens are masked or down-weighted, inducing an implicit curriculum quantified by Token Acceptance Rate (TAR). Positions where teacher-student agreement fails indicate either genuine ambiguity or distributional divergence, and in neither case does the unfiltered gradient carry reliable information.

AdaSwitch (Peng et al., 2025) presents a dynamic switching mechanism between pure exploration and guided distillation. Unlike static filtering that simply attenuates unreliable signals, AdaSwitch maintains a sliding window of recent token-level divergences (KL or JSD between student and teacher logits) and computes a running average $\bar{d}_i = \frac{1}{L}\sum_{j=i-L}^{i-1} d_j$ over a window of length L. When the current token's divergence exceeds a context-adaptive threshold $\tau_i = K \cdot \bar{d}_{i-1}$ (where K is a sensitivity multiplier), the method switches from on-policy exploration to off-policy teacher guidance. This single-switch design preserves semantic coherence (avoiding the jarring token-level alternation of prior mixing methods) while guaranteeing high-quality supervision precisely when the student's generation drifts into high-divergence regions. The threshold's adaptive nature matters here, as a fixed threshold tends to be either too aggressive (switching on easy prefixes where minor divergence is harmless) or too permissive (failing to intervene when the student enters genuinely problematic states). By conditioning on recent divergence history, AdaSwitch calibrates its intervention threshold to the local sequence difficulty.

These four methods span a design spectrum from fine-grained to coarse-grained intervention. TIP applies continuous weighting over every token, SCOPE routes entire rollouts into distinct loss pathways, SelecTKD makes binary accept/reject decisions per token, and AdaSwitch commits to a single trajectory-level mode switch. The progression exposes an underlying tradeoff between signal fidelity (finer granularity preserves more information) and optimization stability (coarser decisions reduce gradient variance). This tradeoff is not merely theoretical. Fine-grained methods like TIP require reliable token-level teacher signals, which degrade precisely when the flawed prefix problem is severe. Coarse-grained methods sacrifice per-token optimality but are robust to local signal corruption because a single bad token cannot contaminate the entire trajectory's gradient. The choice along this spectrum correlates with the dominant failure mode. Token-level methods excel when teacher reliability varies sharply within a sequence (e.g., long reasoning chains where some steps are easy and others require critical insight), while trajectory-level methods are preferred when the primary concern is prefix-quality drift across the generation (e.g., multi-turn dialogue where early errors shift the entire conversation topic).

Sample-level weighting. Beyond token-level filtering, several methods weight entire samples based on their informativeness. The flawed prefix trap manifests most severely on *hard* prompts where the student's pass rate approaches zero. On these prompts, nearly all rollouts contain early catastrophic errors, and the resulting teacher signals are dominated by noise. Methods that uniformly sample prompts waste most of their gradient budget on uninformative hard prompts or trivially correct easy prompts, with the productive learning concentrated on prompts at the boundary of the student's competence.

Failure-mode analysis. Fu et al. (2026) present a systematic empirical analysis of why sampled-token OPD is fragile in long-horizon settings. They identify three failure modes: (1) an imbalanced one-token signal that reduces distribution matching to a single sample, (2) unreliable teacher guidance on student-generated prefixes that deviate from the teacher's training distribution, and (3) distortions caused by tokenizer or special-token mismatch. Their proposed fix, teacher top-K local support matching via truncated Reverse KL with top-p rollout sampling and special-token masking, produces more stable optimization across math reasoning and agentic training.

The token and sample weighting methods above operate *reactively*, attenuating the loss signal *after* the student has already generated a potentially uninformative rollout. A more proactive approach asks whether we can select prompts *before* generation to maximize the expected information gain per rollout, motivating the curriculum methods below.

6.2 Curriculum and Difficulty Adaptation

The sample weighting problem motivates a more principled solution. *Curriculum design* actively selects prompts matching the student's current competence level. Where weighting operates on the loss *after* generation, curriculum design intervenes *before* generation by choosing which prompts to present. The design space spans multiple axes, including prompt difficulty, temporal depth within multi-turn interactions, semantic granularity of supervision, and initialization strategy, with each axis targeting a distinct failure mode of naive uniform sampling.

Competence-boundary sampling. PACED (Xu et al., 2026a) models prompt difficulty dynamically using a Beta-kernel distribution centered on the student's current pass rate. The curriculum shifts toward harder prompts as the student improves, while avoiding prompts where the pass rate is near zero (where gradient SNR vanishes) or near one (where the student has already mastered the content). This "frontier sampling" strategy concentrates the gradient budget on the narrow band of prompts where learning is most efficient.

The mathematical justification connects to the gradient SNR analysis. For a prompt with pass rate p, the expected gradient magnitude under binary reward vanishes at both extremes ($p \to 0$ and $p \to 1$) and peaks at intermediate difficulty. PACED formalizes this through a general Beta-kernel weight family $w(p) = p^{\alpha}(1-p)^{\beta}$, where the symmetric case $\alpha = \beta = 1$ yields the familiar $p(1-p)$ with peak at $p = 0.5$, but asymmetric choices ($\alpha \neq \beta$) shift the peak to prioritize harder or easier prompts depending on the training phase. This generalization matters in practice because the optimal curriculum is rarely symmetric, as early training benefits from emphasizing moderate-difficulty prompts ($\alpha > \beta$, peak shifted left) while late training benefits from focusing on the remaining hard cases ($\alpha < \beta$). Concretely, PACED maintains a running estimate of per-prompt difficulty $\hat{p}_i$ (computed from the last m rollouts) and samples each prompt with probability proportional to $\text{Beta}(\hat{p}_i; \alpha, \beta)$ where (α, β) control both the width and asymmetry of the competence frontier. Early in training, wide kernels cast a broad net. As training converges, narrow kernels focus on the diminishing frontier of unsolved problems. This concentrates rollouts on problems in the productive learning range and reduces wasteful sampling on already-mastered or intractable problems, with compute savings proportional to the curriculum efficiency.

This SNR analysis has a deeper implication that extends beyond curriculum design, as it explains *why* on-policy training naturally outperforms off-policy on reasoning tasks without explicit curriculum engineering. In off-policy settings, the problem difficulty distribution is fixed by the dataset. As the student improves, an increasing fraction of training data falls into the "too easy" zone where gradients are negligible, leading to diminishing returns. On-policy training, by contrast, automatically adjusts the effective difficulty because the student's rollouts reflect its current capabilities, naturally generating sequences at the boundary of its competence. PACED makes this automatic adjustment explicit and tunable, but even naive on-policy training captures much of the benefit simply by sampling from p_θ rather than p_{data}.

Self-adjusting curricula. Several methods from earlier sections supply implicit curriculum mechanisms without explicit difficulty estimation. π-Play's multi-agent framework (Section 5.3.2) automatically generates harder tasks as the student strengthens, PAINT (Tan & Hong, 2026) (Section 5.3.2) adapts its partial-solution disclosure based on student-teacher overlap, and PRISM (Wang et al., 2026d) (Section 5.2) shapes the policy landscape for subsequent RL through a structured multimodal warm-up. These complement PACED's explicit difficulty estimation with emergent difficulty adaptation.

Dual-perspective curriculum. Where PACED treats the student as a black box whose pass rate fully characterizes prompt difficulty, Uni-OPD (Hou et al., 2026) argues that pass rate alone conflates two distinct failure modes: problems the student cannot solve (low pass rate due to capability gaps) and problems the student solves incorrectly in subtle ways (moderate pass rate but low-quality solutions). Separating these requires examining not just *whether* the student succeeds but *how* it fails. From the *student* perspective, Uni-OPD implements difficulty-aware and correctness-aware data balancing that promotes exploration of informative states during rollout sampling, extending PACED's frontier-sampling insight from a single pass-rate metric to a dual-criterion that jointly considers problem difficulty and rollout correctness. From the *teacher* perspective, it tackles a subtler failure mode. Even when the student generates informative rollouts, aggregated token-level supervision can become inconsistent with outcome-level reward signals. A trajectory that receives higher token-level teacher scores may actually produce a worse final answer. Uni-OPD's outcome-guided margin calibration restores order consistency between correct and incorrect trajectories, so that the token-level gradient direction aligns with the sequence-level quality ordering. This dual-perspective framework generalizes across LLMs and MLLMs (5 domains, 16 benchmarks) and accommodates both single-teacher and multi-teacher settings, implying that the curriculum and supervision reliability problems are largely orthogonal and can be solved independently.

Temporal curriculum for multi-turn distillation. PACED, Uni-OPD, and the self-adjusting methods all operate along the *difficulty* axis, selecting *which prompts* to train on. TCOD (Wang et al., 2026c) opens an orthogonal axis by asking not which prompts but *how deep* within a single prompt's trajectory to apply supervision, tackling a failure mode unique to multi-turn agent distillation. In standard OPD applied to multi-turn tasks, the student generates entire interaction trajectories and receives teacher supervision at every turn. TCOD identifies "Trajectory-Level KL Instability," where as conversation depth increases, KL divergence between student and teacher distributions escalates exponentially because errors in early turns shift the state distribution, making subsequent teacher supervision increasingly misaligned with the student's actual trajectory context. The proposed temporal curriculum progressively expands the horizon of teacher supervision rather than applying it uniformly across all turns from the start. Two scheduling variants target complementary failure modes.

Forward-to-Backward (F2B) begins supervision at early turns and progressively extends to later turns, allowing the student to first establish reliable early-turn behavior before tackling the compounding-error regime. Backward-to-Forward (B2F) uses the teacher to generate early turns (supplying a clean state distribution) and applies supervision only at later turns, progressively requiring the student to handle earlier turns independently as training progresses. On ALFWorld, WebShop, and ScienceWorld, TCOD delivers gains of up to +18 points over vanilla multi-turn OPD, with the F2B variant showing particular strength on tasks where early-turn errors are most consequential. The temporal curriculum insight connects directly to the gradient SNR framework developed above. Just as PACED identifies that prompts with near-zero pass rates yield negligible gradients, TCOD identifies that later turns in long trajectories suffer analogous gradient degradation because the student's state has drifted too far from where the teacher's supervision is meaningful.

Off-policy cold start. Li et al. (2026d) identify that OPD can fail catastrophically when the student's initial policy is too far from the teacher's distribution (the "thinking-pattern mismatch" problem). Their proposed solution is an off-policy cold-start phase, a brief period of standard SFT on teacher-generated data that reduces the pattern gap before switching to on-policy training. This two-phase approach, off-policy warmup followed by on-policy refinement, has become a common industrial pattern (Section 8).

Retaining knowledge during on-policy training. A complementary concern is that on-policy training can cause *catastrophic forgetting* of capabilities the student acquired during pre-training. Chen et al. (2025) ("Retaining by Doing") demonstrate that on-policy data generation itself serves as an implicit rehearsal mechanism. By generating sequences from its own distribution, the student naturally revisits and reinforces its existing capabilities while learning new ones. This insight suggests that on-policy generation delivers a dual benefit, both exposing the student to its own error states (allowing correction) and rehearsing its existing capabilities (preventing forgetting). The observation offers a theoretical explanation for why on-policy methods empirically suffer less from catastrophic forgetting than sequential fine-tuning approaches.

Semantic bootstrapping. Semantic Soft Bootstrapping (Mitra & Ulukus, 2025) offers a different perspective on curriculum design. Rather than adapting the *difficulty* of prompts, it adapts the *semantic granularity* of supervision. The method supplies the model with both correct and incorrect rollouts as in-context exemplars that shape the next generation, progressively refining from coarse semantic alignment to fine-grained token-level matching. This posts +10.6% on MATH-500 over GRPO, showing that in-context self-supervision can complement gradient-based optimization.

The hybrid SFT+OPD pipeline. The off-policy cold start (Li et al. (2026d)) combined with on-policy refinement has emerged as a common industrial recipe. Empirically, a useful transition point from off-policy to on-policy training appears to be when the student's generation distribution reaches sufficient overlap with the teacher's. Transitioning too early can waste on-policy compute on uninformative rollouts (the student is too weak for meaningful self-generation). Transitioning too late tends to leave the student stuck near the off-policy ceiling. Qwen3 (Yang et al., 2025a) follows this two-phase approach, using off-policy distillation to establish basic reasoning and mode-switching capabilities before transitioning to on-policy distillation where the student generates its own sequences for logit-level alignment. DeepSeek-R1 (DeepSeek-AI et al., 2025) uses 100% off-policy (no transition), which works well at extreme teacher capability but may leave performance on the table for reasoning tasks where the student could exceed the teacher through on-policy exploration.

6.3 Compute Optimization

Token weighting (Section 6.1) reduces the effective gradient budget wasted on uninformative tokens, and curriculum design (Section 6.2) avoids generating uninformative rollouts altogether. Together they improve *sample efficiency*, extracting more learning per rollout. Yet neither eliminates the underlying *systems-level* bottleneck. OPD requires generating student rollouts *and* scoring them with the teacher at every training step, creating substantial compute overhead relative to off-policy SFT regardless of how efficiently each rollout is utilized. The methods below attack this orthogonal cost dimension.

The cost breakdown for a typical white-box OPD step consists of

1. **Student rollout**: autoregressive generation of $B \times K$ completions (batch size B, K rollouts per prompt), memory-bound due to KV cache growth. In practice this component typically dominates wall-clock time, since generation is sequential and cannot be trivially parallelized across tokens.
2. **Teacher scoring**: forward pass through the teacher to obtain full-vocabulary logits on student-generated sequences, compute-bound. When the teacher is considerably larger than the student, this pass can rival or exceed the student's backward pass in cost.
3. **Student update**: standard backward pass computing gradients of the distillation loss, dominated by optimizer state memory. This component is usually the cheapest of the three but sets the lower bound on memory usage.

Each component admits independent optimization, and the methods below attack different bottlenecks.

Prefix truncation. Fast OPD (Zhang et al., 2026a) observes that the useful distillation signal (measured by per-token reverse-KL loss) is concentrated in the prefix of student-generated sequences, because the student is weakest at high-level planning decisions captured in early tokens. Later tokens contribute diminishing marginal signal as the generation becomes increasingly constrained by the prefix context. Exploiting this observation, Fast OPD truncates on-policy rollouts to a prefix of length k and reverts to off-policy KD for the remainder, reaching comparable quality at 2–47× reduced training FLOPs depending on truncation ratio and task. The optimal truncation point k depends on the information density profile of the task. For mathematical reasoning, where the critical planning decisions (proof strategy, decomposition) are concentrated in the first few tokens, short prefixes ($k \leq 64$) suffice. For open-ended generation, where stylistic and semantic coherence require longer-range on-policy signals, larger k is needed.

Offline teacher caching. Lightning OPD (Wu et al., 2026) decouples student optimization from teacher inference entirely by precomputing teacher distributions into an offline memory bank and asynchronously refreshing them. When teacher distributions are *consistent* (the teacher's output depends primarily on the prompt and only weakly on the specific student prefix), the cached distributions remain valid even as the student's policy evolves. Lightning OPD shows that offline distillation under teacher consistency can match online OPD at 4× lower cost, substantially lowering the barrier for academic research.

These two methods expose a deeper structural point. The "on-policy" requirement is not binary but admits degrees of approximation. Fast OPD shows that partial rollouts preserve most of the distributional alignment benefit. Lightning OPD shows that stale teacher signals are acceptable when the student's policy drift is bounded. Together, they define a *fidelity–efficiency frontier* along which the degree of on-policy approximation trades off against compute cost, with full on-policy training at one extreme (maximum alignment, maximum cost) and fully offline SFT at the other (minimum cost, exposure bias ceiling).

Speculative knowledge distillation. Speculative KD (Xu et al., 2025c) adapts speculative decoding to the distillation setting, where the student generates candidate tokens that the teacher *verifies and selectively replaces*, amortizing the teacher's inference cost across multiple student tokens per teacher forward pass. DistillSpec (Zhou et al., 2024) inverts this approach. Instead of using speculation to speed up training, it uses on-policy KD to improve the draft model for speculative decoding at *inference* time. The expected speedup:

$$E[S] = \frac{\mathbb{E}[K_{\text{accepted}}] + 1}{1 + c} \quad \text{where } c = \frac{\text{cost(student forward)}}{\text{cost(teacher forward)}} \tag{23}$$

The result is a virtuous cycle. Better distillation produces better draft models, which enable faster speculative decoding, which in turn permits more efficient distillation, steadily reducing the computational barrier.

GPU memory management. Beyond FLOPs, the most severe constraint for white-box OPD is GPU memory. Holding a 70B teacher alongside a 7B student requires the teacher's weights (∼140 GB in BF16), the student's weights and optimizer states (∼84 GB), and the full-vocabulary logits tensor $[B, T, |V|]$ for distillation. Peak memory can easily exceed an 8×80 GB H100 node. Practitioners alleviate this via (1) *teacher quantization*, serving the teacher in FP8 or INT4 for a 2–4× memory reduction at negligible precision loss, (2) *logit offloading*, immediately streaming logits to CPU RAM or retaining only the top-k entries (since 99% of the probability mass is typically concentrated in fewer than 100 tokens), and (3) *aggressive gradient checkpointing* to minimize student activation memory during the backward pass. Combining these techniques is typically required for viable white-box OPD on standard clusters.

Concrete cost example. To ground these formulas, consider distilling a 70B teacher into a 7B student on 8×H100 GPUs (representative estimates based on scaling from reported benchmarks). Off-policy over 1B tokens involves three stages. The teacher generates the dataset offline (∼200 GPU-hours), the student trains for ∼100 GPU-hours, totaling ∼300 GPU-hours. On-policy over 1B tokens, each step requires (1) student generation (∼3× forward-pass cost due to autoregressive decoding), (2) teacher scoring (one 70B forward pass), and (3) student backward pass, yielding ∼1,200–1,500 GPU-hours, a 4–5× overhead consistent with empirical measurements at smaller scales (Wu et al., 2026). For reasoning tasks, on-policy methods consistently outperform off-policy SFT, and the off-policy performance ceiling is lower because static datasets cannot cover the combinatorial space of valid reasoning paths. This "ceiling gap" widens with task complexity, providing the strongest cost-benefit justification for on-policy methods. Section 7.4 formalizes this cost-quality tradeoff into a decision framework.

Asynchronous generation-training decoupling. NPD (Rang et al., 2026) attacks the most critical compute bottleneck of OPD, the synchronous coupling between student generation and gradient updates that forces the GPU to alternate between memory-bound decoding and compute-bound backpropagation. NPD reformulates on-policy distillation as a three-stage asynchronous pipeline. Student rollouts are batch-generated via vLLM, teacher top-k logits are computed through parallel prefill with sequence packing, and the student trains with a composite CE+KD loss on packed sequences. The key challenge in async OPD is policy lag, where the generation policy drifts from the training policy across async iterations, potentially degrading into off-policy learning. NPD counters this with Δ-IFD filtering, which estimates teacher-student discrepancy via an Instruction-Following Difficulty metric and rejects extreme out-of-distribution samples that would push updates outside a "safe proximal learning zone." The resulting framework records 8.1× throughput speedup over synchronous on-policy baselines while outperforming SFT by +8.09% on math benchmarks, with the 1B student (openPangu-Embedded-1B, 68.73%) surpassing the larger Qwen3-1.7B (63.69%). NPD also shows that on-policy distillation produces superior initializations for subsequent RL, narrowing the exploration space for GRPO and delivering the best NPD→RL synergy among tested pipelines. The relationship between NPD and the other compute methods is one of orthogonal scope. Fast OPD reduces the *length* of each rollout, Lightning OPD eliminates the *freshness* requirement for teacher signals, and NPD eliminates the *synchronization* requirement between generation and training. All three can be composed within a single system.

The training efficiency stack. The three subsections above form a layered efficiency stack. Token weighting (Section 6.1) eliminates noise at the gradient level, so that each rollout contributes maximally to learning. Curriculum design (Section 6.2) eliminates waste at the sampling level, so that each rollout targets the student's competence frontier. Compute optimization (this section) eliminates overhead at the systems level, reducing the cost of producing and scoring each rollout. These layers are complementary rather than substitutive. PACED's frontier sampling generates maximally informative prompts, TIP's token weighting extracts maximal signal from the resulting rollouts, and Fast OPD's prefix truncation minimizes the generation cost of those rollouts. Deploying all three layers simultaneously can compound their individual 2–4× efficiency gains, potentially reducing the effective cost gap between on-policy and off-policy training from 4–5× to near parity while retaining the quality ceiling advantage.

7 Understanding OPD: Theory, Failure Modes, and Cost

The preceding sections addressed the mechanics of on-policy distillation, covering what to optimize (Section 4), where the signal comes from (Section 5), and how to stabilize training (Section 6). This section turns to the equally important questions of *why* and *when*: under what conditions does OPD succeed, how does it fail, what theoretical frameworks unify the methods surveyed above, and when is OPD preferable to simpler off-policy pipelines?

These four dimensions of understanding are not independent but form a cumulative argument. Identifying *when* OPD works (Section 7.1) establishes the preconditions that the theoretical framework must explain. Cataloging *how* it fails (Section 7.2) motivates the specific design choices, adaptive divergences, curriculum weighting, and hybrid RL objectives, whose theoretical grounding Section 7.3 supplies. Both the success conditions and the failure modes then inform the practical cost-benefit analysis in Section 7.4, which distills the theoretical insights into engineering guidelines. Resolving these questions moves OPD from an empirical art toward a principled engineering discipline with predictable outcomes (Zhang et al., 2026c; Li et al., 2026d).

7.1 Success Conditions

Li et al. (2026d) present the most systematic investigation of OPD training dynamics, identifying two necessary conditions for effective distillation. First, the student and teacher must share compatible reasoning patterns (operationalized as high overlap in their top-k token distributions). Even a superior teacher fails to transfer knowledge if its thinking style is structurally mismatched with the student's (e.g., a non-thinking teacher distilling into a thinking student), because low initial distributional overlap cannot be recovered through training alone. Second, the teacher must supply genuinely new capabilities beyond what the student has already acquired. When both models are trained on the same data and recipe, they converge to similar distributions at their respective scales, leaving the teacher with little transferable signal, as reverse distillation experiments show that same-family 1.5B and 7B teachers become distributionally indistinguishable from the student's perspective.

At the token level, successful OPD is characterized by progressive alignment on *high-probability overlap tokens*, a small shared token set that concentrates the bulk of the probability mass. When these conditions fail, recovery strategies include off-policy cold start (SFT initialization that reduces the pattern gap) and teacher-aligned prompt selection. Critically, the quality of OPD's dense supervision degrades systematically with trajectory depth, with instability originating at later tokens and propagating backward, raising open questions about OPD's applicability to long-horizon reasoning.

These findings point toward a general principle. OPD's benefit scales with the *exploitable gap* between teacher and student. When this gap is too small (same-recipe models), the teacher offers no new information. When too large (thinking-pattern mismatch), the student cannot absorb the supervision. The productive regime lies in between, and identifying it is the central diagnostic challenge.

Diagnostic checklist. Based on the surveyed literature, four pre-training diagnostics emerge as predictors of OPD benefit:

1. *Token overlap ratio*: Compute top-k overlap between teacher and student on a held-out set. If too low, apply off-policy cold start first.
2. *Pass rate on target prompts*: If the student cannot solve any problems in the target domain, OPD gradients will vanish. Use curriculum pacing (Xu et al., 2026a) or easier warm-up tasks.
3. *Teacher calibration*: Verify that the teacher's confidence correlates with correctness on student-generated prefixes (Zhang et al., 2026b). Poorly calibrated teachers inject noise.
4. *Length stability*: Monitor output length during early training. Monotonic growth indicates the KL gradient asymmetry identified by Luo et al. (2026). Apply Stable-OPD corrections preemptively.

OPSD as compression, not correction. Kim & Lee (2026) sharpen the characterization of *what* on-policy self-distillation actually accomplishes in thinking-enabled mathematical reasoning. By applying OPSD separately to correct-only versus incorrect-only rollout groups, they isolate two potential mechanisms. Correct-only OPSD preserves accuracy while substantially shortening reasoning traces (compression), whereas incorrect-only OPSD damages accuracy (correction fails). Three alternative explanations (richer teacher context, mid-trace feedback reinjection, and extended training) are tested and ruled out, supporting the interpretation that the hindsight-conditioned self-teacher can identify redundancy in long thinking traces but does not reliably supply better alternative reasoning steps. This finding refines the success conditions above. OPSD's productive regime appears to be "making the model express known solutions more efficiently" rather than "making the model solve harder problems". The practical implication is a refined pipeline ordering. SFT establishes format compliance, RLVR expands the set of reachable correct trajectories, and OPSD compacts them for cheaper inference. Reversing the last two stages (compressing before exploration) tends to remove the raw material that exploration would produce, suggesting why post-RL placement is preferred. Across Qwen3-8B, DeepSeek-R1-Distill-7B, and AceReason-7B, the post-RLVR + correct-only OPSD configuration reaches the best position in the accuracy-length plane in their experiments, consistent with OPSD's mechanistic role rather than implementation details.

7.2 Failure Modes

Complementing the success conditions, a growing body of work catalogs the specific ways OPD can fail even when structural preconditions hold, as artifacts of the on-policy training dynamics themselves. Understanding these failure modes helps debug OPD pipelines in practice, and each mode has a corresponding mitigation strategy developed in the methods sections above.

We organize failure modes by their *root cause* instead of their symptom, as multiple failure modes can produce similar surface-level degradation (e.g., accuracy collapse) through different mechanisms.

The flawed prefix trap. As detailed in Section 6.1, when the student generates an erroneous prefix, the teacher's conditional distribution over subsequent tokens is poorly calibrated because the teacher was never trained on such out-of-distribution inputs. Fu et al. (2026) formally characterize three specific failure modes, including (1) an imbalanced one-token signal that reduces distribution matching to a single sample, (2) unreliable teacher guidance on out-of-distribution student prefixes, and (3) tokenizer mismatch distortions. The theoretical analysis shows a fundamental bias-variance tradeoff, where token-level OPD is biased relative to sequence-level Reverse KL but enjoys tighter worst-case variance bounds.

Two subtler failure modes compound the prefix trap. *Gradient SNR collapse* occurs when the student's pass rate on a prompt approaches zero. In this state, all rollouts contain early errors, teacher signals are dominated by noise, and the gradient SNR vanishes. This explains why hard prompts yield no learning signal in standard OPD and motivates PACED's competence-boundary curriculum (Section 6.2). *Epistemic suppression* (Kim et al., 2026a) manifests in self-distillation. When the process shortens reasoning traces, it disproportionately removes hedging phrases and uncertainty markers that serve as implicit attention flags for difficult sub-problems. This produces shorter but less calibrated reasoning where the model confidently commits to flawed steps. This calibration degradation connects directly to the teacher uncertainty challenge, where an overconfident student produces unreliable self-play targets, creating a self-reinforcing cycle that compounds with each iteration.

Self-play saturation (the Ouroboros problem). In self-distillation settings, the student optimizes against a target sharing its own inductive biases and architectural limitations, causing the hypothesis space to gradually collapse. If the model discovers a syntactic "hack" or a highly confident but flawed reasoning heuristic, no external signal penalizes it. The model self-reinforces this flawed trajectory, driving $p_\theta(y_{\text{flawed}}|x) \to 1$. Once entirely certain of its own hallucinations, gradients vanish, exploration ceases, and the policy is trapped. This is distinct from exposure bias (which arises from train-test distribution mismatch in off-policy settings). Saturation arises from the absence of distributional diversity in fully on-policy self-play, bearing structural similarities to GAN mode collapse, where a generator locks onto a narrow output manifold once the discriminator signal becomes non-informative.

The precision-recall tradeoff. Cha & Cho (2025) formalize a core tension in distillation. As the student concentrates mass on high-quality outputs (improving precision), it necessarily reduces coverage of the teacher's full output distribution (reducing recall). In the OPD setting, this manifests as the "diversity collapse" problem. Aggressive Reverse KL distillation produces a student that scores highly on Pass@1 (precision) but catastrophically on Pass@k (recall), because the student has collapsed onto a single reasoning strategy per prompt. On-policy generation can shift this tradeoff by allowing the student to explore its own distribution instead of being constrained to teacher-demonstrated paths, potentially improving recall without sacrificing precision. Temperature-scaled sampling during on-policy generation, as used in GKD and G-OPD, offers a direct control knob: higher sampling temperatures encourage distributional exploration (increasing recall) at the cost of noisier gradients (potentially reducing precision).

The calibration-capability gap. CaOPD (Zhang et al., 2026b) identifies a "Scaling Law of Miscalibration". While OPD reliably improves task accuracy, the authors report that it tends to leave models in a state of severe overconfidence. They attribute the root cause to an information mismatch. Teacher supervision is formed under privileged context available during training (full reasoning traces, multiple samples), whereas the deployed model must report confidence using only its deployment-time context. By decoupling capability improvement from calibration degradation, CaOPD argues that the distilled model becomes more capable but less aware of its own uncertainty boundaries. This finding has potential implications for deployment. A distilled model that scores higher on benchmarks but does not reliably indicate when it might be wrong is arguably less safe than a weaker but better-calibrated baseline. The "illusion of certainty" connects to the saturation analysis above. Both indicate that OPD can produce students that lose metacognitive capabilities (uncertainty awareness, epistemic verbalization) even as task-level performance improves.

Agentic collapse in multi-turn OPD. The preceding failure modes all assume a single-turn response. Extending OPD to multi-turn tool-using agents surfaces a qualitatively different failure family, in which the interaction between teacher dynamics and trajectory structure produces pathologies unseen in single-turn settings. TT-OPD (Jeong, 2026) diagnoses three such pathologies on Healthcare AI Gym, and each maps onto a distinct structural mismatch that other agentic OPD methods also confront from complementary angles. *First, teacher-dynamics collapse.* Periodic hard-copy teacher resets cause a KL collapse (divergence drops abruptly from its accumulated value to near zero, e.g., $2.637 \rightarrow 0.343$ at a single copy event with period T=30), erasing the distillation gradient and driving accuracy monotonically downward. This is the multi-turn analogue of TCOD's "Trajectory-Level KL Instability" (Section 6.2), but surfaced at the teacher-update granularity rather than the turn-depth granularity, and shows that stable teacher dynamics (EMA rather than hard copy) are a prerequisite for any other structural fix. *Second, trajectory-structure erosion.* Even with EMA teachers that eliminate KL collapse, the multi-turn structure itself erodes, as the student learns that verbose single-turn monologues maximize the sparse terminal reward more easily than coordinated tool-use sequences (turns collapse from 7.65 to 5.52 per episode under periodic resets, and still erode from 7.82 to 6.23 even with EMA teachers). This mirrors the reward-granularity mismatch that motivates Skill-SD's skill-level decomposition and MAD-OPD's step-level sampling (Section 8), indicating that sparse terminal rewards alone cannot sustain multi-turn structure regardless of which granularity is chosen. *Third, reward-hint runaway.* Adding outcome-conditioned teacher hints without length control triggers a response explosion, where positive hints reinforce ever-longer reasoning until responses saturate the context limit and accuracy collapses from 54.5% to 49.0%. This is the multi-turn extension of the epistemic-suppression phenomenon documented in Kim et al. (2026a), as both arise when informative supervision signals that are helpful *on average* are applied without regularizers on the shape of the student's response.

Taken together, these pathologies imply a general principle for agentic OPD. Stable teacher dynamics, explicit regularizers on trajectory structure, and granularity-matched credit assignment appear jointly necessary based on current evidence, and each existing method (TCOD, Skill-SD, MAD-OPD, TT-OPD) targets a different subset of the three.

7.3 Unified Theoretical Perspectives

The success conditions and failure modes above share deeper mathematical roots than their diverse surface manifestations suggest.

The unified f-divergence framework of Wen et al. (2023) subsumes Forward KL, Reverse KL, JSD, and α-divergences as special cases parameterized by a single convexity function f. Since all these divergences can be expressed as expectations under the student's on-policy distribution, the choice of divergence is ultimately a regularization decision. The generator function f determines the implicit weighting of likelihood ratios $p_T(y)/p_\theta(y)$, with Forward KL up-weighting regions where the student underestimates the teacher (coverage) and Reverse KL up-weighting regions of overestimation (precision). Wu et al. (2025) refine this picture for discrete LLM distributions, showing that under practical epoch budgets the mode-seeking/covering dichotomy is better understood through head-tail focus. Forward KL concentrates learning on the head of the teacher distribution while Reverse KL emphasizes the tail, explaining why adaptive methods that switch between divergences based on local geometry consistently outperform any fixed choice. At a more foundational level, Cha & Cho (2025) identify the minimal conditions under which a student can provably benefit from teacher supervision versus learning from data alone. The benefit scales with the information gap between the teacher's distribution and the data distribution as seen by the student, providing a theoretical foundation for the "exploitable gap" identified empirically by Li et al. (2026d). A complementary unification comes from Li et al. (2026b), who show that GRPO and self-distillation tackle complementary failure modes and can be composed within a single framework via sample routing. Correct samples follow GRPO's reward-aligned reinforcement path while failed samples receive dense logit-level correction from a feedback-conditioned self-teacher, suggesting that the two paradigms target distinct learning signals instead of being redundant approaches to the same objective.

Length inflation (OPD students' outputs growing progressively longer during training) is a persistent practical concern. Luo et al. (2026) identify the root cause as an asymmetry in the KL gradient, where positive KL gradients (pushing toward teacher-preferred tokens) are stronger than negative ones (pulling away from dispreferred tokens), creating a net bias toward longer sequences. Their Stable-OPD framework adds a reference divergence term anchoring the student's length distribution to a pre-distillation baseline, combined with rollout mixing that blends on-policy student generations with off-policy teacher generations in a decaying ratio. These stabilization mechanisms post +7.2% over vanilla OPD by breaking the self-amplification cycle, and are composable with any divergence choice.

G-OPD's equivalence derivation (Section 4.3) supports the view that standard OPD can be cast as a special case of dense KL-constrained RL, bridging the distillation and RL communities and making it more likely that advances in either field transfer to the other. This connection extends to DPO-based methods. As shown in Section 4.3, substituting the teacher's log-probability as DPO's implicit reward produces the same geometric mixture targeted by GKD's λ-mixing, relating token-level KL distillation, preference optimization, and reward-regularized RL as three optimization trajectories toward the same family of target distributions parameterized by divergence choice and supervision density.

The unified KD+RL framework of Li et al. (2025) makes this complementarity precise by decomposing the hybrid gradient into a dense KD component (stable, analytically computable) and a Monte Carlo RL component (high variance, reward-driven). KD dampens variance in policy gradient estimation while RL prevents collapse onto suboptimal teacher modes, with the optimal KD-to-RL weighting ratio decreasing over training as the student's policy stabilizes. Song et al. (2026) further show that textual feedback from a teacher can expand the effective capabilities of RL beyond what outcome-only rewards attain, offering theoretical grounding for the verbal feedback methods (Lion, OVD, ThinkTuning) surveyed in Section 5.2. Text feedback carries a denser reward signal than binary outcomes, allowing the student to learn from partial successes that binary verification would discard.

Together, these perspectives converge on a central insight. On-policy distillation, reinforcement learning, and preference optimization are not separate training approaches but rather points on a continuous spectrum parameterized by the density of supervision, the choice of divergence, and the source of the training signal.

7.4 On-Policy vs. Off-Policy: A Decision Framework

The theoretical frameworks above establish that on-policy training offers principled advantages over off-policy alternatives. But theory alone cannot settle the engineering question of when this advantage justifies the additional infrastructure cost. This section distills the accumulated evidence into a practical decision framework.

The DeepSeek-R1 anomaly. DeepSeek-R1 (DeepSeek-AI et al., 2025) represents a prominent off-policy distillation success, training students from 1.5B to 70B parameters on $\sim$800K chain-of-thought traces via purely off-policy SFT. On AIME 2024 (pass@1), R1-Distill-Qwen-7B reaches 55.5% and R1-Distill-Qwen-32B reaches 72.6%, clearly outperforming direct RL at the same scale (GRPO on Qwen2.5-32B-Base scores only 47.0%). Three factors plausibly explain this anomaly. An exceptionally strong 671B MoE teacher covers the problem space densely, diverse reasoning traces include backtracking and verification strategies, and relatively small student models maintain manageable effective conditional complexity under off-policy learning. The implication is that off-policy distillation often suffices when a massively capable teacher generates sufficiently diverse traces for a relatively small student. On-policy becomes more attractive as student capacity grows and diverges from the static training distribution.

The off-policy ceiling. DDT (Zhang et al., 2026c) supplies formal evidence for the generalization limits of off-policy approaches. The key factor driving on-policy generalization advantage is that generating training data from the current policy keeps learning aligned with the model's evolving distribution. Off-policy performance is bounded by distributional mismatch that accumulates as the student's policy drifts from the static training data, because fixed datasets cannot cover the combinatorial space of valid reasoning paths. This gap widens with task complexity (more reasoning steps create more paths to cover), output diversity requirements (creative tasks vs. mathematics), and model capability (stronger students explore further from training data).

Distillation vs. direct RL. The choice between knowledge distillation and direct RL depends on three factors that interact non-trivially. Teacher quality relative to the reward signal determines whether dense per-token supervision outweighs sparse outcome rewards. Student scale modulates this tradeoff: smaller students ($\leq$7B) benefit more from dense teacher supervision because their limited capacity cannot efficiently explore reward landscapes, while larger students ($\geq$32B) have sufficient capacity for RL exploration. Target ceiling determines whether the goal is to match the teacher (distillation-bounded) or surpass it (RL-enabled). The hybrid approaches surveyed in Section 4.3 (G-OPD, KDRL, RLAD) resolve this tension by combining teacher logits as a dense variance-reducing baseline with sparse outcome rewards that drive exploration beyond the teacher's ceiling. Empirically, this combination dominates either approach alone across all student scales.

The hybrid pipeline as standard recipe. Qwen3 (Yang et al., 2025a), DeepSeek-V4 (DeepSeek-AI, 2026), and MiMo-V2 (Xiaomi LLM-Core Team et al., 2026) converge on a two-phase pattern in which off-policy SFT establishes a strong distributional foundation and on-policy OPD or RL closes the remaining gap between the student's generation distribution and the optimal policy. The transition point depends on the student's initial capability. Weaker students benefit from extended off-policy warmup to reduce the teacher-student pattern gap (Li et al., 2026d), while stronger students can transition earlier because their on-policy rollouts already occupy high-density regions of the teacher's distribution.

Compute-quality tradeoff. The expected compute cost for N tokens under each regime:

$$C_{\text{off}} \approx N \times (F_{\text{teacher}} + F_{\text{student}} + B_{\text{student}}) \tag{24}$$

$$C_{\text{on}} \approx N \times (G_{\text{student}} + \rho F_{\text{teacher}} + F_{\text{student}} + B_{\text{student}}) \tag{25}$$

where F, B denote forward and backward FLOPs, G is the autoregressive generation cost, and $\rho \in (0, 1]$ is the teacher supervision refresh rate (the fraction of steps requiring a fresh teacher forward pass). Because $G_{\text{student}} \gg F_{\text{student}}$, on-policy training incurs a substantial multiplier. For generic knowledge transfer where the off-policy ceiling is high, this overhead is rarely justified. For reasoning, code generation, and complex mathematics where the off-policy ceiling is low and marginal on-policy returns are high, the investment typically pays off. In practice, the most effective industrial pipelines tend to front-load cheap off-policy learning and reserve expensive on-policy compute for the final quality push (Section 8.5).

Summary decision rule. Use off-policy SFT alone when the teacher-student capacity ratio exceeds $10\times$ and the task admits bounded reasoning depth (factual QA, translation, summarization). Switch to on-policy OPD when any of three conditions holds: (1) the student exceeds $\sim$7B parameters and begins to explore regions uncovered by static teacher traces, (2) the target task involves multi-step reasoning where intermediate errors compound, or (3) the off-policy loss plateaus while held-out reward continues to improve under on-policy rollouts. In all other cases, the hybrid pipeline (off-policy warmup followed by on-policy refinement) yields the most robust cost-quality tradeoff.

8 Applications, Systems, and Emerging Domains

The preceding sections developed OPD from theory through methods to unified understanding. This section examines how these algorithmic advances translate into practice. The shift from research methods to industrial deployment suggests that the choice of OPD method is often shaped less by theoretical elegance than by practical constraints, including available compute, teacher access level, the need for multi-teacher coordination, system-level throughput bottlenecks, and domain-specific challenges such as cross-modal transfer and privacy. We organize industrial applications by their *deployment pattern* and emerging domains by the new challenges they introduce.

8.1 Industrial Deployment

Recent foundation models have increasingly incorporated OPD into their post-training pipelines, with four distinct deployment patterns that align with the theoretical progression surveyed above.

Two-phase distillation pipelines. The most common industrial pattern combines off-policy cold-start with on-policy refinement. Qwen3 (Yang et al., 2025a) employs strong-to-weak distillation where larger models (32B, 235B-A22B) supply logit-level supervision on student-generated on-policy sequences, with the student generating responses in either thinking or non-thinking mode. The central architectural choice is “mode fusion,” where the student must learn to produce both short, direct answers (non-thinking mode) and extended chain-of-thought reasoning (thinking mode) by distilling from teachers operating in each mode independently. On-policy generation plays a central role here because the student’s mode-selection behavior (when to think vs. when to answer directly) is itself a learned policy that is difficult to capture with static datasets. Qwen3 reports that on-policy distillation outperforms direct reinforcement learning at roughly one-tenth of the GPU hours, and further improves pass@64 on AIME benchmarks where reinforcement learning from the same off-policy checkpoint does not. Gemma 2 (Gemma Team et al., 2024) embeds knowledge distillation directly into pre-training to bridge performance gaps across its model family, distilling the 27B teacher into both the 9B and 2B students independently (parallel distillation instead of cascading). This represents a more aggressive integration where OPD is not a post-training add-on but a core training ingredient from the start, with the architectural insight that parallel teacher-student pairs avoid compounding approximation errors that cascade topologies would introduce. MiMo-V2 (Xiaomi LLM-Core Team et al., 2026) combines distillation from multiple domain-specialized teachers with RL-based training, reaching strong performance on mathematical reasoning and agentic tasks with a 309B MoE model (15B active parameters).

OPD as model consolidation. A second pattern uses on-policy distillation for *merging* multiple specialized experts into a single deployable model. DeepSeek-V4 (DeepSeek-AI, 2026) entirely replaces the mixed RL stage of its predecessor (DeepSeek-V3.2) with pure multi-teacher OPD, consolidating more than ten domain-specific experts (mathematics, coding, agent, instruction-following) into a unified 1.6T-parameter model through full-vocabulary Reverse KL distillation. Computing the full $|V|$-dimensional KL at each position (rather than the sampled-token approximation) ensures more stable gradients when merging experts with diverse specializations, while hidden-state caching with on-the-fly logit reconstruction makes this feasible across trillion-parameter teachers (Section 4.1). KAT-Coder-V2 (Li et al., 2026a) decomposes agentic coding into five expert domains, each undergoing independent SFT and RL training, and consolidates them via on-policy distillation, scoring 79.6% on SWE-bench Verified. Nemotron-Cascade 2 (Yang et al., 2026d) applies the same principle to a 30B MoE model with only 3B activated parameters, reaching Gold Medal-level performance on IMO, IOI, and ICPC World Finals with 20× fewer parameters than DeepSeek-V3.2-Speciale. CoPD (Gu et al., 2026) identifies a key limitation of sequential expert-then-distill pipelines, namely that large behavioral pattern gaps between fully-trained teachers and untrained students prevent effective capability absorption. The root cause is that experts trained in isolation develop idiosyncratic reasoning strategies that are mutually incompatible when merged. CoPD's solution is *co-evolution*, running parallel RLVR expert training with interleaved bidirectional OPD so that experts serve as mutual teachers during their own training rather than after completion. This maintains consistent behavioral patterns among experts while preserving complementary knowledge. The result is a single model integrating text, image, and video reasoning capabilities that surpasses both mixed RLVR and sequential multi-teacher OPD (MOPD). The deeper insight is architectural. Co-evolution imposes a soft constraint on teacher diversity, preventing the specialization-consolidation gap that plagues post-hoc distillation from independently trained experts.

OPD for multi-budget reasoning. ORBIT (Liang et al., 2026) targets the deployment challenge of variable reasoning depth, where users need different compute budgets depending on query difficulty. It produces stage-wise expert policies through RLVR under successively tighter context budgets ($L_{k+1} = L_k/2$), then fuses them into a single model through mode-aware Reverse KL:

$$\mathcal{L}_{\text{fusion}}(\phi) = \mathbb{E}_{k\sim U(1,K)}\mathbb{E}_{\substack{q\sim\mathcal{D} \\ o\sim\pi_\phi(\cdot|q,p_k)}}\left[\sum_t \log \frac{\pi_\phi(o_t|q,p_k,o_{<t})}{\pi_{\theta_k}(o_t|q,p_k,o_{<t})}\right] \tag{26}$$

where p_k is the mode-specific prompt and $\{\pi_{\theta_k}\}$ are the frontier experts. Model merging initializes the student to mitigate the cold-start problem identified by Jang et al. (2026). Without initialization near the expert manifold, the student's on-policy rollouts rarely reach the teachers' high-density regions, starving the distillation objective of useful signal. Across DeepSeek-Distill-Qwen-1.5B, Qwen3-4B, and Openmath-Nemotron-7B, ORBIT produces models that smoothly interpolate between four reasoning modes (2K to 32K token budgets) with competitive performance at each budget level, matching the initial model's accuracy at the highest budget while delivering significant token savings at lower budgets. The connection to the consolidation pattern above is instructive. Where KAT-Coder-V2 and Nemotron-Cascade 2 merge experts specialized by *domain*, ORBIT merges experts specialized by *compute budget*, suggesting that on-policy distillation is emerging as a general-purpose multi-teacher fusion tool for any axis of model specialization.

Agentic distillation. Extending OPD from single-turn generation to multi-turn agentic tasks introduces a distinct challenge: *error compounding*. In single-turn settings, a flawed token affects only subsequent tokens within the same sequence. In multi-turn agent settings, a flawed action at step t shifts the entire environmental state, causing all subsequent teacher supervision to operate on a distribution far from what the teacher encountered during its own training. This error compounding problem motivates three complementary solutions that attack it at different granularities.

At the *trajectory level*, TCOD (Wang et al., 2026c) prevents compounding by controlling the temporal depth of teacher supervision. Its temporal curriculum progressively expands the horizon, initially supervising only short interaction prefixes (where compounding is minimal) and gradually extending to full trajectories as the student's early-turn reliability improves. This directly counters the "Trajectory-Level KL Instability" that TCOD identifies empirically, gaining up to +18 points over vanilla multi-turn OPD on ALFWorld, WebShop, and ScienceWorld.

At the *step level*, MAD-OPD's OPAD variant (Wang et al., 2026b) offers an orthogonal decomposition. Rather than limiting *which* steps receive supervision, it treats each agentic step as an independent distillation unit, breaking the error propagation chain by preventing gradients from flowing across step boundaries. This step-level isolation is complemented by the multi-teacher debate ensemble, which targets a *second* agentic failure mode, namely single-teacher unreliability at later trajectory steps. When individual teachers err on complex sequential decisions, the collective intelligence of debating teachers yields more robust supervision than any single teacher can. Combined with task-adaptive divergence selection (JSD for agentic stability, Reverse KL for code), OPAD lifts the agentic average by +2.4% over single-teacher OPD.

At the *skill level*, OpenClaw-RL (Wang et al., 2026e) uses hindsight-guided OPD with a Process Reward Model that delivers step-level credit assignment, training a Qwen3-4B agent across personal, terminal, GUI, and SWE environments. Skill-SD (Wang et al., 2026a) applies importance-weighted Reverse KL with dynamic teacher synchronization for skill-conditioned learning, posting +14.0% over GRPO on AppWorld and +10.9% on Sokoban.

The progression from trajectory-level control (TCOD) through step-level isolation (MAD-OPD/OPAD) to skill-level decomposition (Skill-SD) exposes a general design principle. Effective agentic distillation requires matching the granularity of the distillation unit to the granularity at which errors compound. Coarse-grained approaches (full-trajectory OPD) fail because compounding is local. Fine-grained approaches (token-level) waste compute on positions where the agent's behavior is already correct. The emerging consensus suggests that the optimal granularity is the *decision boundary*, the point at which the agent commits to an action whose consequences are difficult to reverse. This parallels findings in hierarchical RL (Ross et al., 2011), where option boundaries similarly define the natural unit of credit assignment, and suggests that future agentic OPD methods may benefit from automated decision-boundary detection instead of hand-specified decompositions.

These four patterns reflect a broader shift in how industry treats OPD, namely as a quality refinement tool (improving an existing pipeline's outputs), an architectural simplification tool (collapsing a multi-expert system into a single deployable model), an inference-time control mechanism (allowing a single model to serve multiple latency-accuracy operating points), and an enabler for capability domains where traditional training falls short (agentic, multi-turn, multi-modal). All four patterns rely on on-policy generation rather than static dataset distillation, consistent with the theoretical advantages of distributional alignment surveyed above.

8.2 Emerging Domains

Multimodal OPD. Vision-language models face unique exposure bias challenges because multimodal interleaved generation must maintain coherence across modalities, and high-quality visual reasoning data is scarce compared to text. A natural question arises: can on-policy distillation transfer knowledge *across* modality boundaries, and if so, what structural conditions make this feasible?

VOLD (Bousselham et al., 2025) answers affirmatively by showing that a text-only teacher can guide a VLM student through on-policy distillation without ever observing the visual input. The teacher supplies reasoning supervision on student-generated visual reasoning traces, leveraging text-based reasoning structure as a modality-agnostic scaffold. The authors report that an SFT cold-start phase is important for cross-modal distributional alignment. Without it, the student's on-policy rollouts in the visual reasoning space are too far from the teacher's text-based distribution for distillation to converge. On MMMU-Pro, VOLD improves Qwen2.5-VL-3B from 27.1% to 32.0%, consistent with the hypothesis that the reasoning structure itself (rather than perceptual grounding) carries much of the transferable knowledge.

This cross-modal principle extends naturally along two axes: temporal depth and modality self-alignment. Video-OPD (Li et al., 2026c) tackles temporal video grounding, where the student must localize events within video sequences, adding a temporal dimension that static image reasoning lacks. X-OPD (Cao et al., 2026) and CORD (Hu et al., 2026) then push beyond teacher-student pairs entirely, realizing that different modality modes of the *same* model can serve as mutual teachers. X-OPD distills a text-mode Qwen3-Omni into its own speech mode by generating on-policy speech responses and aligning them with its text-mode reasoning. CORD formalizes this with weighted Reverse KL between audio and text modes:

$$\mathcal{L}_{\text{CORD}} = \mathbb{E}_{x \sim \mathcal{D}} \mathbb{E}_{\hat{y} \sim \pi_{\text{audio}}(\cdot|x)} \left[\sum_t w_t \cdot D_{\text{KL}}(\pi_{\text{audio}}(\cdot|x, \hat{y}_{<t}) \parallel \pi_{\text{text}}(\cdot|x, \hat{y}_{<t})) \right] \tag{27}$$

where the audio mode generates on-policy rollouts and the text mode supplies token-level targets, combined with GRPO for outcome reward maximization. CORD reports strong speech reasoning performance, suggesting that cross-modal self-distillation can serve as a general mechanism for aligning weaker modality modes to stronger ones within a single model.

KEPO (Yang et al., 2026b) identifies a prerequisite for all these multimodal OPD methods: when initial solve rates approach zero (common in visual mathematics), sparse RLVR fails due to exploration collapse before any meaningful on-policy data can be generated. By conditioning preference optimization on teacher knowledge, KEPO supplies the dense bootstrapping signal that allows subsequent on-policy training to converge. VISD (Lin et al., 2026) extends the paradigm to video reasoning through structured self-distillation, where a video-aware judge decomposes reasoning quality into multiple diagnostic dimensions and a direction-magnitude decoupling mechanism combines RL stability with fine-grained OPD credit assignment, recording $2\times$ faster convergence than RLVR baselines (see Section 5.3.1 for details). The progression from VOLD through CORD to KEPO and VISD thus maps a complete pipeline: dense bootstrapping (KEPO) $\rightarrow$ cross-modal transfer (VOLD) $\rightarrow$ intra-model self-alignment (X-OPD, CORD) $\rightarrow$ structured self-distillation for temporal reasoning (VISD).

Embodied intelligence and physical reasoning. Moving beyond language-only domains, OPD is increasingly applied to settings where the output space is continuous and errors are physically irreversible. These applications form a spectrum from high-level spatial reasoning through planning to low-level motor control, each demanding increasingly tight distributional alignment.

At the reasoning end of this spectrum, HY-Embodied-0.5 (Tencent Robotics X et al., 2026) uses multi-stage OPD to compress a 32B expert VLM into a 2B embodied MoT model for spatial reasoning, outperforming similarly sized models on 16 out of 22 benchmarks. The task here is cognitive (understanding spatial relations), not physical, and OPD's role is primarily model compression with minimal distributional mismatch.

Moving toward physical planning, OPD-AV (Afsharrad et al., 2026) applies GKD with $5\times$ compression for autonomous driving on nuScenes, distilling a Qwen3-8B SFT model into a 1.7B student. The challenge intensifies because planning errors compound through time. A suboptimal lane-change decision at t constrains all future maneuvers, making on-policy training essential to expose the student to the consequences of its own imperfect decisions. GUI-SD (Zhang et al., 2026f) tackles an analogous compounding problem in graphical interfaces, where the student generates click trajectories while a privileged version of itself (with access to bounding boxes and Gaussian soft masks) supplies token-level KL supervision. Its entropy-guided distillation selectively weights tokens where teacher-student divergence is highest, focusing compute on the decision points that most affect downstream trajectory quality and reaching state-of-the-art on 6 GUI grounding benchmarks.

At the motor control end, VLA-OPD (Zhong et al., 2026) applies Reverse KL on self-generated robot trajectories, distilling expert demonstrations into a Vision-Language-Action student. Because the action space is continuous rather than discrete, distributional mismatch manifests as physically unsafe trajectories (collisions, drops) instead of mere quality degradation. The student must learn precise motor control from the teacher's dense token-level supervision on its own trajectories, attaining better sample efficiency than pure RL and better robustness than offline SFT.

The consistent finding across this spectrum is that on-policy generation becomes *more* critical as we move from cognitive to physical tasks. The diversity of valid outputs (multiple correct grasping trajectories, multiple valid temporal localizations, multiple acceptable spatial descriptions) makes off-policy data coverage impractical, and the consequence of distributional mismatch escalates from lower benchmark scores to physical failure. This gradient suggests that embodied OPD will increasingly require tighter integration with environment simulators that can supply the dense rollout data necessary for safe policy transfer.

Medical agents and tool-augmented clinical reasoning. Clinical decision making poses a different set of challenges as an emerging domain. The task is multi-turn by nature (history taking, test ordering, result interpretation, treatment selection), tool libraries are large and heterogeneous (medical knowledge bases, imaging interpreters, drug interaction checkers), and the terminal reward is sparse relative to the episode length. Healthcare AI Gym (Jeong, 2026) offers the first gymnasium-compatible environment at this scale, covering 10 clinical domains, 3.6K+ tasks, 135 domain-specific tools, and 828K medical passages, and uses it to systematically study how OPD behaves in tool-augmented clinical reasoning. Beyond benchmarking, the study exposes an agentic-textual transfer gap. RL-based OPD reliably improves procedural competence on multi-turn benchmarks but does not translate into gains on text-only knowledge-recall benchmarks (MedMCQA, MMLU-Med), indicating that agentic training specializes the model for interactive reasoning at the cost of static parametric recall. The proposed TT-OPD method (discussed in Sections 5.3.1 and 7.2) reaches the best score on 10 of 18 benchmarks, with an average +3.9 percentage-point improvement over a non-RL baseline, while sustaining stable multi-turn structure throughout training. Taken together, Healthcare AI Gym suggests that high-stakes specialist domains are not merely *scaled-up* agentic environments but require distillation objectives that explicitly decouple procedural competence from parametric recall, a distinction absent in current single-domain agentic benchmarks.

Privacy-preserving OPD. DP-OPD (Khadem et al., 2026) injects differential privacy noise (DP-SGD) into the student's gradients during on-policy training while keeping the teacher frozen. The study shows that on-policy querying can extract useful utility while providing formal differential-privacy guarantees on the teacher's training data, tackling a growing concern as distillation from proprietary models becomes standard practice. Without such guarantees, on-policy querying carries a risk of memorizing and reproducing private training examples from the teacher.

8.3 System-Level Integration

The compute optimization methods surveyed in Section 6.3 target algorithmic efficiency, but industrial-scale OPD also requires systems-level solutions that span the full hardware-software stack.

Production OPD pipelines must orchestrate three concurrent workloads, namely student rollout generation (autoregressive, memory-bound), teacher scoring (compute-bound forward pass), and student gradient update (compute-bound backward pass). Frameworks like OpenRLHF (Hu et al., 2024) and veRL (Sheng et al., 2024) decompose these into separate process groups with distinct parallelism strategies. Teacher inference uses tensor parallelism across 4–8 GPUs for low latency, student generation uses pipeline parallelism for throughput, and the optimizer uses ZeRO-3 for memory efficiency. The main engineering challenge is synchronization. Since the student's policy changes after each gradient step, all pending rollouts become stale. Asynchronous architectures that tolerate slight policy staleness gain higher throughput at the cost of noisier gradients. High-throughput serving engines (vLLM (Kwon et al., 2023), TensorRT-LLM (NVIDIA, 2024)) serve the generation phase, with continuous batching and PagedAttention allowing efficient rollout generation across large prompt pools. White-box OPD additionally requires exposing the full vocabulary logits rather than just the sampled token. The communication cost is substantial.

A 70B teacher with 7B student on 8×H100 must transmit logit data exceeding 16 GB per batch ($[B, T, |V|] \times 2$ bytes in BF16, e.g., B=16, T=4096, $|V|$=128K), feasible within a single node via NVLink (900 GB/s) but requiring careful tensor slicing or top-k sparsification for multi-node setups.

DeepSeek-V4 (DeepSeek-AI, 2026) offers the most detailed public account of full-vocabulary OPD systems engineering at trillion-parameter scale, tackling the specific challenges of distilling from 10+ teachers each comprising trillions of parameters. Rather than materializing the full $|V| > 100$K logit tensor for each teacher (which is prohibitive when distilling from 10+ teachers simultaneously), the framework caches only the last-layer hidden states in a centralized buffer. It reconstructs logits on-the-fly through the corresponding prediction head at training time, virtually eliminating memory pressure while incurring negligible recomputation cost. Training samples are ordered by teacher identity within each mini-batch, so that at most one teacher's prediction head resides in GPU memory at any time. All weight loading and offloading proceed asynchronously without blocking the critical path. The framework further integrates FP4 quantization-aware training during the OPD phase itself, quantizing MoE expert weights and attention QK activations so that the distilled model is natively adapted to deployment-time precision from the start rather than requiring post-hoc quantization.

8.4 When to Use On-Policy vs. Off-Policy

The theoretical framework above produces concrete engineering implications. On-policy training is not universally superior; its benefit depends on whether the specific setting satisfies conditions that justify the additional compute overhead.

The choice between off-policy and on-policy distillation tends to reduce to a small set of conditions. Off-policy SFT is often sufficient when the target distribution is well-covered by the teacher's beam-search outputs (instruction following, factual QA) and three conditions hold simultaneously, including an exceptionally strong teacher (>500B), diverse reasoning traces, and a relatively small student (≤70B). DeepSeek-R1 (DeepSeek-AI et al., 2025) exemplifies this regime. The transition to on-policy becomes attractive when (1) the student exhibits clear exposure bias (performing well on teacher-style but not user-generated prompts), (2) the task involves multi-step reasoning where compounding errors degrade quality (Ross et al., 2011; Gudibande et al., 2023), or (3) the student needs to exceed the teacher via reward-guided exploration (Yang et al., 2026c). A rough formal criterion is whether the inference-time state visitation $d_{\pi_\theta}(s)$ diverges substantially from training-time $d_{\mathcal{D}}(s)$.

Within on-policy methods, the choice between logit-level and reward-level supervision is influenced by student scale. Smaller students (<7B) often benefit from dense logit supervision (GKD, DistiLLM) to bootstrap competence, while larger students (≥7B) with reliable verifiers tend to benefit from reward-guided OPD (KDRL (Xu et al., 2025a), G-OPD (Yang et al., 2026c)) that allows discovery beyond the teacher ceiling. The hybrid approach combining off-policy SFT warm-up with on-policy refinement consistently performs well across the surveyed literature (Qwen3 (Yang et al., 2025a), MiMo-V2 (Xiaomi LLM-Core Team et al., 2026)). At a deeper level, OPD and RLHF share much of their computational structure, including on-policy rollouts, per-trajectory signals, and policy gradient updates. The main distinction is the signal source (teacher distribution vs. reward model preference). Methods like G-OPD and REOPOLD (Ko et al., 2026) formalize this relationship, with practical consequence. For reward-guided OPD, much of the existing RLHF infrastructure applies directly (the rollout-score-update loop is essentially the same), which helps explain why frameworks like OpenRLHF (Hu et al., 2024) serve both communities. White-box logit-level OPD additionally requires teacher co-hosting and full-vocabulary logit transfer, an extension that existing frameworks support through their teacher-inference process groups.

8.5 The Distillation Tax: Compute Budget Allocation

Based on the compute analysis in Section 6.3 and the staged pipelines described in recent industrial reports (DeepSeek-AI et al., 2025; Yang et al., 2025a), a practical rule of thumb is observable for compute budget allocation. Many of the more effective industrial pipelines reported so far follow a three-stage pattern:

1. **Off-policy warm-up** (majority of budget): Standard SFT on teacher-generated data supplies cheap, high-bandwidth learning that establishes the student's basic capabilities and aligns its distribution close enough to the teacher for effective on-policy training.
2. **On-policy logit distillation** (largest on-policy share): Token-level OPD with adaptive divergences (GKD, DistiLLM, ToDi) closes the distribution gap on the student's own generation trajectories, mitigating exposure bias where it matters most.
3. **Reward-guided refinement** (final smaller share): RL-augmented objectives (G-OPD, KDRL, RLAD) push the student beyond the teacher's capability ceiling on reasoning tasks, using sparse outcome rewards to discover new solution paths.

This staged approach front-loads cheap, high-bandwidth learning and reserves expensive on-policy compute for the final quality push. Qwen3 (Yang et al., 2025a) exemplifies this pattern, with its multi-stage post-training pipeline (cold-start SFT $\rightarrow$ domain RL for math and code $\rightarrow$ mode-fusion SFT $\rightarrow$ general RL) allocating gradually more compute to on-policy methods as training progresses, while distillation from larger models (32B, 235B-A22B) serves as the primary transfer mechanism for smaller variants. On-policy training tends to be most useful after off-policy warm-up has brought the student close to the teacher's distribution, since this reduces the wasted rollouts that occur when the student-teacher gap is too large (Section 7.1).

9 Open Problems and Future Directions

The methods surveyed above follow a recognizable trajectory, moving from fixed objectives to adaptive ones, from external teachers to self-generated signals, and from single-turn generation to multi-turn trajectories. Continuing along this trajectory, distillation, reinforcement learning, and alignment look less like separate training stages than facets of a single optimization process in which the model generates, evaluates, and refines its own behavior under evolving constraints. The open problems below are organized along this direction.

Distillation scaling laws. While parameter-scaling laws for pre-training are well-established (Kaplan et al., 2020; Hoffmann et al., 2022), the study of distillation-specific scaling has only recently begun. Busbridge et al. (2025) proposed the first distillation scaling laws under an off-policy setting, exposing a non-monotone relationship between compute budget and optimal teacher size: as compute increases, the optimal teacher first grows until it slightly exceeds the student, then plateaus, and eventually *decreases* because inference cost with large teachers dominates the compute budget. Meanwhile, optimal student distillation tokens continue to scale as a power law. DeepSeek-R1 (DeepSeek-AI et al., 2025) offers complementary off-policy evidence on the student-size axis (Section 7.4): on AIME 2024, performance scales as 28.9% $\rightarrow$ 55.5% $\rightarrow$ 69.7% $\rightarrow$ 72.6% for student sizes 1.5B $\rightarrow$ 7B $\rightarrow$ 14B $\rightarrow$ 32B, with the steepest gain between 1.5B and 7B and sharply diminishing returns beyond 14B, indicating that student capacity follows a power-law saturation pattern even under off-policy distillation.

Yet no equivalent framework exists for on-policy distillation, where the generation cost of student rollouts introduces an additional compute axis absent from off-policy settings. By analogy with these off-policy findings, a natural conjecture is that the on-policy distillation loss follows a joint power-law of the form

$$L(N_S, N_T, D_{\text{on}}) = E + \frac{A}{N_S^{\alpha}} + \frac{B}{N_T^{\beta}} + \frac{C}{D_{\text{on}}^{\gamma}} + f(N_S, N_T) \tag{28}$$

where E is the irreducible task entropy, the first three terms capture independent scaling of student size, teacher size, and on-policy rollout budget, and $f(N_S, N_T)$ models the capacity-gap interference between student and teacher. While no study has yet validated this functional form for on-policy settings, the off-policy evidence above constrains its shape: the non-monotone teacher-size finding implies that B/N_T^{β} alone is insufficient and the interaction term f must capture the regime where teacher inference cost overwhelms quality gains. Qwen3 (Yang et al., 2025a) further shows that distilling from multiple specialized teachers yields better scaling than from a single monolithic teacher of equivalent total parameters, indicating that f must additionally account for teacher diversity. The critical open question is whether the on-policy rollout term C/D_{on}^{γ} interacts synergistically or antagonistically with the teacher-size term, that is, whether more on-policy data can compensate for a weaker teacher, or whether teacher quality sets a hard ceiling regardless of rollout volume. Future work should conduct controlled grid searches that independently vary N_S, N_T, and D_{on} under on-policy training to disentangle these effects, addressing fundamental allocation questions such as "Given a fixed GPU budget, is it more effective to invest in a larger teacher or in more student rollouts?" Establishing such scaling laws would put OPD compute allocation on the same predictable footing as pre-training scaling.

Uncertainty-aware feedback. Even with perfect scaling laws, a second foundational gap remains. Teachers currently supply point-estimate probabilities. When the teacher itself is uncertain about a token, its logits carry little useful information, yet the student treats all teacher signals equally. Future OPD frameworks should incorporate the teacher's *epistemic uncertainty*, allowing the student to discount feedback on queries where the teacher is guessing. Process reward models (Lightman et al., 2024) offer a potential path by scoring each reasoning step independently, allowing identification of precisely where teacher supervision is reliable versus speculative.

The connection to the failure-mode analysis of Section 7.2 is direct. The flawed prefix trap is at its core a problem of unrecognized teacher uncertainty, and explicitly modeling this uncertainty would yield a principled solution. The core intuition is to suppress the distillation gradient at positions where the teacher's output entropy approaches its maximum, effectively treating such positions as uninformative. TIP (Xu et al., 2026b) offers empirical support for this direction, showing that teacher-student divergence-based filtering outperforms entropy-only schemes, suggesting that *relative* uncertainty (how surprised the teacher is by the student's continuation) matters more than the teacher's absolute confidence.

The next frontier is *predictive* uncertainty. Rather than measuring the teacher's entropy on the student's current prefix, the goal would be to estimate what the teacher's reliability would be several tokens ahead, allowing the student to avoid committing to reasoning paths that will lead to regions of teacher incompetence. Systematically avoiding these paths before the student wastes compute exploring them represents a major unexploited efficiency gain.

Agent-level, continual, and lifelong distillation. The uncertainty problem intensifies when distillation extends beyond single-turn generation. Extending to multi-turn agentic settings requires new credit assignment mechanisms, as the student must learn long-horizon tool use, environmental state tracking, and error recovery across entire interaction trajectories. OpenClaw-RL (Wang et al., 2026e), Skill-SD (Wang et al., 2026a), TCOD (Wang et al., 2026c), and MAD-OPD's OPAD variant (Wang et al., 2026b) represent early steps, but the systematic degradation of teacher supervision with trajectory depth (Li et al., 2026d) remains a hard open problem. MAD-OPD's step-level sampling strategy offers a partial mitigation by treating each agentic step as an independent distillation unit instead of propagating errors across the full trajectory, while TCOD's temporal curriculum controls how much of the trajectory receives teacher supervision at each training stage. Together, these methods suggest that the multi-turn distillation frontier requires *both* temporal decomposition (breaking trajectories into manageable units) and progressive exposure (gradually increasing horizon complexity).

Three unsolved sub-problems define this frontier. *First, environment non-stationarity*. Agentic environments change in response to actions, making direct trajectory comparison meaningless unless the teacher delivers counterfactual evaluations conditioned on the student's actual state. *Second, tool-use combinatorics*. Modern agent frameworks expose dozens of tools with structured argument schemas, suggesting distillation may need to operate at the tool-call level instead of individual tokens, with credit assignment mechanisms attributing trajectory success to specific tool selections. *Third, safety-critical exploration*. A single incorrect file write in coding agents or misguided click in web-browsing agents can trigger irreversible actions, requiring safety constraints that prevent exploring catastrophically dangerous action sequences.

On a separate axis, production models undergo repeated fine-tuning cycles (safety patches, capability updates), raising the question of how distillation objectives should adapt when the teacher itself evolves. SDFT (Shenfeld et al., 2026) and OEL (Ye et al., 2026a) offer initial answers, but *lifelong distillation*, attaining re-alignment without catastrophic forgetting, without retraining from scratch, and without accumulating distributional drift, connects to continual learning with the added complication that both teacher and student distributions are non-stationary.

Efficiency frontiers. The agentic and multi-turn settings above amplify what is already a central systems bottleneck of OPD. The considerable compute overhead of on-policy training remains a barrier. While Fast OPD (Zhang et al., 2026a), Lightning OPD (Wu et al., 2026), and Speculative KD (Xu et al., 2025c) have made progress, the theoretical minimum cost of on-policy distillation (the information-theoretic lower bound on how much on-policy data is needed to reach a given quality level) remains unknown. A promising underexplored direction is *selective teacher inference*. MiniPLM (Gu et al., 2025) shows that selecting training instances based on the log-probability discrepancy between teacher and a reference model ("Difference Sampling") substantially improves data efficiency under off-policy KD. Transplanting this principle to on-policy rollouts, using teacher-reference divergence to prioritize which student-generated sequences deserve dense teacher feedback, could yield significant compute savings by skipping expensive teacher inference on uninformative rollouts where the student has already converged.

Latent-space and cross-modal distillation. Efficiency improvements tackle the *cost* of OPD, but a separate limitation constrains its *scope*. Current OPD operates exclusively in vocabulary space, where cross-architecture alignment requires expensive projections (DSKD, Cross-Tokenizer KD). A more radical approach would distill in a *shared latent space*, bypassing the vocabulary bottleneck entirely and allowing distillation between architecturally dissimilar models that share no structural overlap. The extension to multimodal settings raises additional open questions. How does exposure bias manifest when multiple modalities must maintain coherence? What are the optimal divergence choices when teacher and student operate in different modality spaces? Can the unified f-divergence framework be extended to handle continuous signal spaces (images, audio) alongside discrete tokens? VOLD (Bousselham et al., 2025), X-OPD (Cao et al., 2026), and CORD (Hu et al., 2026) offer empirical starting points, but the theoretical foundations for cross-modal OPD remain to be established.

Privacy and evaluation methodology. As OPD becomes more powerful and more widely deployed, questions of trust, both in the training process and in its evaluation, grow increasingly urgent. DP-OPD (Khadem et al., 2026) offers the first principled framework for differentially private distillation, with an asymmetric design applying DP-SGD only to the student while keeping the teacher frozen. The on-policy formulation is well-suited to the DP setting because training on self-generated prefixes reduces compounding errors from DP noise. Open questions include scaling DP-OPD to reasoning-capable LLMs and integrating DP with self-distillation. On the evaluation front, Fu et al. (2026) demonstrate that token-level OPD can appear successful on standard benchmarks while failing on distribution-shifted prompts, highlighting a gap between benchmark performance and true generalization. The precision-recall framework of Cha & Cho (2025) shows distillation induces a deep trade-off between sample precision and distributional coverage, suggesting that evaluation protocols need to probe whether the student has learned underlying causal mechanisms rather than superficial correlations.

Diagnostic tools for failure modes. A natural extension of uncertainty-aware training is developing tools to detect when distillation breaks down entirely. As the scale and complexity of OPD increase, the failure modes diagnosed in Section 7.2 demand more systematic diagnostic tools. While current literature identifies isolated traps, such as the flawed prefix trap and epistemic suppression (Kim et al., 2026a), the field lacks a unified framework to detect and quantify these pathologies during training. Real-time diagnostic probes that track gradient signal-to-noise ratios, representation collapse, and teacher-student divergence dynamics without incurring the full forward-pass cost would fill this gap. Drawing inspiration from gradient profiling in pre-training, such tools could automatically surface when a student enters the Ouroboros self-play saturation or when a multi-turn agent begins to suffer from trajectory-structure erosion (Jeong, 2026). Such probes would shift OPD debugging from post-hoc benchmark failure analysis to proactive mid-training intervention.

Cross-architecture scalability. A structural barrier intertwined with the efficiency frontier is the scalability of cross-architecture distillation. As explored in Section 5.1, mapping between heterogeneous model families introduces significant friction. While DSKD (Zhang et al., 2025b) and Cross-Tokenizer KD (Boizard et al., 2025) offer initial solutions via dual-space projection and optimal transport, these methods have primarily been validated on relatively small scale differences. The viability of these mappings when the capacity gap spans orders of magnitude, such as distilling a 400B MoE teacher into a 1B dense student, remains largely unproven. At such extremes, the representational bottleneck may preclude simple linear projections or vocabulary-level alignments. Developing non-linear, hierarchical alignment mechanisms that bridge massive architectural divides without prohibitive compute overhead remains an open challenge with significant practical implications.

Unified scheduling of OPD and RLVR. The convergence of distillation and reinforcement learning highlighted throughout this survey surfaces a critical scheduling challenge. While joint optimization objectives (G-OPD (Yang et al., 2026c), KDRL (Xu et al., 2025a), RLAD (Zhang et al., 2026h)) show the value of combining dense teacher guidance with sparse outcome rewards, the temporal scheduling of these signals remains largely heuristic. The co-evolutionary framework of CoPD (Gu et al., 2026) points toward continuous interleaving, but the fundamental question of how to allocate compute between imitating a teacher and exploring beyond it across the training lifecycle is still unanswered. A rigorous theoretical framework for this allocation would depend on the student's relative competence, implying that the optimal schedule is not a static hyperparameter but a dynamic policy that transitions from pure distillation (when the capability gap is large) to pure RL (when the student matches the teacher). Formulating this schedule as an active learning problem represents a highly promising frontier.

Closing the distillation-RL loop. The open problems above (scaling, uncertainty, agents, efficiency, cross-modal transfer, privacy) are individually significant, but they share a common resolution pathway. The field has historically treated distillation and RL as separate, linearly staged processes. Static distillation saturates when the student exhausts the teacher's knowledge, while pure RL diverges when reward signals are sparse. Chen et al. (2025) show that RL retains more pre-training capabilities than SFT because on-policy rollouts maintain distributional proximity to the student's prior. KDRL (Xu et al., 2025a) jointly optimizes KD and RL, reducing the catastrophic forgetting common to sequential pipelines. REOPOLD (Ko et al., 2026) offers a further instantiation, with its relaxed on-policy framework combining dense KD gradients for early-stage stability with RL-style reward signals for late-stage exploration. CoPD (Gu et al., 2026) shows that the sequential pipeline (train experts first, then distill) can be replaced entirely by *co-evolutionary* training where RLVR and bidirectional OPD are interleaved (Section 8.1), maintaining behavioral proximity that enables continuous knowledge absorption instead of post-hoc transfer. A natural extrapolation of this co-evolution principle is *simultaneous* optimization in which distillation and RL operate as complementary forces throughout training. Principled methods for interleaving these objectives, deciding when to switch, and preventing RL-discovered capabilities from being overwritten during subsequent distillation phases remain open.

Toward self-improving systems. A natural endpoint of the trends above, from external teachers to self-distillation, from static datasets to on-policy generation, from single-shot training to iterative refinement, is a system that continues to improve with limited human intervention. π-Play (Zhang et al., 2026g) and SSD (Zhang et al., 2026d) supply early existence proofs, showing models that improve through self-generated curricula and self-filtered training data. The theoretical ceiling of such systems, whether they can exceed the capability of any fixed teacher given sufficient compute and environmental grounding, connects to open questions about the limits of self-play in imperfect-information games. The saturation analysis of Section 7.2 identifies the boundary conditions: pure self-play converges to the model's own prior distribution unless grounded by external verification or internally generated privileged information. Whether it is possible to raise this ceiling without access to a stronger oracle, attaining genuine capability *creation* rather than mere capability *redistribution*, remains an open question at the intersection of distillation, RL, and learning theory. Progress in this direction is likely to depend on verification mechanisms that scale without human labeling.

10 Conclusion

This survey has systematically examined on-policy distillation for large language models, synthesizing contributions across foundational formulations, algorithmic innovations, theoretical analyses, and industrial deployments. We organize this fast-growing literature along three core design axes rather than surface-level categories. *Objective function* design progresses from fixed divergences through adaptive routing to RL-augmented objectives that lift the teacher ceiling. *Signal source* architecture spans from dense white-box logits through black-box API access to self-distillation that eliminates the external teacher entirely. *Training dynamics* encompasses token weighting, curriculum adaptation, and compute optimization.

Key Findings. Three patterns recur across the methods surveyed. *First*, shifting from off-policy to on-policy training distributions, from $y \sim p_{\text{data}}$ to $y \sim p_\theta$, has been associated with consistent accuracy gains across mathematical reasoning, code generation, and instruction following (Tables 2 and 3). The improvement is commonly attributed to reduced exposure bias, as the student learns to recover from its own errors instead of memorizing idealized trajectories, with the performance gap reportedly widening for longer reasoning chains where error compounding is more severe (Gu et al., 2024; Agarwal et al., 2024). *Second*, divergence selection appears to benefit from being adapted to both task and token position rather than fixed globally. Reverse KL has been observed to do well on reasoning tasks where mode-seeking can prevent hallucination, while Forward KL preserves the diversity often needed for open-ended generation. Adaptive methods such as ToDi (Jung et al., 2025) and Entropy-Aware OPD (Jin et al., 2026), which select divergences based on local distributional geometry, represent the current frontier. *Third*, the boundary between distillation and reinforcement learning continues to narrow. G-OPD (Yang et al., 2026c), RLAD (Zhang et al., 2026h), and REOPOLD (Ko et al., 2026) suggest that combining dense teacher supervision with sparse outcome rewards can be effective, with KD providing gradient stability while RL permits exploration beyond the teacher's capability. The convergence is not only conceptual but also yields a shared toolkit (trust regions, advantage estimation, policy gradient variance reduction) that both communities can draw on.

Practical Takeaways. With white-box teacher access, token-level methods (GKD (Agarwal et al., 2024), DistiLLM (Ko et al., 2024)) tend to offer a favorable quality-compute tradeoff for general instruction following. For reasoning-intensive tasks, sequence-level methods (MiniLLM (Gu et al., 2024)) or hybrid approaches (Fast OPD (Zhang et al., 2026a), PACED (Xu et al., 2026a), Lightning OPD (Wu et al., 2026)) are often preferred despite higher gradient variance inherent to REINFORCE-style estimators (Gu et al., 2024). With only API access, GAD (Ye et al., 2025) and OVD (Xiong et al., 2026) can support on-policy learning through adversarial and verbal formulations, respectively. Self-distillation methods (SPIN (Chen et al., 2024), OPSD (Zhao et al., 2026b), SD-ZERO (He et al., 2026)) remove the need for a separate teacher entirely, which makes them attractive under resource constraints. The substantial compute overhead of on-policy generation can often be reduced via speculative decoding (Xu et al., 2025c), prefix truncation (Zhang et al., 2026a), or offline precomputation (Wu et al., 2026).

Looking Ahead. The progression from off-policy imitation to on-policy self-correction echoes a broader pattern in machine learning. Classical supervised learning assumes a fixed target distribution, whereas on-policy distillation treats the learner's own distribution as part of the training objective. The methods surveyed here progressively relax the assumptions under which distillation was originally formulated as a compression technique. Fixed objectives are replaced by adaptive ones, external teachers by self-generated signals, and single-turn optimization by multi-turn trajectory learning. A natural continuation of these trends is a training regime in which models generate, evaluate, and update their own behavior more continuously rather than only during a one-shot training run. The absence of distillation scaling laws (Section 9), the open problem of teacher uncertainty quantification, and the challenge of lifelong adaptation without catastrophic forgetting (Kirkpatrick et al., 2017) appear to be among the main barriers between the current state of the art and such a regime. As LLMs evolve from static text generators into more interactive reasoning agents, the principles of on-policy alignment surveyed here seem likely to become a more integral component of the LLM training stack rather than an optional enhancement.

Limitations. This survey focuses exclusively on methods that involve on-policy generation from the student during training. We deliberately exclude general knowledge distillation (e.g., feature-based, attention transfer), model compression techniques orthogonal to the training regime (pruning, quantization), and methods that use student-generated data only at inference time (self-consistency, majority voting). Our coverage is current through April 2026. Given the field's rapid pace (more than ten new OPD papers per month through early 2026), some concurrent work may be absent. The method selection considerations in Section 3.3 reflect current empirical evidence and hardware constraints; as compute costs decrease and training frameworks mature, the cost-benefit calculus will shift toward more aggressive on-policy methods.

Reproducibility challenges. A significant concern across the OPD literature is the difficulty of fair comparison. Methods are evaluated on different base models, different training compute budgets, different benchmark versions (e.g., MATH-500 vs. full MATH), and with different numbers of rollouts per prompt. Few papers control for all these variables simultaneously, making cross-method comparison unreliable from published numbers alone. Our Tables 2 and 3 report original-paper results rather than controlled reproductions, and readers should interpret cross-row comparisons with appropriate caution. We advocate for standardized OPD benchmarking protocols that fix the base model, compute budget, and evaluation suite, analogous to what HELM (Liang et al., 2023) offers for general LLM evaluation.

Broader impact. OPD carries immediate implications for AI accessibility, safety, and environmental sustainability.

Regarding accessibility, effective distillation allows smaller, more efficient models that run on consumer hardware, democratizing advanced reasoning capabilities that would otherwise require expensive API access to frontier models. The self-distillation methods surveyed here (OPSD, SD-ZERO, SSD) stand out in this regard, as they permit capability improvement without any external teacher, allowing resource-constrained researchers and developers to iterate rapidly.

Regarding safety, OPD introduces both risks and mitigations. The risk is that effective distillation of frontier capabilities lowers barriers to misuse. The mitigation is that the teacher retains a supervisory role during training, offering a natural point for injecting safety constraints, alignment objectives, and capability boundaries. MSD (Qin et al., 2026) concretely instantiates safety-aware OPD, transferring safety alignment from high-resource to low-resource languages through on-policy self-distillation without requiring target-language safety data. By adaptively upweighting safety-critical tokens where teacher confidence and student disagreement are both high, MSD generalizes to unseen languages and more challenging jailbreak attacks, showing that OPD can serve as a scalable mechanism for multilingual safety alignment. As the field matures, we expect safety-aware OPD, where the teacher not only transfers knowledge but also enforces alignment properties during the interactive training loop, to become a critical research direction.

Regarding environmental sustainability, OPD's compute overhead (4–5$\times$ over off-policy SFT, Section 6.3) raises concerns at industrial scale. Yet the resulting smaller, more capable students reduce downstream inference costs by an order of magnitude or more, and the net carbon footprint over a model's lifetime is typically lower when amortized across billions of inference calls. Ultimately, the convergence of distillation, reinforcement learning, and self-play into a unified on-policy training framework suggests that future LLMs are less likely to be trained in one pass and then deployed unchanged, and more likely to be refined continuously through their interactions with users and environments.

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